

Ch.1- Introduction

Contents

- Imaging systems, Objects and images, The digital image processing system, Applications of digital image processing
- Fundamental models of image formation- Kinds of radiation and imaged properties, The imaging system- Point spread function, Imaging filters: Monochromatic, colour, multi-spectral and hyperspectral images, Resolution (pixel, spatial, radiometric/magnitude, spectral, temporal, super resolution)
- Image quality and uncertainties in image formation (digitization, quantum efficiency, metamerism, calibration, CNR, SNR)
- Major imaging modalities- Magnetic Resonance Imaging, Optical Imaging (inc. X-Ray, OCT, NIRS, microscopy, confocal imaging ,one and two photon imaging, fluoroscopy, CT), Electrical and magnetic imaging (inc. EEG/MEG, EMG, ECG, etc), Ultrasound

What is an Image?

- Image is a source of information according to information theory
- An image may be defined as a two-dimensional function $f(x, y)$ where x and y are spatial coordinates and amplitude of f at any pair of coordinates (x, y) is called the intensity or Gray level of the image at that point.

Digital Image

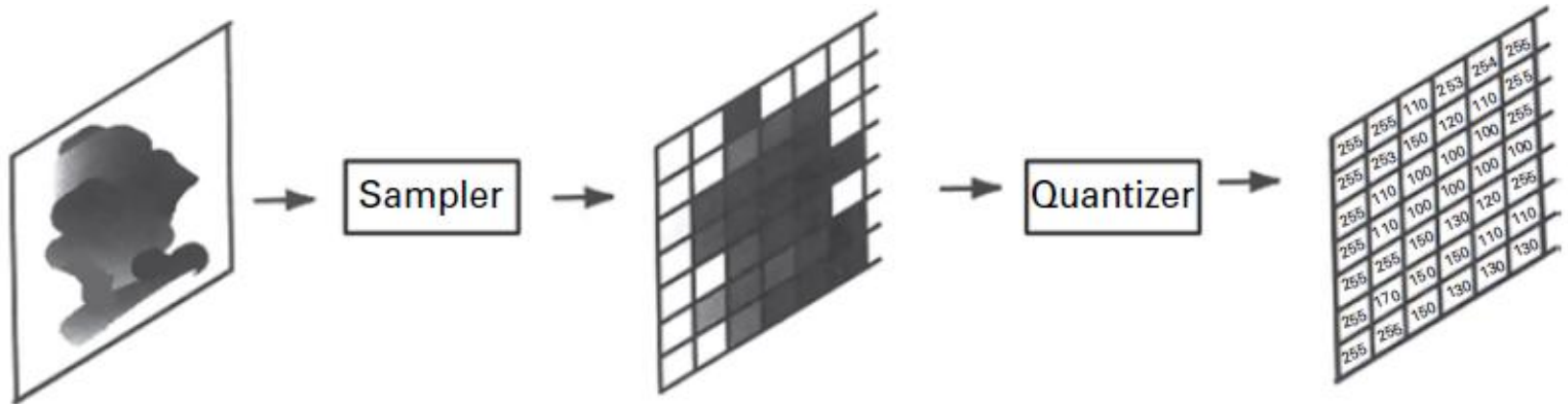
- When x , y and the amplitude values of f are all finite, discrete quantities, we call the image a Digital Image.
- A digital Image is composed of a finite number of elements each of which has a particular location and value
- These elements are referred to as Picture Elements, Image Elements, Pels or Pixels.

Pixel

- In digital imaging, a pixel is the smallest piece of information in an image.
- Pixels are normally arranged in a regular 2-dimensional grid, and are often represented using dots or squares
- The intensity of each pixel is variable; in color systems, each pixel has typically three or four components such as red, green, and blue, or cyan, magenta, yellow, and black

Objects and images

- An imaging system senses or responds to an input signal, such as reflected or transmitted electromagnetic radiation from an object, and produces an output signal or image



The relationship between an analog image and a digitized image.

Objects and images

- An imaging system →
 - continuous-to-continuous system- responding to a continuous input signal and producing a continuous or analog output image
 - continuous-to-discrete system- responding to the continuous input signal by producing a discrete, digital output image.
 - Tomographic images are reconstructed from many, one-dimensional, views or projections collected over the exposure time.
 - X-ray computed tomography (CT) imaging is an example of a continuous-to-discrete imaging system, using computer reconstruction to produce a digital image from a set of projection data collected by discrete sensors.

Image Processing → Image Analysis

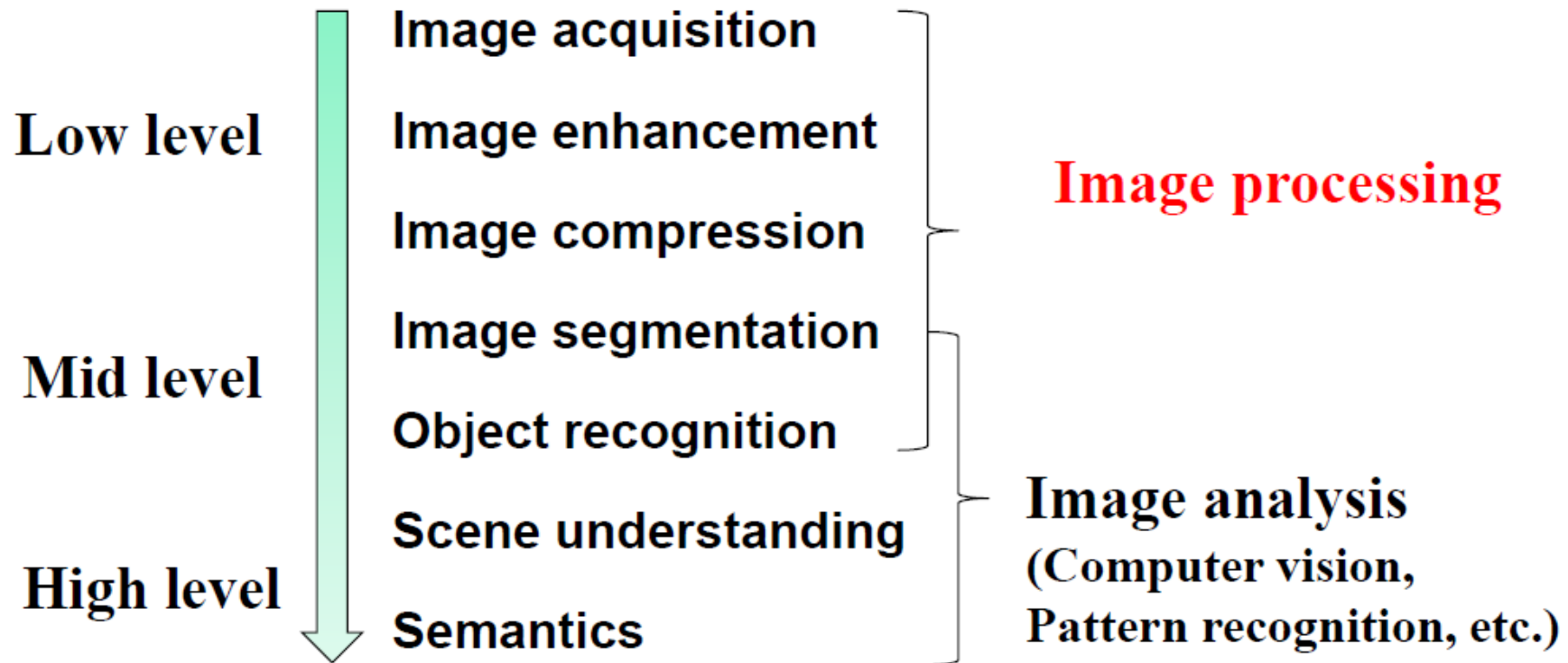


Image Processing → Image Analysis

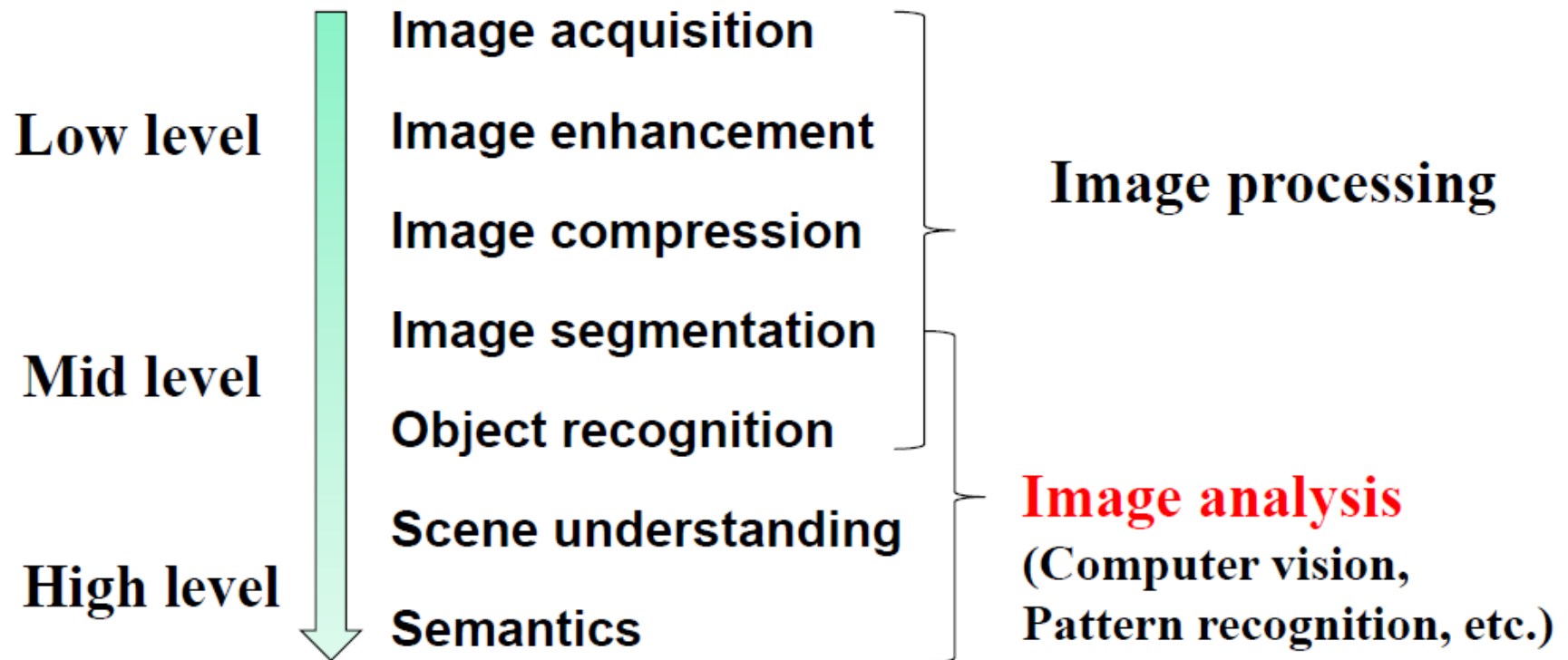


Image Segmentation

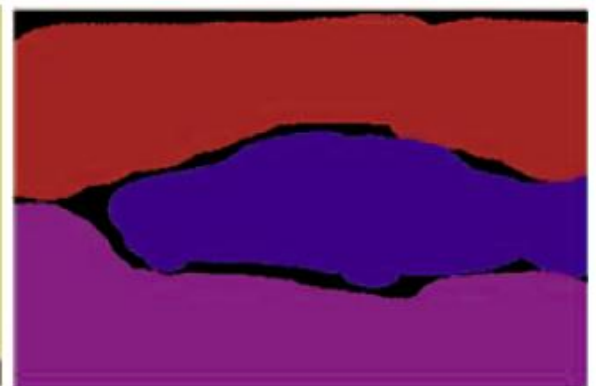
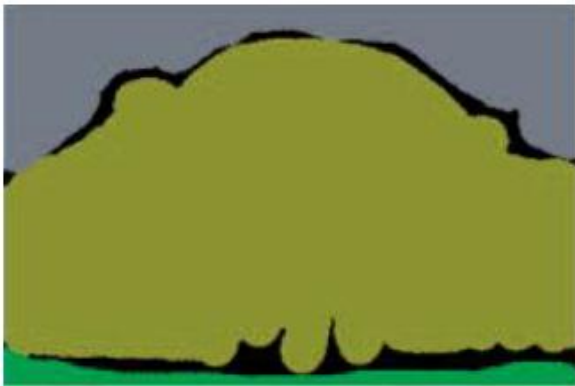


Image Completion

Interactively select objects. Remove them and automatically fill with similar background (from the same image)

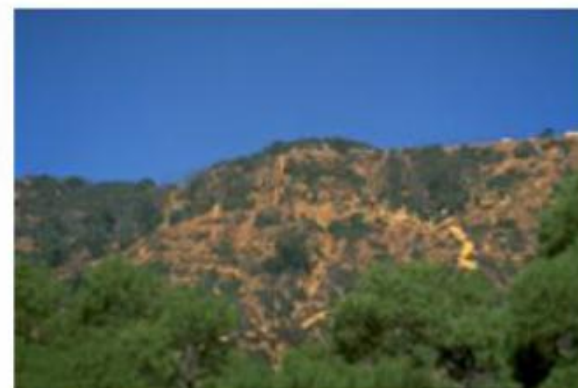


I. Drori, D. Cohen-Or, H. Yeshurun, SIGGRPAH'03

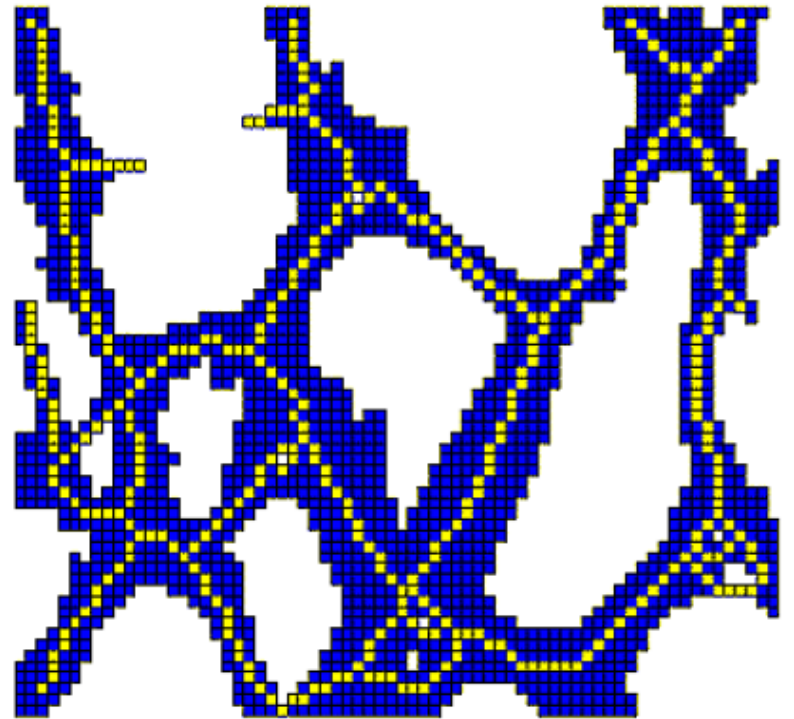
Image Completion



HOLLYWOOD



Morphological Image processing



Why do we Process Images?

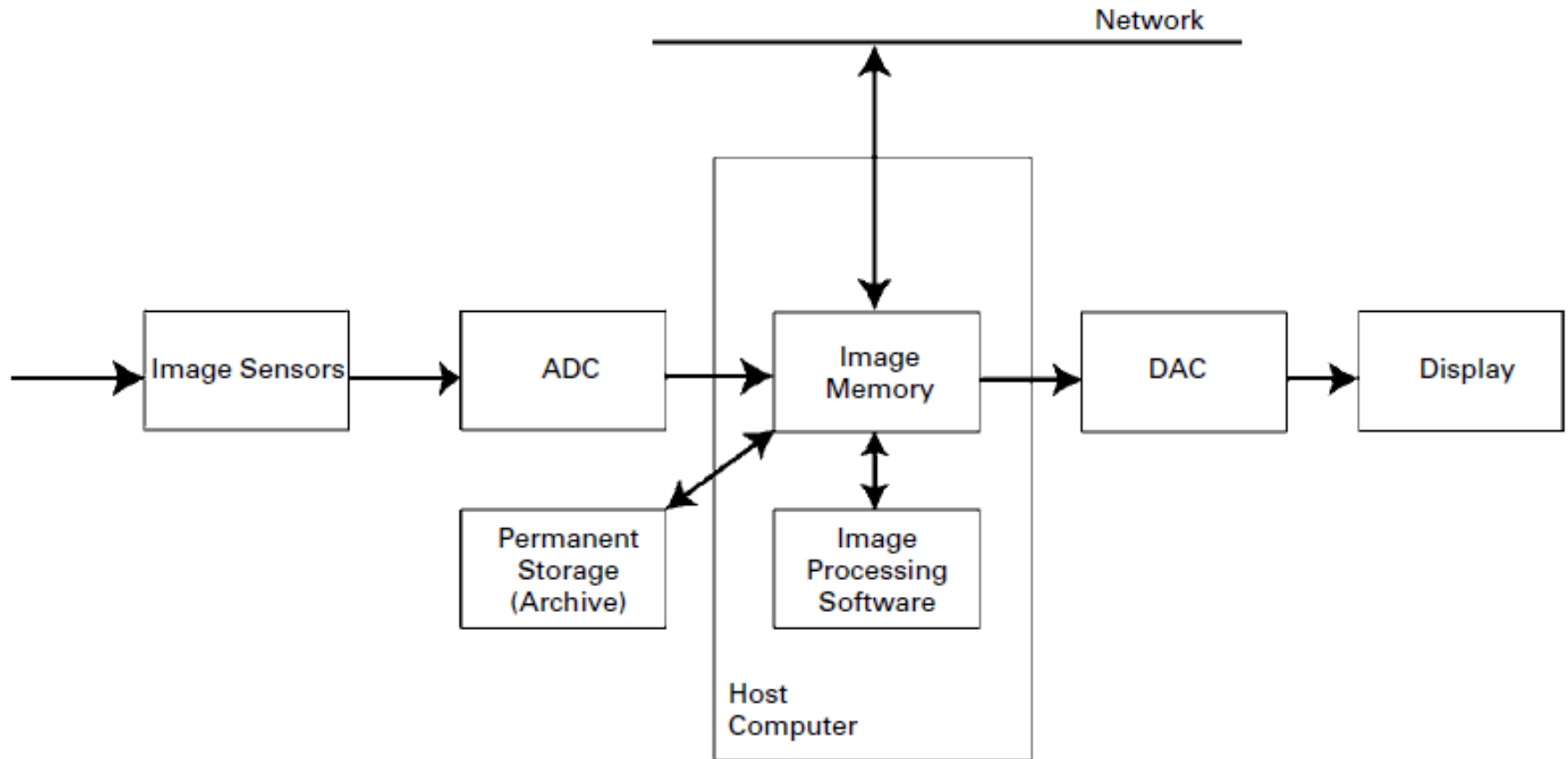
Facilitate picture storage and transmission

Enhance and restore images

Extract information from images

Prepare for display or Printing

The digital image processing system

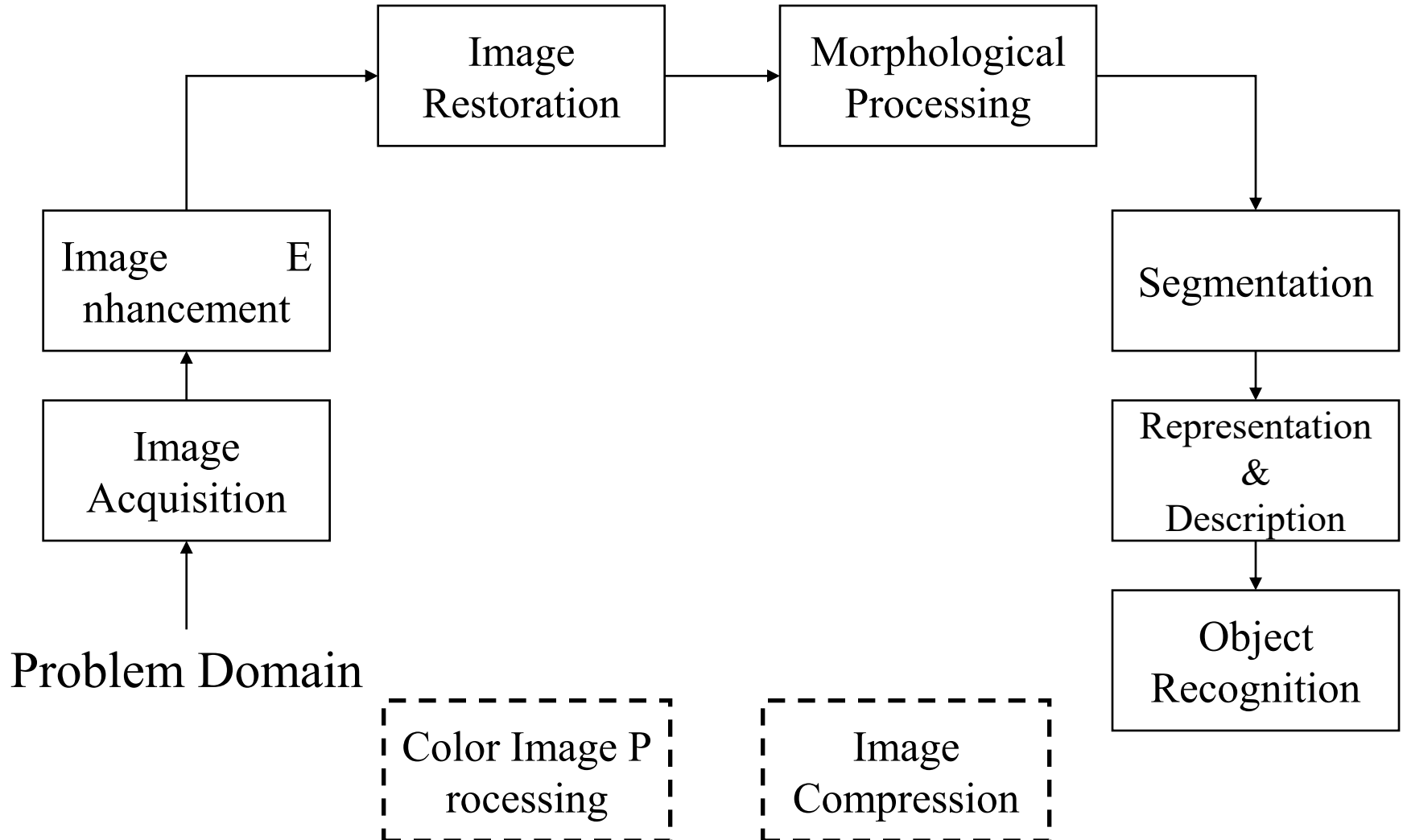


The digital image processing system

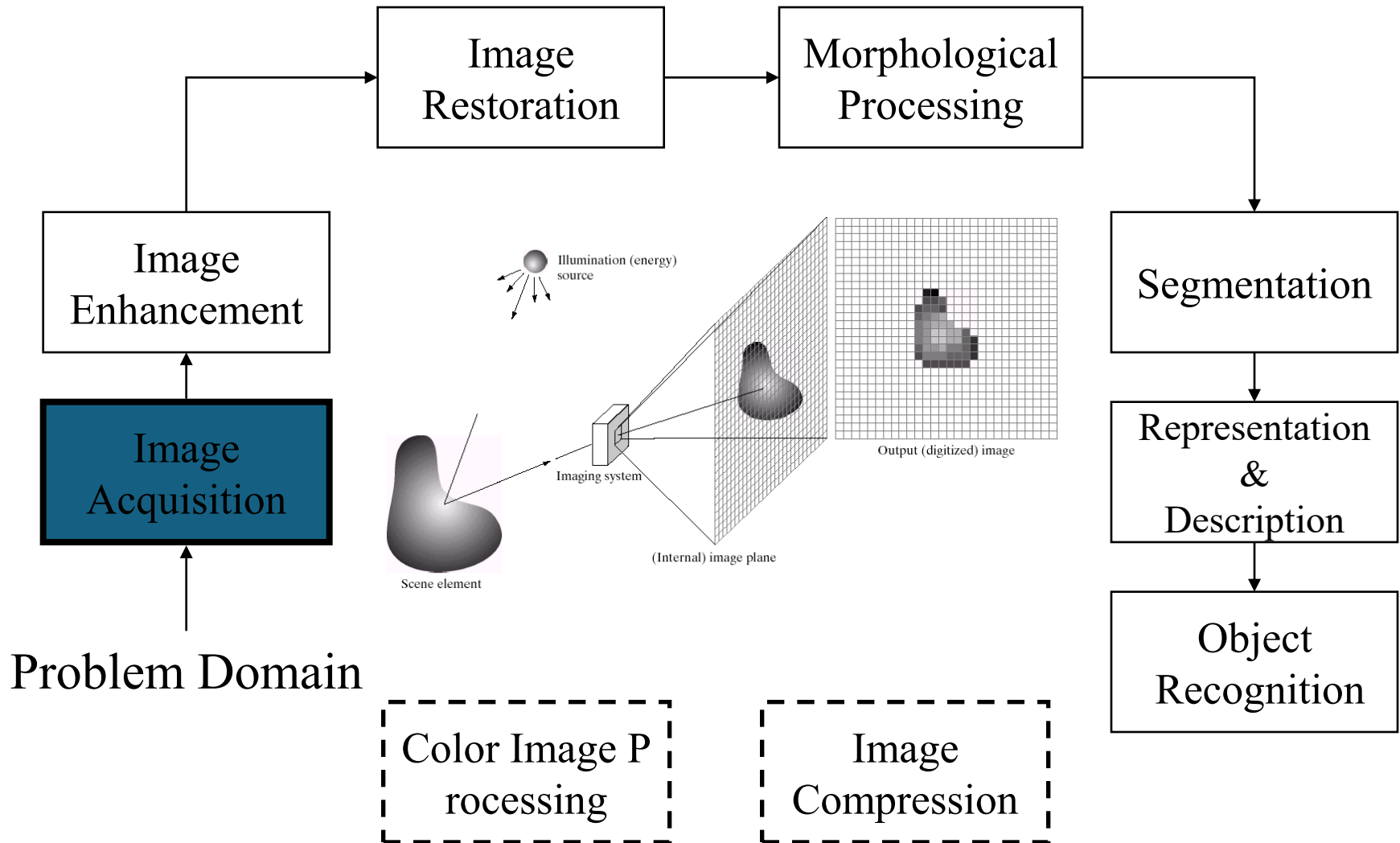
Digital image processing classes and examples of the operations within them.

Classes	Examples of operations
Image enhancement	Brightness adjustment, contrast enhancement, image averaging, convolution, frequency domain filtering, edge enhancement
Image restoration	Photometric correction, inverse filtering
Image analysis	Segmentation, feature extraction, object classification
Image compression	Lossless and lossy compression
Image synthesis	Tomographic imaging, 3-D reconstruction

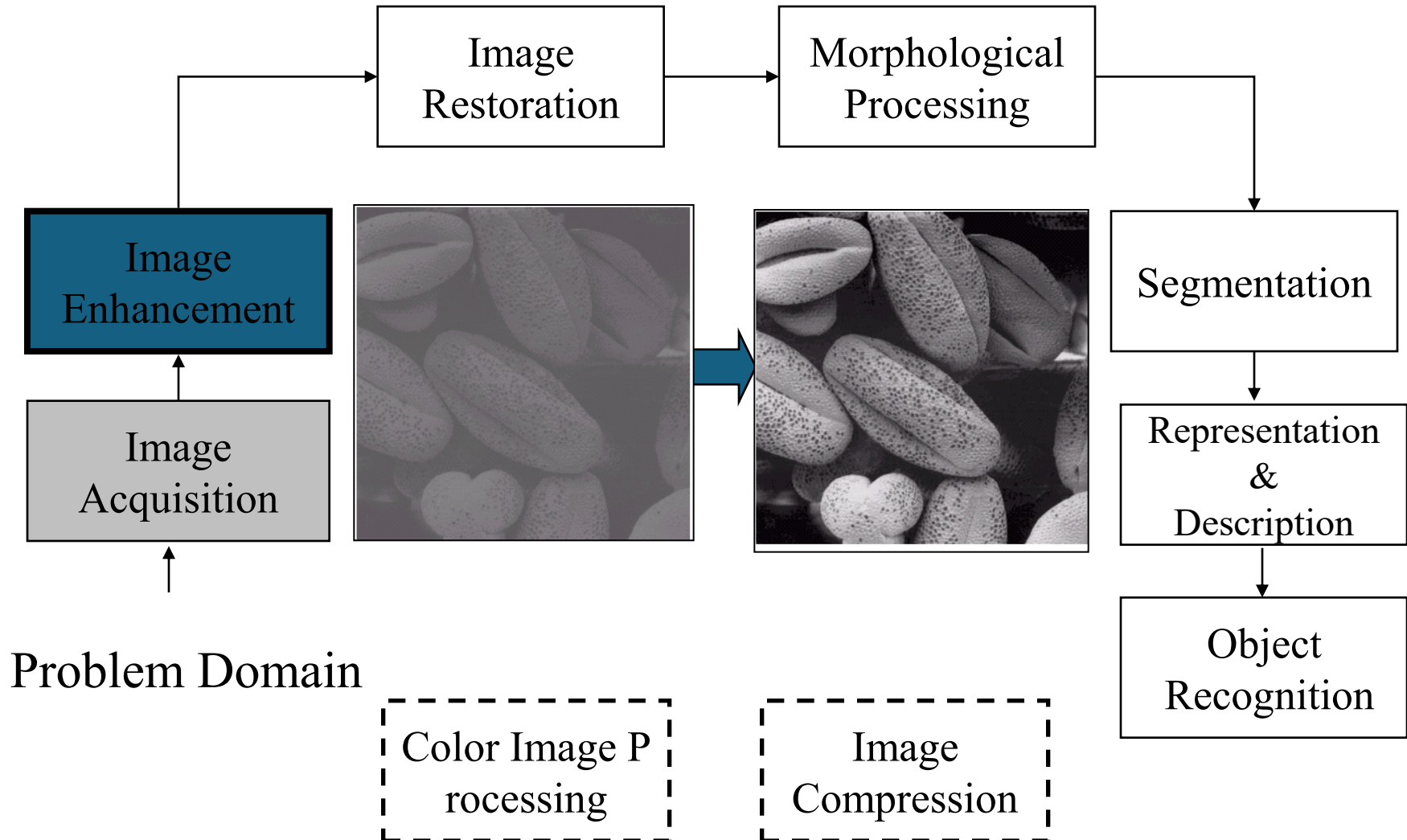
Key Stages in Digital Image Processing



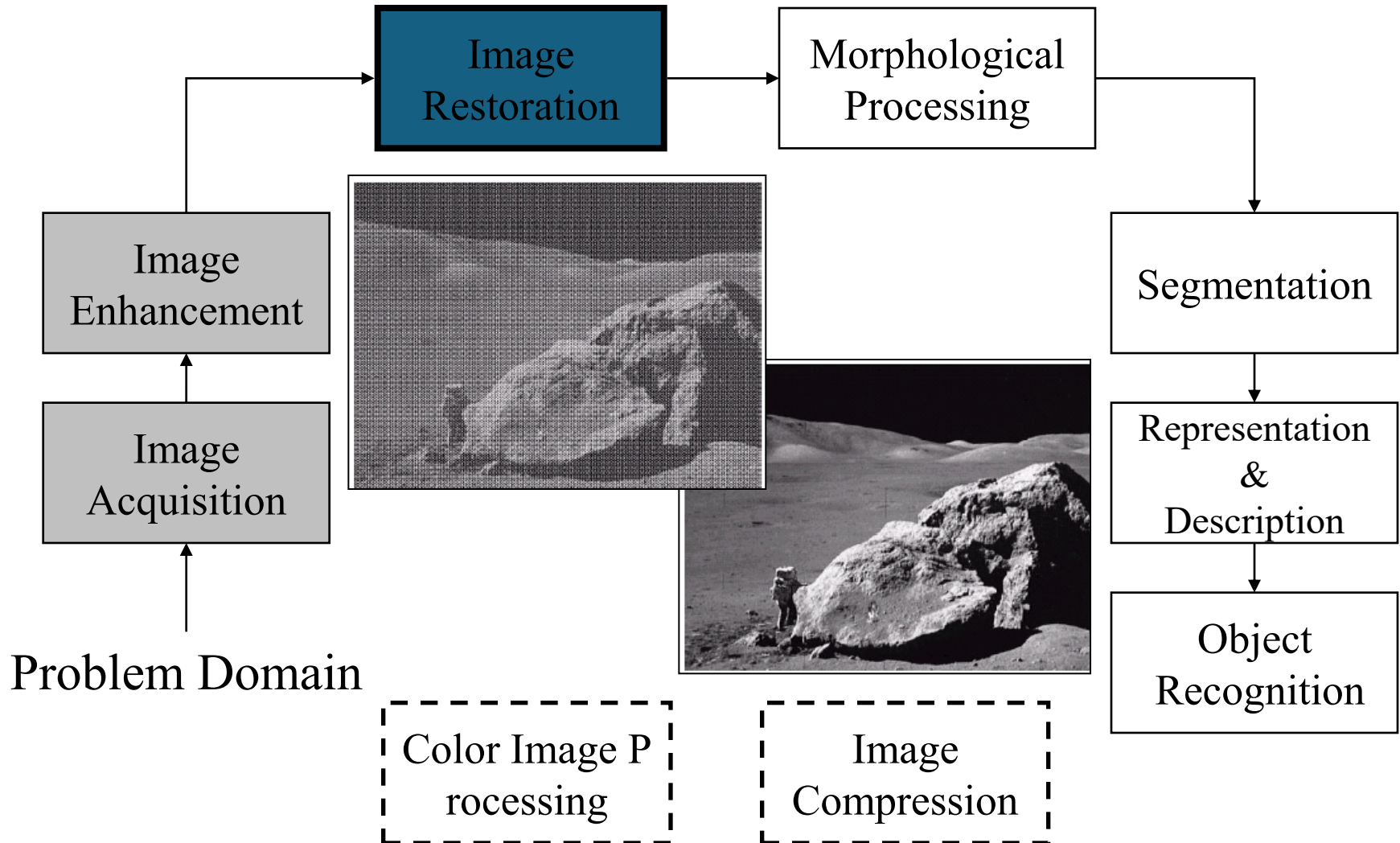
Key Stages in Digital Image Processing: Image Acquisition



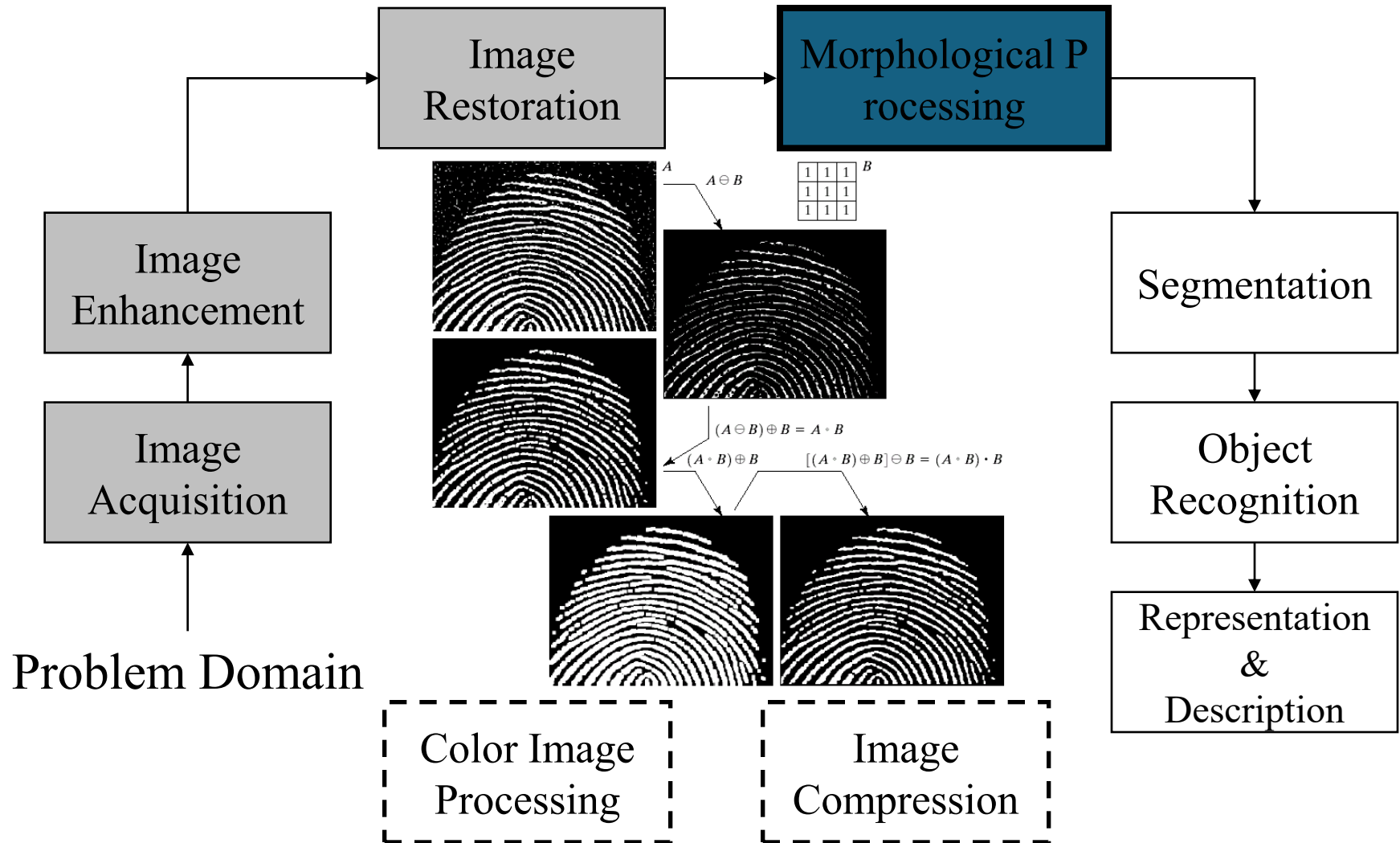
Key Stages in Digital Image Processing: Image Enhancement



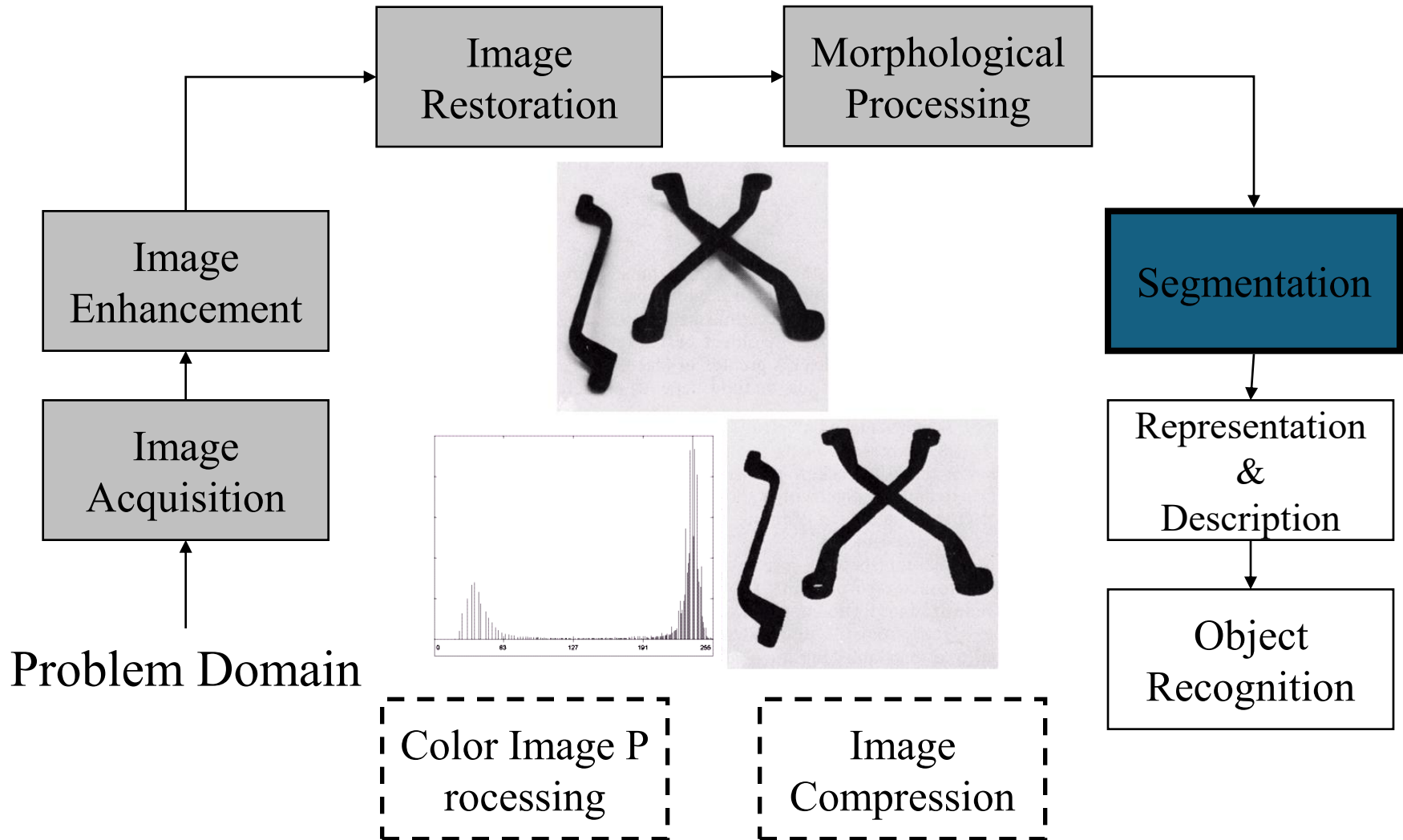
Key Stages in Digital Image Processing: Image Restoration



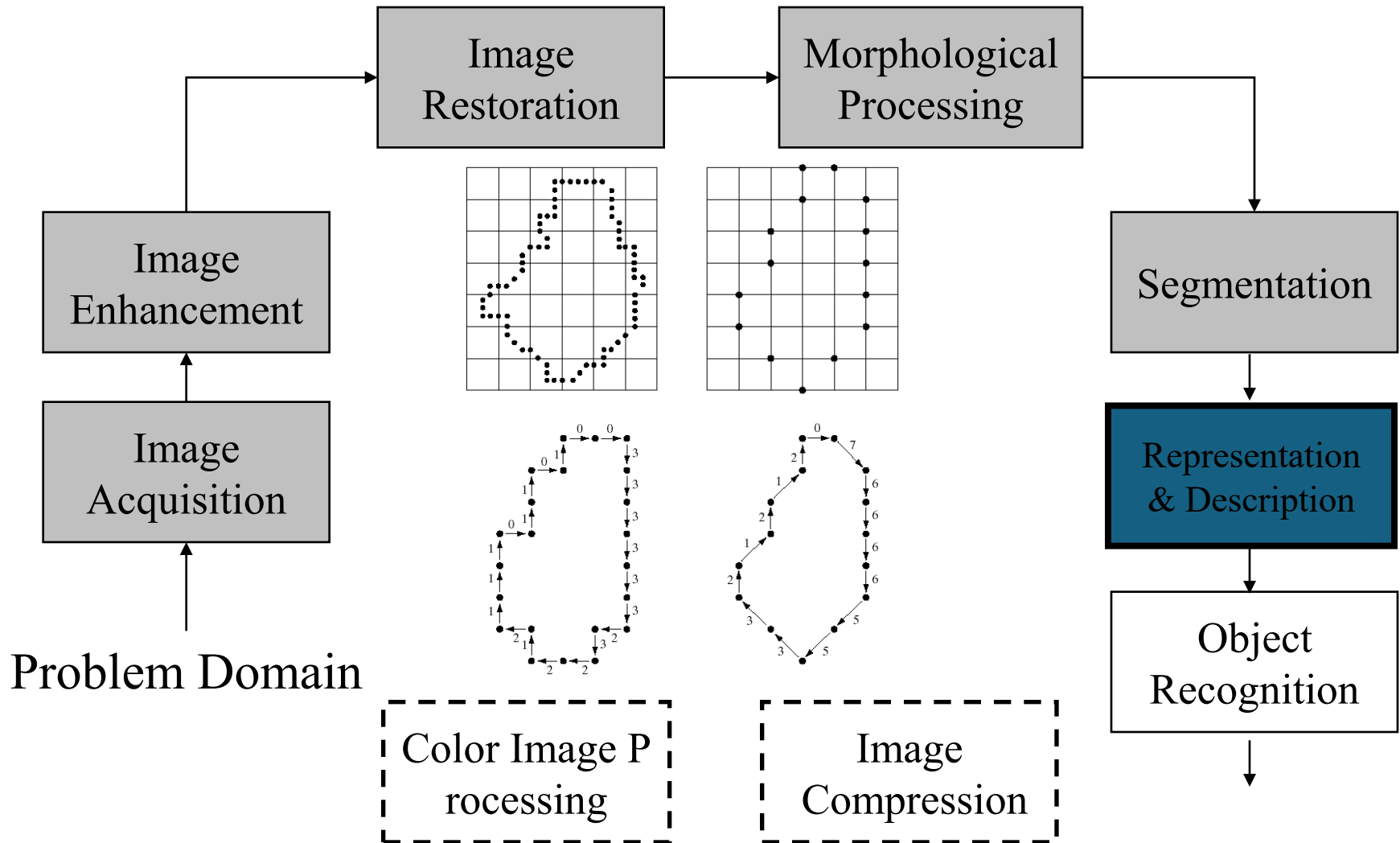
Key Stages in Digital Image Processing: Morphological Processing



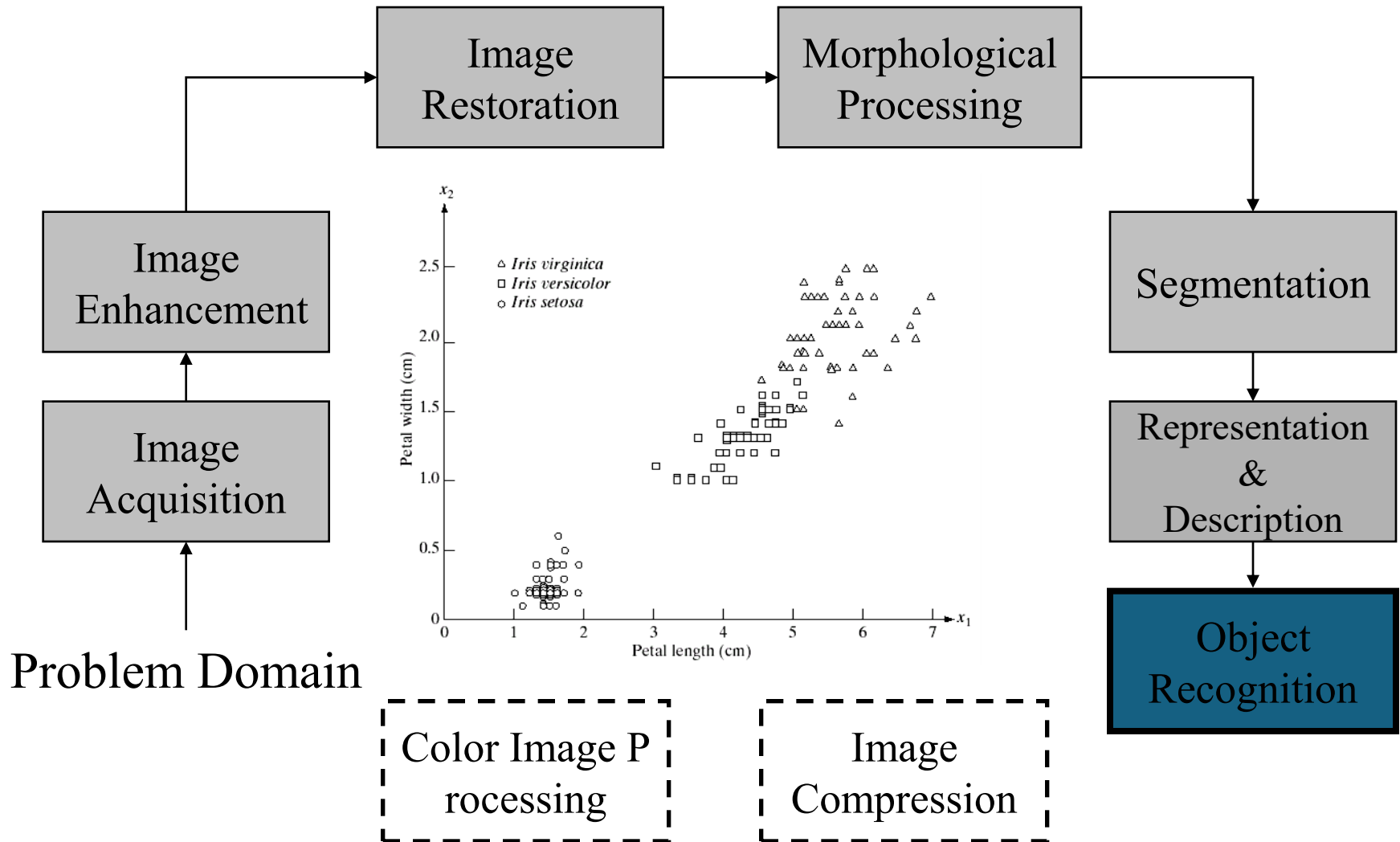
Key Stages in Digital Image Processing: Segmentation



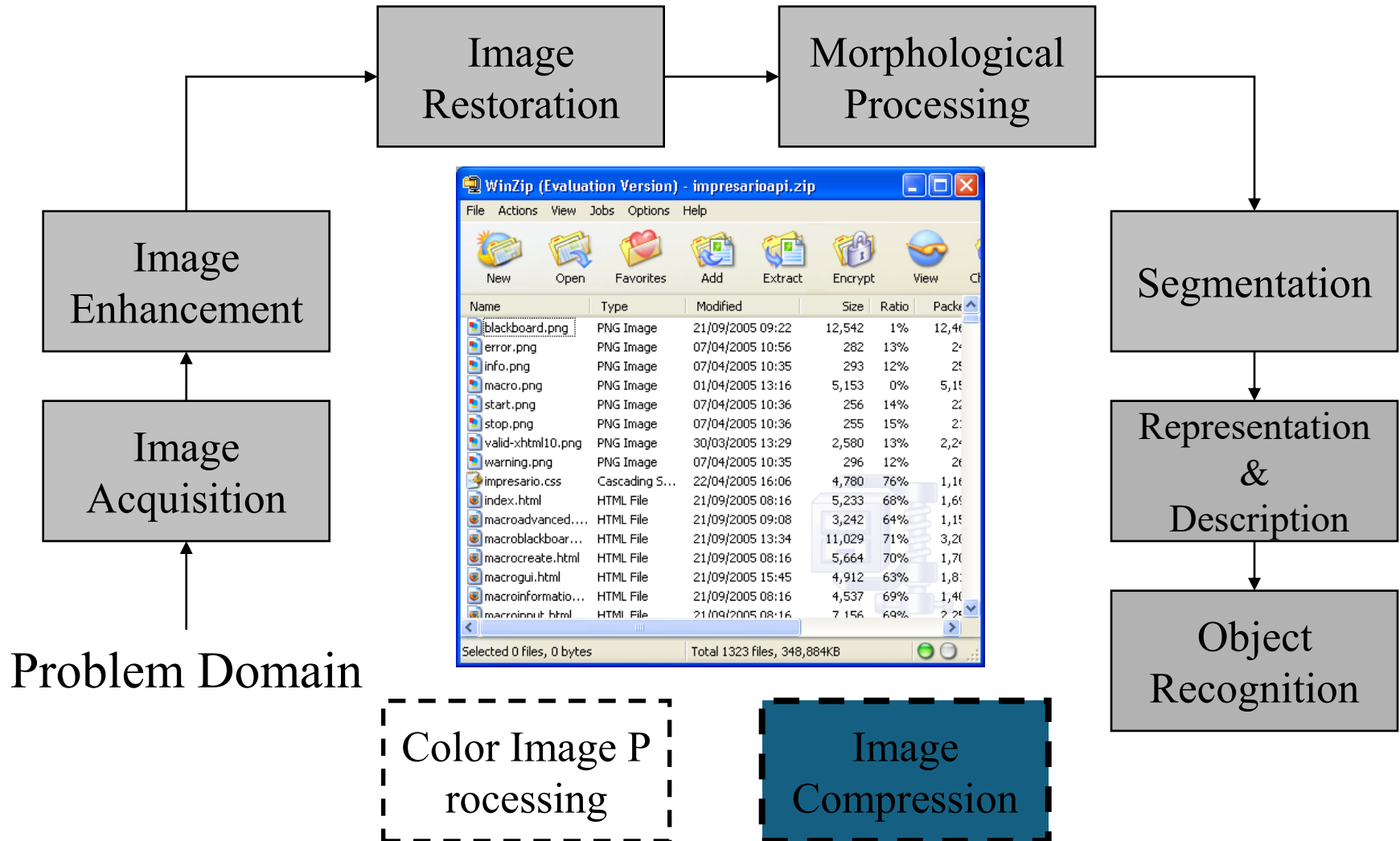
Key Stages in Digital Image Processing: Representation & Description



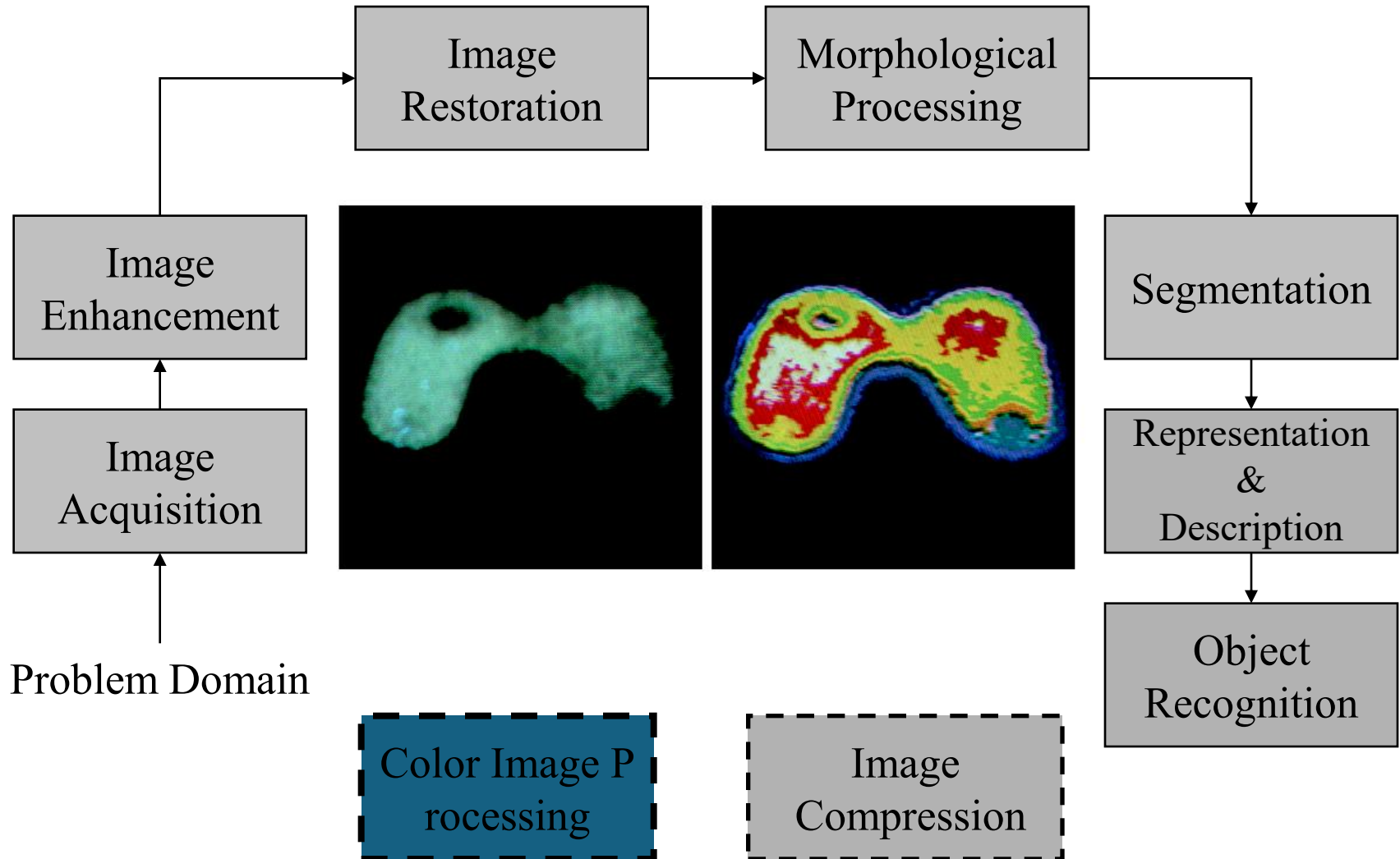
Key Stages in Digital Image Processing: Object Recognition



Key Stages in Digital Image Processing: Image Compression



Key Stages in Digital Image Processing: Colour Image Processing



Applications of Image Processing

1. Image Restoration



Damaged Image



Restored Image

Image Colorization



Original B/W Image



Colorized Image



Original Image



Colorized Image

Image Enhancement



Original Images

Enhanced Images

Face Detection



Face Tracking



Face Morphing



Finger Print Recognition



X



X



✓



Personal Identification Using Iris Recognition



1983



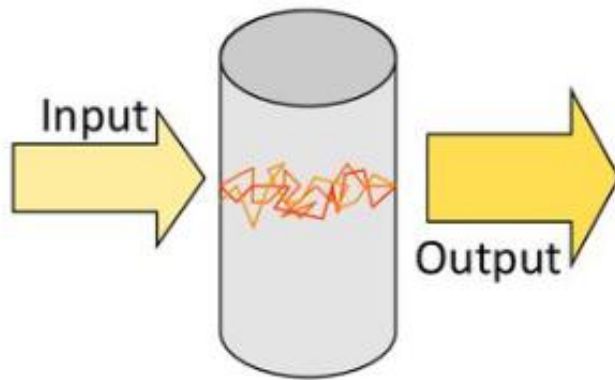
2000



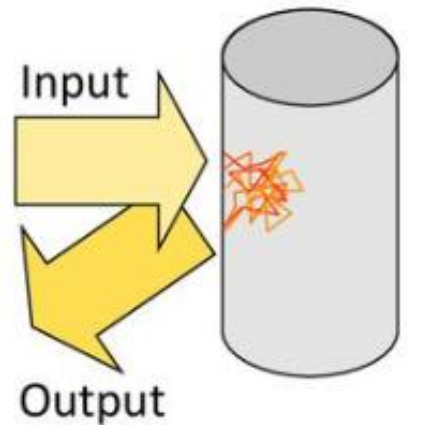
Fundamental models of image formation

Kinds of radiation and imaged properties

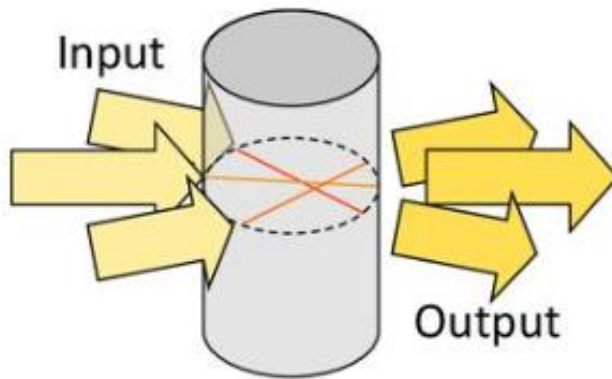
Models of Image Formation



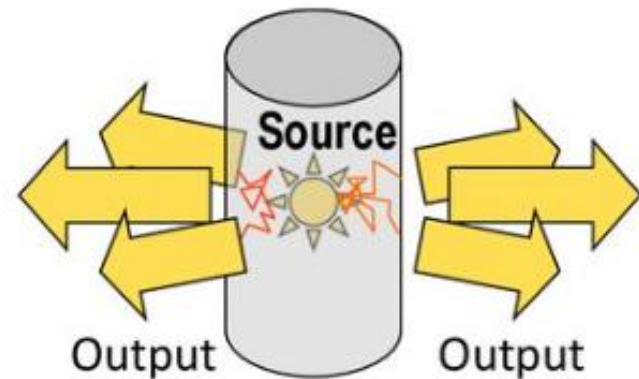
Transmission



Reflectance



Tomography



Emission

Simple Image Formation Model

- An image is defined by two-dimensional function $f(x, y)$. The value or amplitude of f at spatial coordinates (x, y) is a positive scalar quantity.
- When an image is generated from a physical process, its values are proportional to energy radiated by physical source
- Therefore, $f(x, y)$ must be nonzero and finite; that is,

$$0 < f(x, y) < \infty$$

- The function $f(x, y)$ is characterized by two components
 - The amount of source illumination incident on the scene being viewed (Illumination).
 - The amount of illumination reflected by the objects in the scene (reflectance).

Simple Image Formation Model

- A simple image model is

$$f(x, y) = i(x, y) \cdot r(x, y)$$

Where $0 < i(x, y) < \infty$ and $0 < r(x, y) < 1$

$r(x, y) = 0$ means **total absorption**

$r(x, y) = 1$ means **total reflectance or transmittance**

- The intensity of a monochrome image at any coordinates (x, y) is the gray level (l) of the image at that point.
- The interval of l ranges from $[0, L - 1]$.

where $l = 0$ indicates **black** and $l = 1$ indicates **white**.

All the intermediate values are shades of gray varying from black to white.

Simple Image Formation Model

For color image

$$f(x, y) = \{f(x, y)_{green}, f(x, y)_{red}, f(x, y)_{blue}\} \quad (1)$$

Sample values of reflectance $r(x, y)$:

0.01: black velvet

0.65: stainless steel

0.93: snow

Simple Image Formation Model

Sample values of illumination, $i(x, y)$:

90,000 foot-candles(lm/sq.m): sunny day



10,000 foot-candles: cloudy day



0.1 foot-candles: full moon



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[BY-SA](#)

Image Formation Model

- **Geometrical model-**

- Describes how the 3 world dimensions are translated into the dimensions of the sensor.
- In the context of a TV or still camera having a single 2D image plane, perspective projection is the fundamental mechanism whereby light is projected into a single **monocular** view.

Image Formation Model

Geometrical model-

- This type of projection does not yield direct information about the z-coordinate
- **Binocular** imaging uses a system with two viewpoints, in which the eyes do not normally converge, i.e. the eyes are aimed in parallel at an infinite point in the z direction.
- The depth information is encoded by it's different positions (disparity) in the two images.

Image Formation Model

Radiometric model –

- Illustrates the way in which the imaging geometry, the light sources and the reflectance properties of objects influence the light measured at the sensor.
- The brightness of the image at a point, the image intensity or image irradiance where irradiance defines the power per unit area falling on a surface depends on the following factors-
 - First, there is the radiant intensity of the source.
 - Second there is the reflectance of the objects in the scene, in terms of the proportion, spatial distribution and spectral variation of light reflected.

Image Formation Model

The digitizing model-

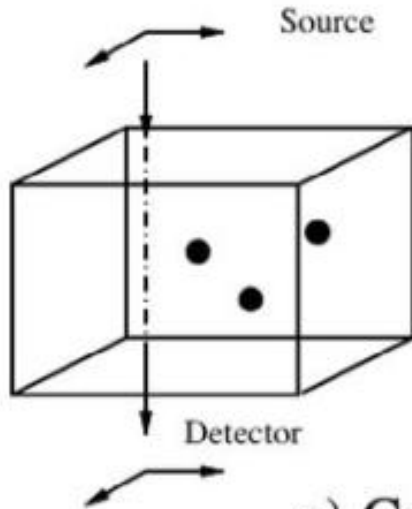
- Implies that the analogue scene data which varies continuously in intensity and space, must be transformed into a discrete representation.
- Digitized images are sampled, i.e. only recorded at discrete locations, quantized , i.e. only recorded with respect to the nearest amplitude level.
 - for example: 256 levels of intensity, and windowed, i.e. only recorded to a finite extent in x and y etc.
- All these processes change fundamentally the world as seen by the camera or other sensor.

Image Formation Model

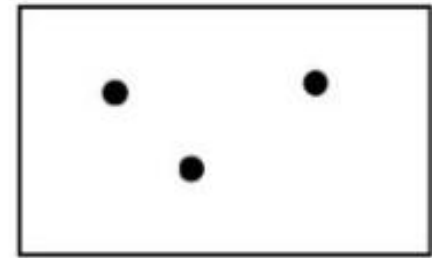
Spatial frequency model-

- Describes how the spatial variations of the image may be characterized in the spatial frequency domain,
- The more rapid the variations in the image, the higher the spatial frequency.
- This type of analysis is fundamental to image processing.

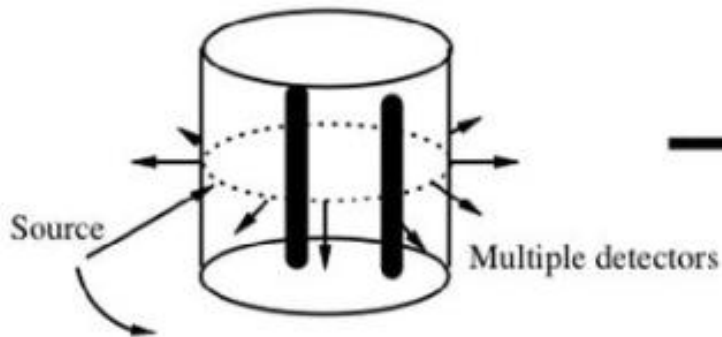
Acquisition geometrics



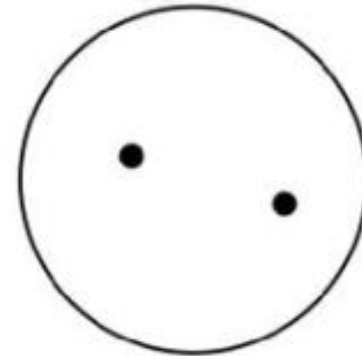
a) Coaxial Scanning



Single projection image



b) Tomographic Sample



Slice Image

Acquisition geometries

- 2D or **topographical images**
 - **Single sensor scanning**: You may get a topographical image with just one sensor by moving it in the horizontal and vertical directions.
- **Sensor arrays**: This is just a matter of having many sensors in a particular geometrical disposition. Sensor arrays may be:
 - in line (unidimensional),
 - in a matrix (bidimensional)
 - Two common sensor arrays are:
 - **CCD** (Charge-Coupled Devices)
 - **CMOS** (Complementary Metal Oxide Semiconductor)
 - or have a more ad-hoc spatial distribution
 - e.g. arrays abiding the EEG international 10/20 systems

Acquisition geometries

■ 3D or tomographical images

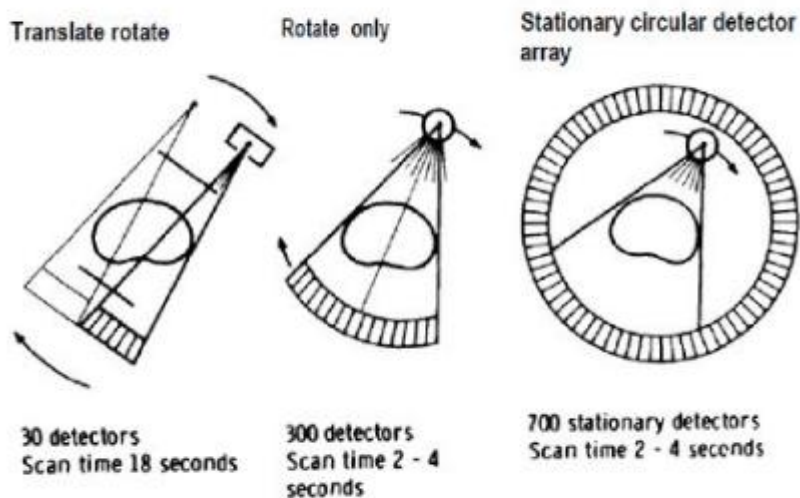
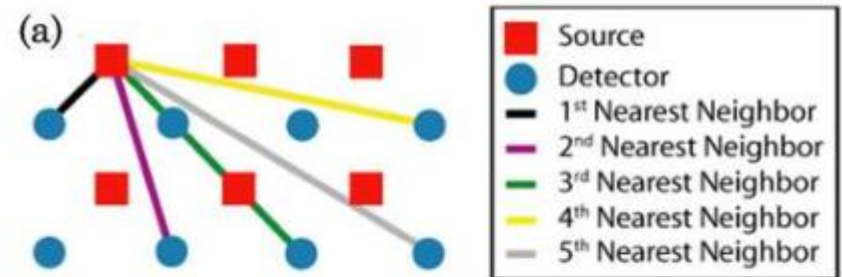


Fig. 7. Three different methods of scanning the patient.

Magnetic resonance of the torso
[HounsfieldGN, Nobel Lecture 1979]



fNIRS multidistance optode array

[Deghhani H et al (2009), Applied Optics 48(10):D137-D143]

Kinds of radiation and imaged properties

Categorization	
Kind of radiation	• Electromagnetic waves • Mechanical waves (e.g. sound) • Particles • Fields
Imaging mechanism	• Reflection • Refraction • Modulation • Shadow casting
Property being imaged	• Reflectance (optical, acoustic, etc) • Attenuation • Source strength • Electromagnetic properties (polarization, diffuseness, etc) • Surface height • Phase
Directedness	• Direct • Indirect

THE IMAGING SYSTEM

Components of imaging system

- **Acquisition unit**

- Generates the radiation and captures the image.

- **Processing unit**

- Prepares the image for storing and rendering.

- **Rendering unit**

- Stores and renders the image and permits its manipulation.

Acquisition unit

- The hardware responsible for generating and sensing the radiation and converting it to electric or mechanical output.
- Composed of:
 - **Radiation source:**
 - Generates radiation to interact with the tissue at a certain range of the spectrum
 - Not necessary if bioluminescence occurs; then the living organism or tissue itself is the source of radiation
 - Also alternatively, in some cases “natural light” can be exploited.
 - **Radiation sensors** or detectors (e.g. photodetectors)
 - Captures radiation from a certain range of the spectrum
 - Produces an analogic output
 - ☞ Those called “digital sensors” actually mean that the processing unit is incorporated in the device. I'm not aware as today of any truly digital sensor.

Processing unit

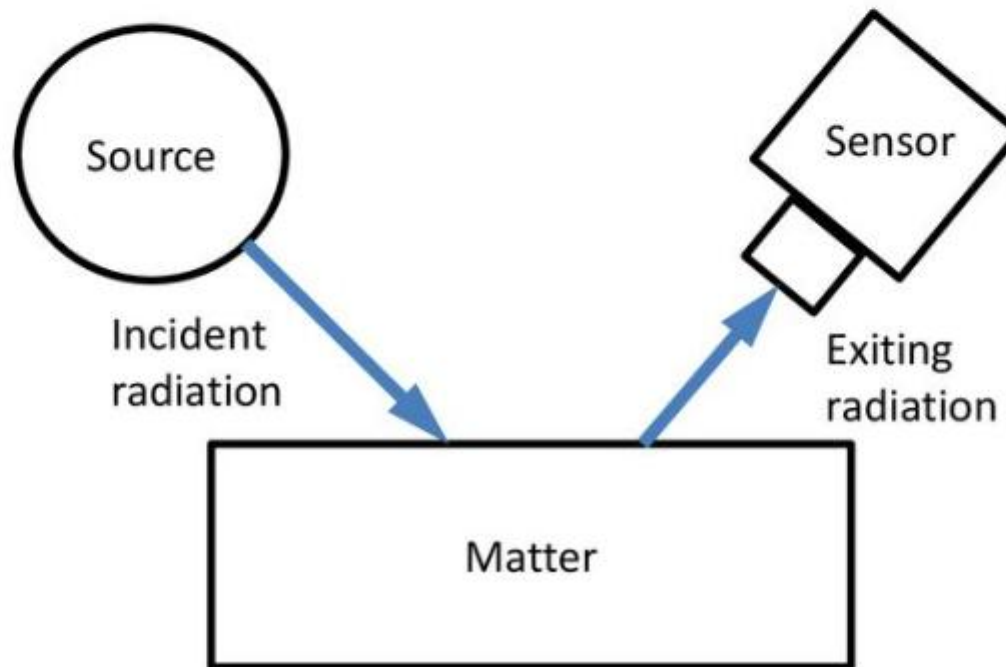
- A specialised hardware to carry out some basic operations such as discretizing time (sampling), and/or digitizing (sampling + quantizing), perhaps even some noise removal.
 - Space is often automatically discretized by the number of sensors, but not always; e.g. films captures continuous locations
- May incorporate:
 - Sample & Hold (S&H):
 - Temporally freezes a continuous input to discretize time.
 - Different S&H may have different strategies e.g. averaging of several frames, keeping an instantaneous value, etc
 - Analogic-Digital Converter (ADC):
 - Digitizes (i.e. samples and also quantizes) the image.
 - Only necessary for digital systems.

Rendering unit

- The hardware (and software) responsible for visualization and storage of the image.
- May incorporate:
 - Computer (CPU)
 - Image rendering software
 - Image rendering hardware (screen, printer)
 - Storage device

Image acquisition

- Radiation from a source interacts with matter
- Part of the radiation is remitted or transmitted and recorded by a sensor.



POINT SPREAD FUNCTION

Point Spread Function

- Intuitively, the **Point Spread Function (PSF)** is the function representing the response of a system to an impulse input
 - whether temporal, spatial or spatiotemporal e.g. [ShmuelA (2007) Neuroimage 35:539–552].
 - Spatially it can spread in 2D or 3D e.g.
- In imaging, this is the response of the imaging system to a point source.
 - Diffraction blurs out any point-like object to a certain minimal size and shape called the PSF
 - In images; point in a system refers to something which in focus occupies 1 pixel wide

Point Spread Function

- A bit more formally; the PSF is the normalized intensity distribution of measured signals in the image $i(x', t')$ related to a point at position x in the object pattern $o(x) = I_o(x, \lambda) * R(x, \lambda)$.
- The image results from the convolution ($\otimes$) between each individual point of the object pattern $o(x)$ and the imaging system characterization i.e. the $PSF = Rn(x, t, \lambda)$.

$$i(x', t') = o(x) \otimes PSF(x - x', t - t')$$

- Adapted from [Senoser M et al (2010) J.Anal.At.Spectrom, 25:1440–1452]

Point spread function

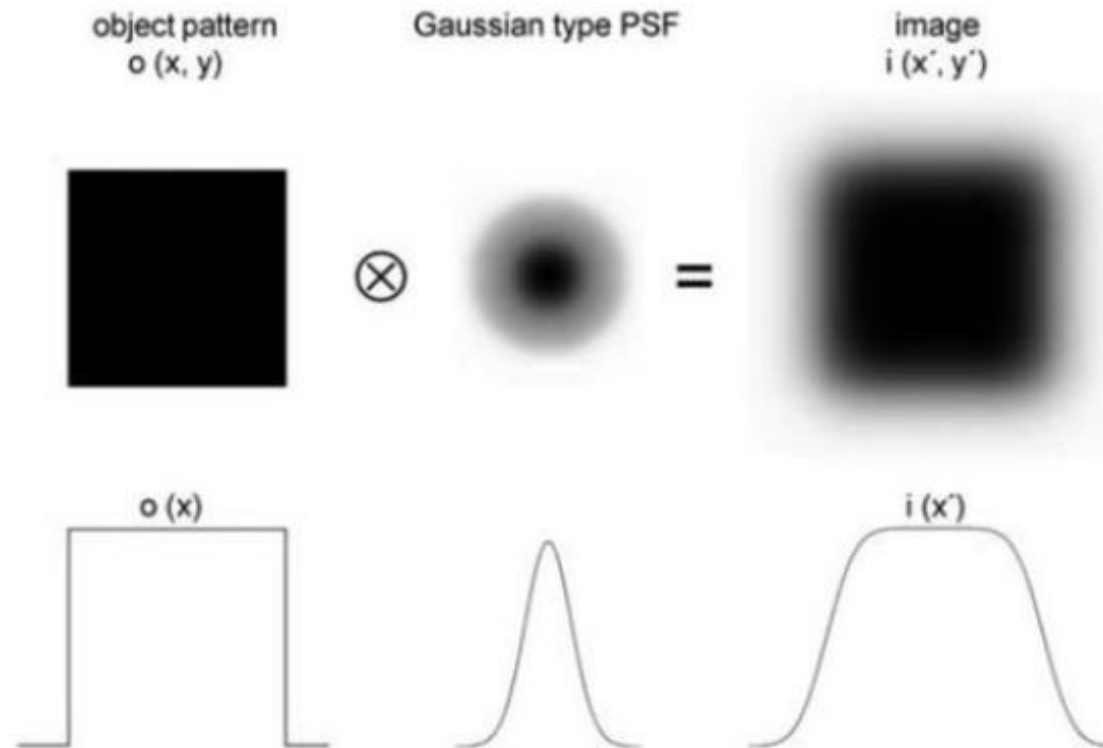


Fig. 1 Image (top) of and line scan (bottom) over a square simulated by the convolution of the square with a Gaussian type PSF. $\otimes$ denotes the convolution operation.

Figure from: [Senoser M et al (2010) J.Anal.At.Spectrom, 25:1440–1452]

Point Spread Function

- The ideal PSF is the 3D diffraction pattern of light emitted from an infinitely small point source in the specimen and transmitted to the image plane through a high numerical aperture (NA) objective
- It is considered to be the fundamental unit of an image in theoretical models of image formation
- [http://www.zeiss.com/microscopy/en_de/campus/basic-microscopy/the-point-spread-function-.html]

Point spread function

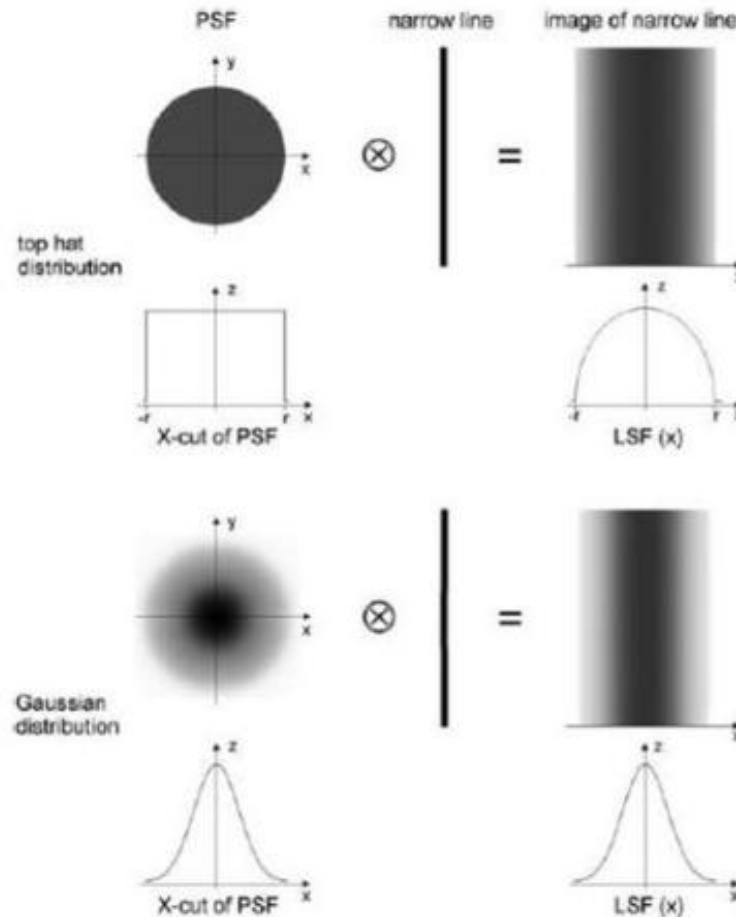


Fig. 2 Determination of the LSF by imaging of a narrow line. Different types of PSF were convoluted with a narrow line. The z-axis denotes the signal intensity within the images (from ref. 34).

Figure from: [Senoser M et al (2010) J.Anal.At.Spectrom, 25:1440–1452]

Point Spread Function

- The PSF:
 - Is not necessarily isotropic
 - Depends on the imaging device (e.g. strength of the radiation source) as well as the imaging modality (nature of the radiation)
 - Depends on the wavelength;
 - shorter wavelengths of light result in a smaller PSF (better resolution), while longer wavelengths result in a larger PSF i.e. worse resolution.
- ...ergo
 - it is of value to determine the resolution performance of an imaging system
 - It is responsible for image blurring

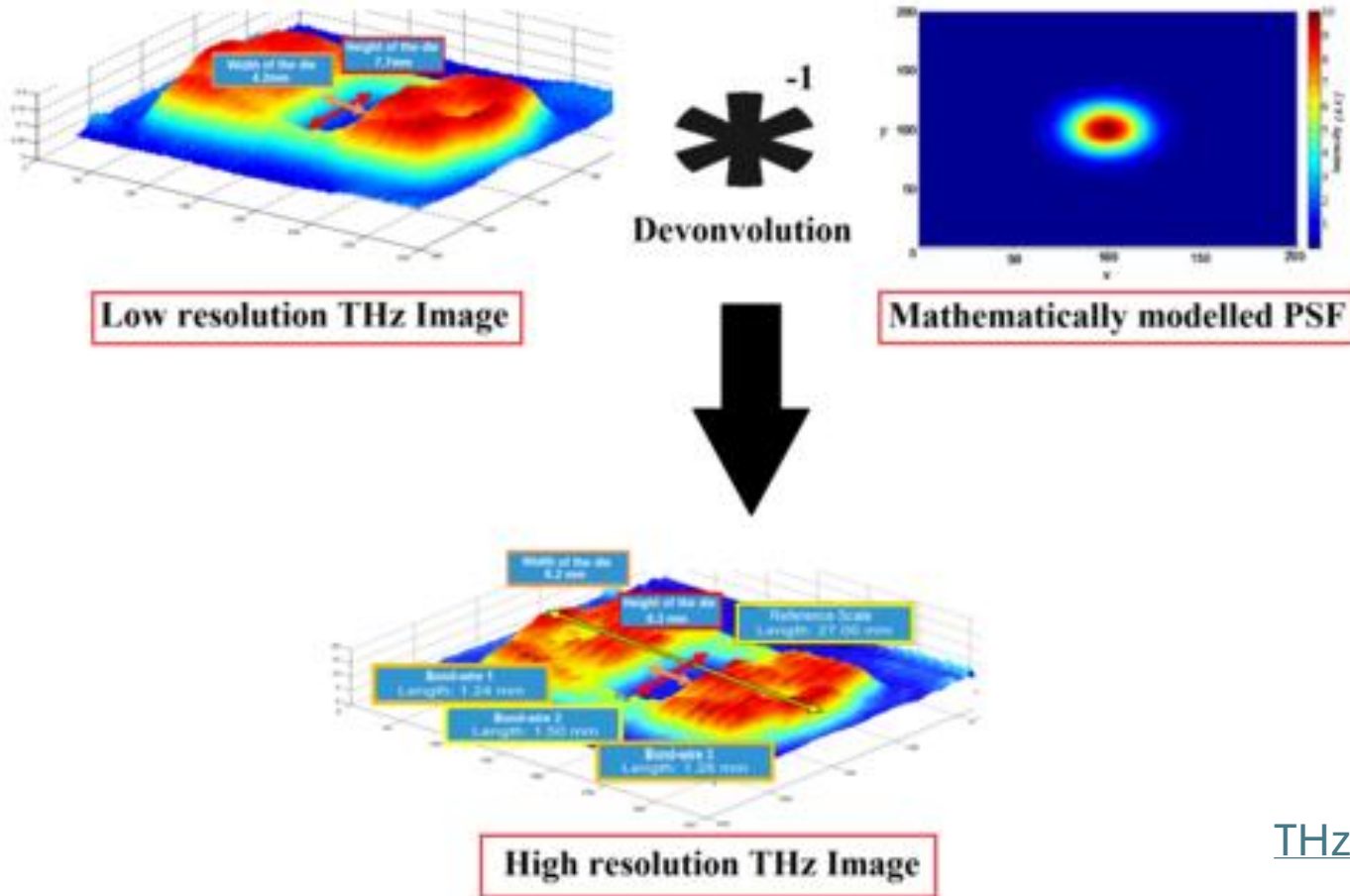
Point Spread Function

- PSF in an imaging system represents the system's response to a point source or point object.
- A measure of how well the system can resolve a single point, and it's often described as a 3D shape or the "resolution cell" of the system.
- Shows how a single point of light is blurred or spread out in the image.

Point Spread Function

- The PSF illustrates the spatial distribution of light in the image plane when a point source is imaged.
- It's a fundamental characteristic of an imaging system, whether it's an optical microscope, telescope, or medical imaging device.

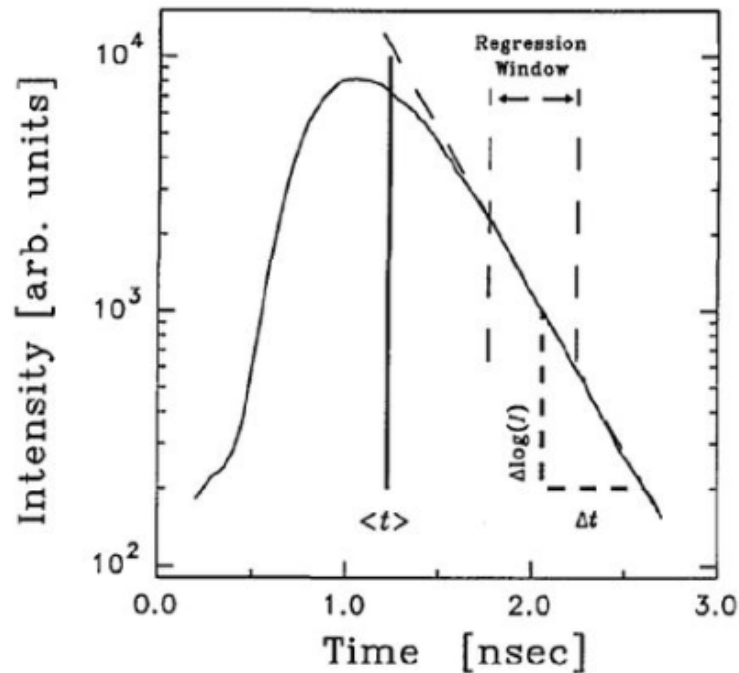
Point Spread Function



Application of PSF: Deconvolution of the mathematically modeled PSF and the low-resolution image enhances the resolution.

Point Spread Function

Temporal dependence



K_{ls} represents the gradient of the decay of the logarithmic intensity

Fig. 2. Typical temporal point spread function measured on the head of an adult showing the mean time $\langle t \rangle$ and the calculation of K_{ls} .

Figure from: [Essenpreis et al (1993) Applied Optics 32(4):418-425]

Point Spread Function

Wavelength dependence

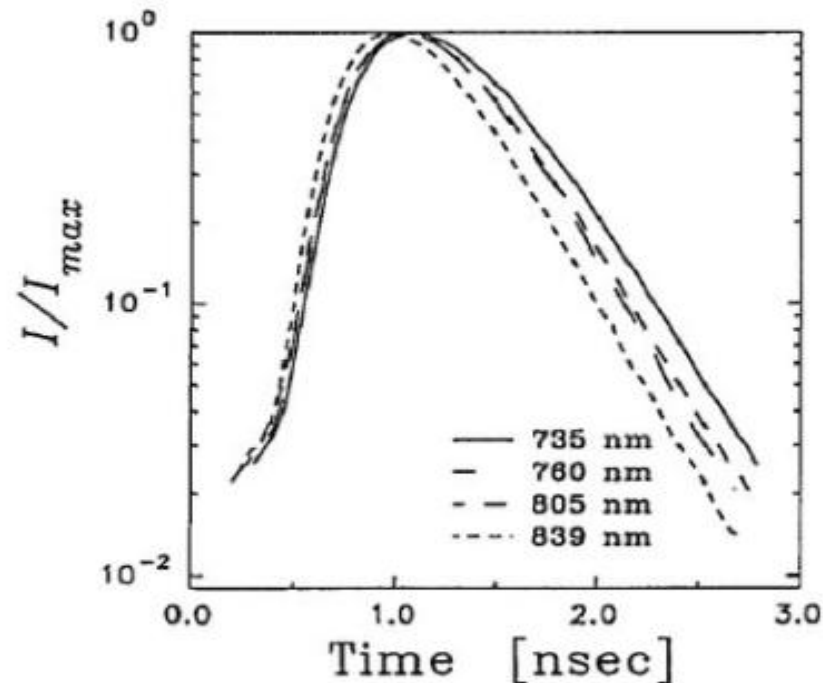


Fig. 3. Temporal point spread functions measured at several wavelengths on the head of one subject (optode spacing 40 mm).



Figure from: [Essenpreis et al (1993) Applied Optics 32(4):418-425]

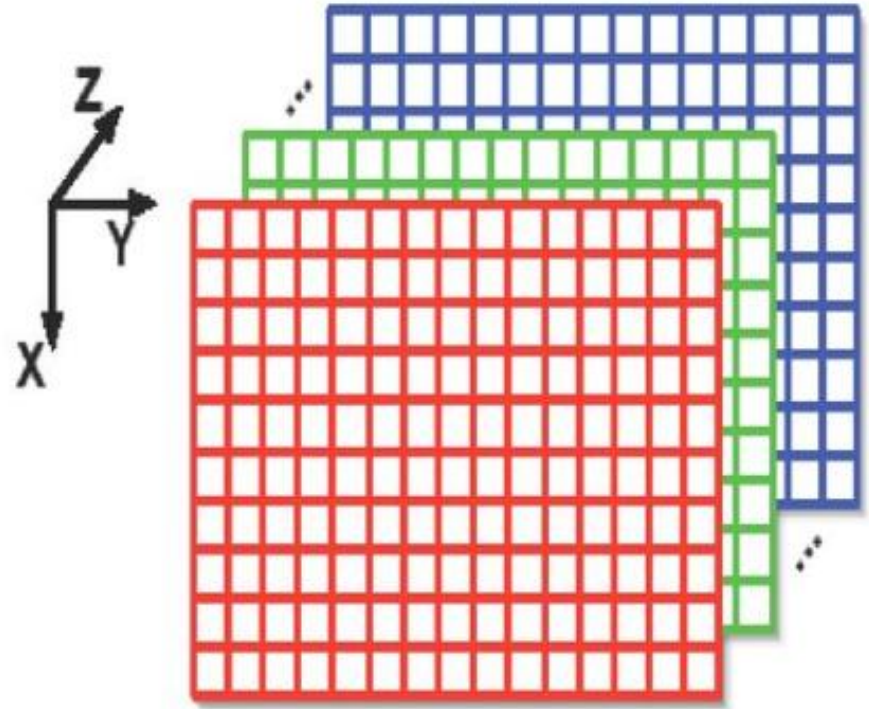
IMAGING FILTERS

Introduction to Imaging Filters

- Imaging filters are techniques used to capture and process images based on the spectral characteristics of light.
- They play a key role in various applications including medical imaging, remote sensing, and machine vision.

Imaging filters

- Arranging sensor arrays (also called filters):
 - Colour (x3)
 - Multispectral (>3)
 - Hiperspectral ($>>3$)



Imaging filters

Table 1 Comparison of monochrome, RGB, spectroscopy, multispectral, and hyperspectral features.

Feature					
	Monochrome	RGB	Spectroscopy	Multispectral	Hyperspectral
Spatial information	Yes	Yes	No	Yes	Yes
Band numbers	1	3	From several dozens to hundreds	3 to 10	From several dozens to hundreds
Spectral information	No	No	Yes	Limited	Yes
Multiconstituent information	No	Limited	Yes	Limited	Yes
Sensitivity to minor components	No	No	No	Limited	Yes

Figure from: [Li et al (2013) JBO 18(10):100901]

Monochromatic Imaging

- Captures images in a single wavelength or band of light
- Typically black and white or grayscale
- Utilizes differing amounts of light instead of different colors to capture and represent images.
- Monochrome photography takes only one single color and uses a range of tones of that color.
- High contrast and sharp detail
- Commonly used in X-ray, CT scans, and microscopy



Colour Imaging

- - Captures images in multiple bands corresponding to visible light (RGB)
- - Each channel (Red, Green, Blue) captures specific wavelengths
- - Used in digital cameras, medical diagnostics, and television
- - Enables visual realism and color differentiation

Multi-spectral Imaging



Captures data at specific wavelength bands across the electromagnetic spectrum



Typically 3 to 10 bands

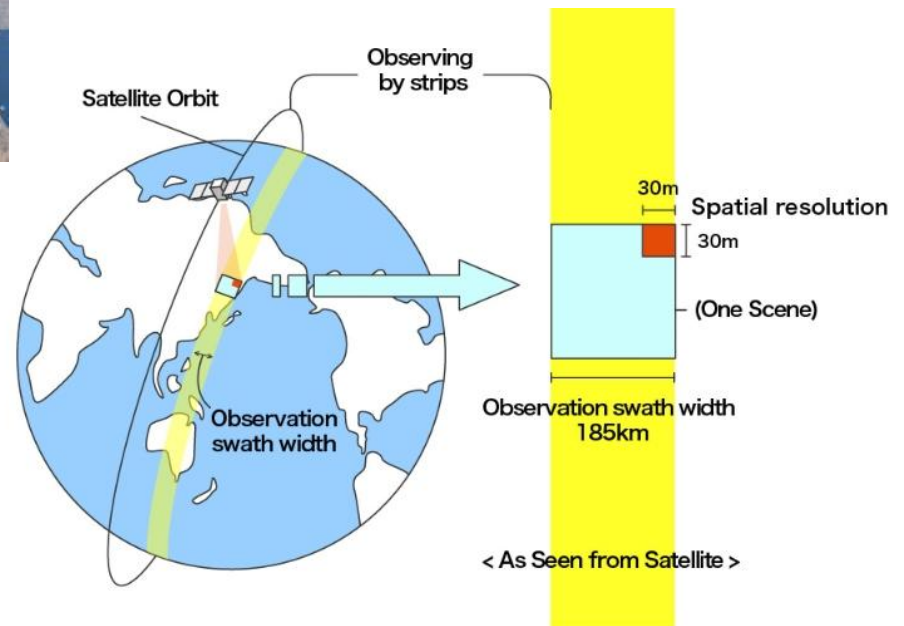
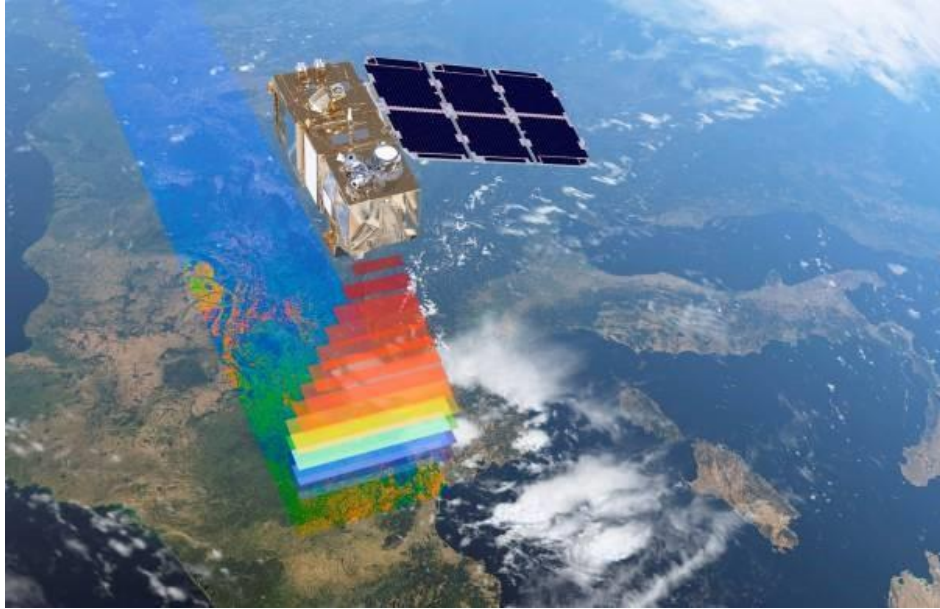


Used in agriculture, environmental monitoring, and military surveillance

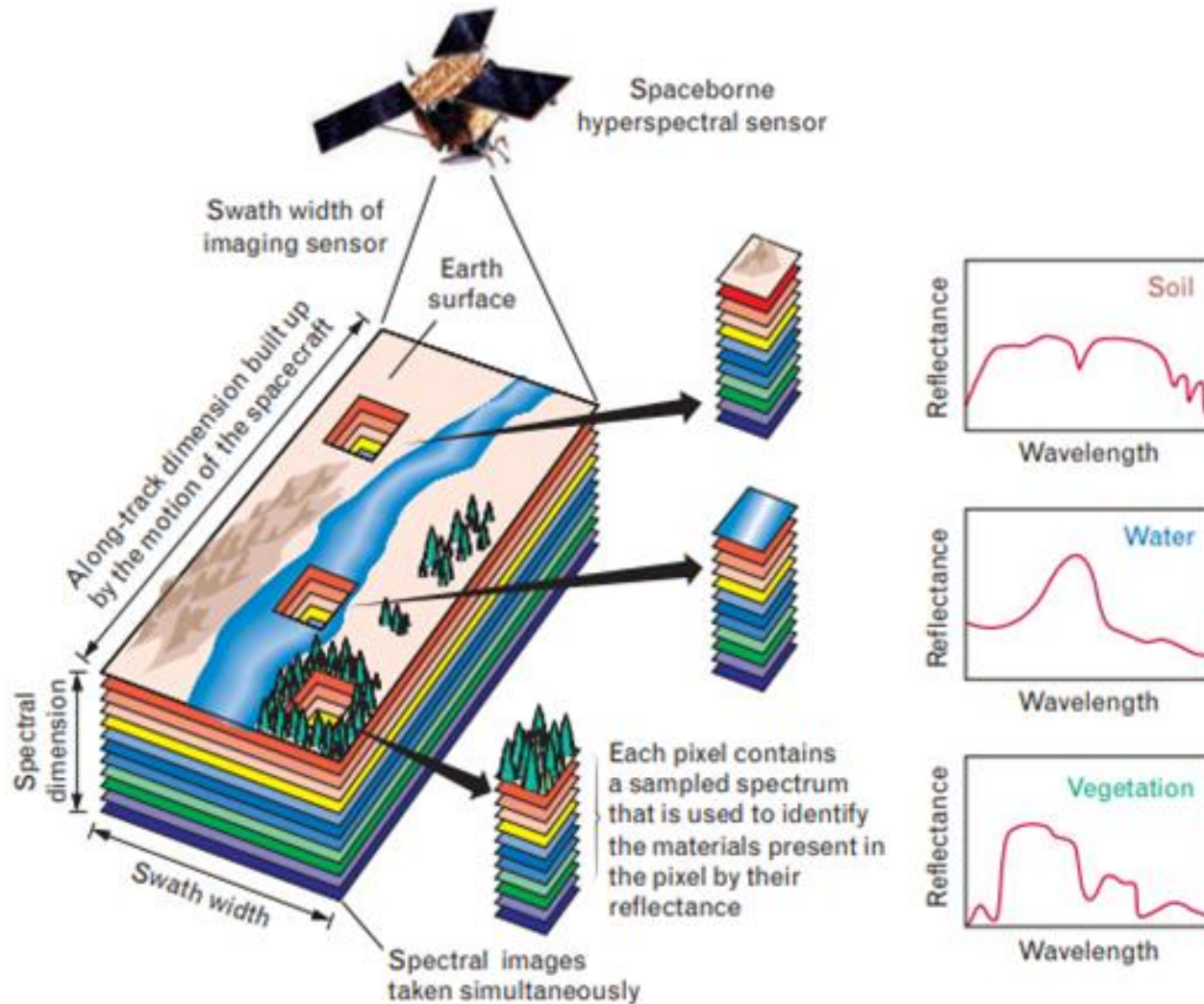


Provides more information than visible imaging

Multi-spectral Imaging



Multi-spectral Imaging

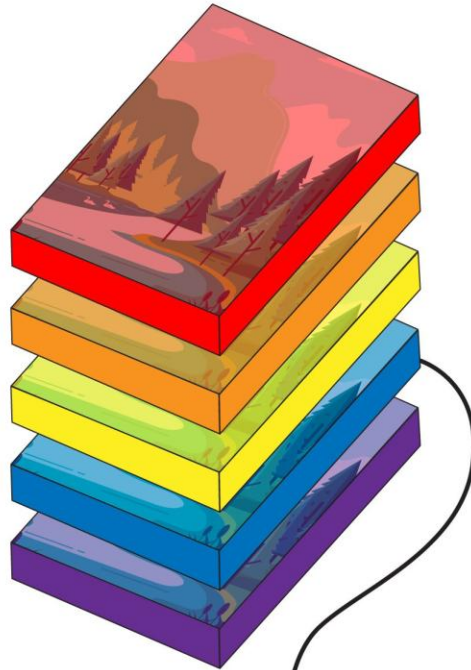


Hyperspectral Imaging

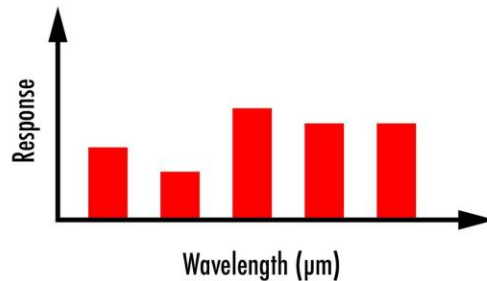
- Hyperspectral imaging (HSI) is a process used to obtain high spectral resolution imagery by dividing light into many narrow, contiguous spectral bands.
- Captures images in hundreds of narrow and contiguous spectral bands
- High spectral resolution
- Allows material identification and detailed spectral analysis
- Used in mineralogy, medical diagnostics, and food quality control

MULTISPECTRAL/ HYPERSPECTRAL COMPARISON

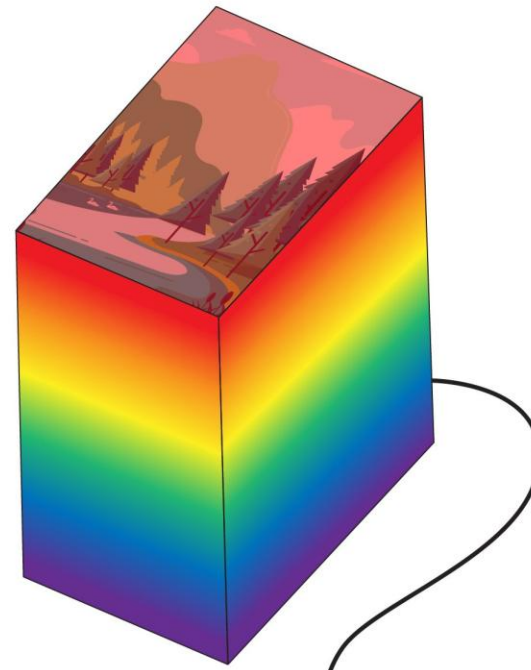
Multispectral



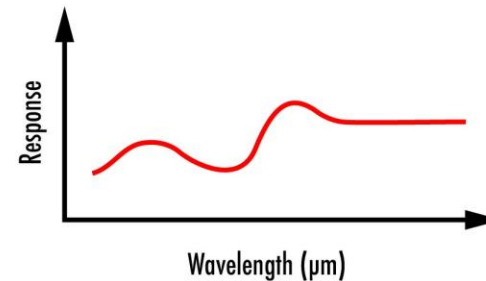
Several Spectral Bands



Hyperspectral



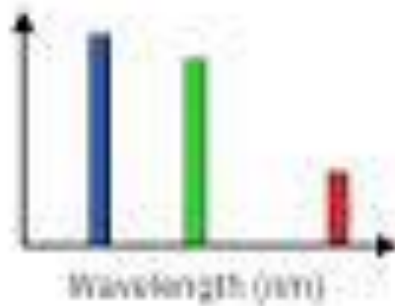
Continuous Spectrum



Comparison

RGB

3 separated bands



MULTISPECTRAL

N separated bands



HYPERSPECTRAL

Continuous Spectrum



Comparison of Imaging Techniques

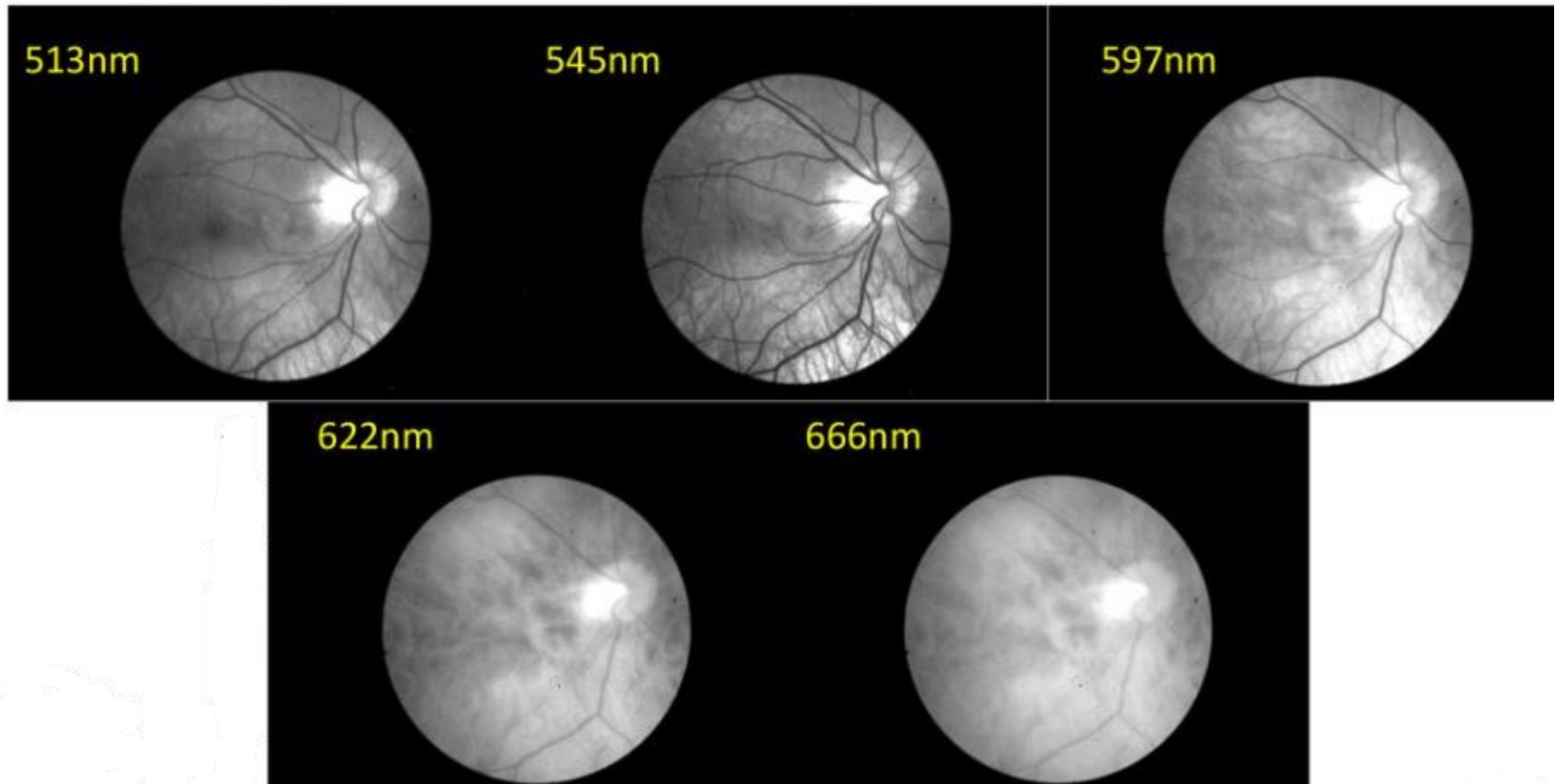
- Monochromatic: Single band, grayscale
- Colour: 3 bands (RGB), visible light
- Multi-spectral: 3–10 bands, discrete non-continuous
- Hyperspectral: 100+ bands, continuous spectrum

Applications of Imaging

- - Medical imaging (e.g., MRI, PET, dermatology)
- - Remote sensing and Earth observation
- - Agriculture and crop monitoring
- - Industrial inspection and quality control
- - Military and defense surveillance

Imaging filters

- A fundus image set acquired with 5 different filters.
 - Images have been registered to account for acquisition time



Motivation: Noise reduction

- Given a camera and a still scene, how can you reduce noise?



Take lots of images and average them!

What's the next best thing?

Image processing



(a) original image



(b) Increased contrast



(c) change in hue



“posterized” (quantized colors)



(e) blurred

Image Filters

- Neighborhood operators can be used to filter images in order to add soft blur, sharpen details, accentuate edges, or remove noise

$$g(i, j) = \sum_{k, l} f(i + k, j + l) h(k, l)$$

$$g = f \otimes h.$$

Image Filters

45	60	98	127	132	133	137	133
46	65	98	123	126	128	131	133
47	65	96	115	119	123	135	137
47	63	91	107	113	122	138	134
50	59	80	97	110	123	133	134
49	53	68	83	97	113	128	133
50	50	58	70	84	102	116	126
50	50	52	58	69	86	101	120

$f(x,y)$

*

0.1	0.1	0.1
0.1	0.2	0.1
0.1	0.1	0.1

=

69	95	116	125	129	132
68	92	110	120	126	132
66	86	104	114	124	132
62	78	94	108	120	129
57	69	83	98	112	124
53	60	71	85	100	114

$g(x,y)$

Neighborhood filtering (convolution): The image on the left is convolved with the filter in the middle to yield the image on the right

Image Filters



(a)



(b)



(c)



(d)

Some neighborhood operations: (a) original image; (b) blurred; (c) sharpened; (d) smoothed with edge-preserving filter;



(e)



(f)



(g)



(h)

(e) binary image; (f) dilated; (g) distance transform; (h) connected components.

Image Filters

Moving average

- Let's replace each pixel with a *weighted* average of its neighborhood
- The weights are called the *filter kernel*
- What are the weights for a 3x3 moving average?

$$\frac{1}{9}$$

1	1	1
1	1	1
1	1	1

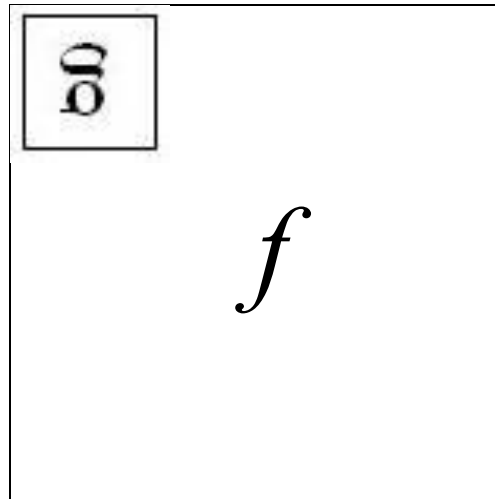
“box filter”

Image Filters

Defining convolution

- Let f be the image and g be the kernel. The output of convolving f with g is denoted $f * g$.

$$(f * g)[m, n] = \sum_{k, l} f[m - k, n - l] g[k, l]$$



- Convention: kernel is “flipped”
- MATLAB: conv2 vs. filter2 (also imfilter)

Image Filters

Key properties

- **Linearity:** $\text{filter}(f_1 + f_2) = \text{filter}(f_1) + \text{filter}(f_2)$
- **Shift invariance:** same behavior regardless of pixel location: $\text{filter}(\text{shift}(f)) = \text{shift}(\text{filter}(f))$
- **Theoretical result:** any linear shift-invariant operator can be represented as a convolution

Image Filters

Properties in more detail

- Commutative: $a * b = b * a$
 - Conceptually no difference between filter and signal
- Associative: $a * (b * c) = (a * b) * c$
 - Often apply several filters one after another: $((a * b_1) * b_2) * b_3$
 - This is equivalent to applying one filter: $a * (b_1 * b_2 * b_3)$
- Distributes over addition: $a * (b + c) = (a * b) + (a * c)$
- Scalars factor out: $ka * b = a * kb = k(a * b)$
- Identity: unit impulse $e = [..., 0, 0, 1, 0, 0, ...]$,
 $a * e = a$

Practice with linear filters



Original

0	0	0
0	1	0
0	0	0

?

Practice with linear filters



Original

0	0	0
0	1	0
0	0	0



Filtered
(no change)

Practice with linear filters



Original

0	0	0
0	0	1
0	0	0

?

Practice with linear filters



Original

0	0	0
0	0	1
0	0	0



Shifted left
By 1 pixel

Practice with linear filters



Original

 $\frac{1}{9}$

1	1	1
1	1	1
1	1	1

?

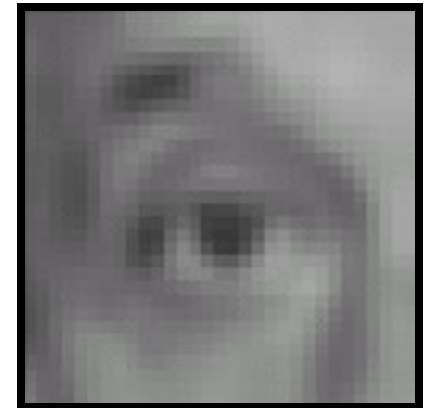
Practice with linear filters



Original

$$\frac{1}{9}$$

1	1	1
1	1	1
1	1	1



Blur (with a
box filter)

Practice with linear filters



Original

0	0	0
0	2	0
0	0	0

—

$\frac{1}{9}$

1	1	1
1	1	1
1	1	1

?

(Note that filter sums to 1)

Practice with linear filters



Original

0	0	0
0	2	0
0	0	0

—

$\frac{1}{9}$

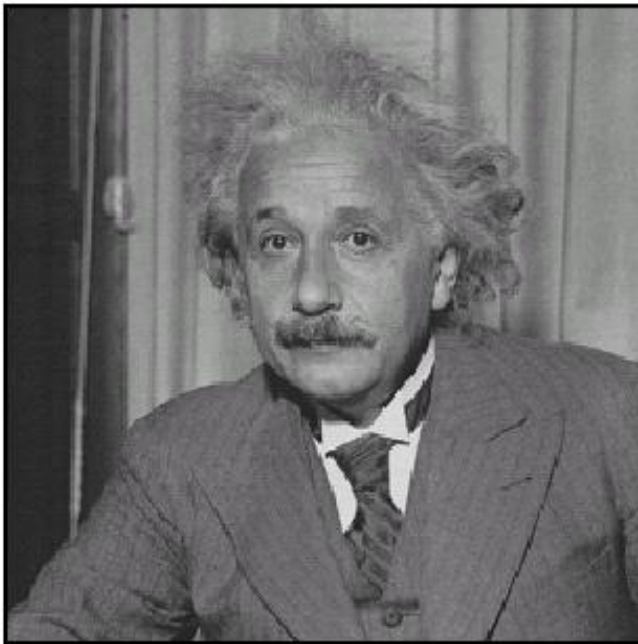
1	1	1
1	1	1
1	1	1



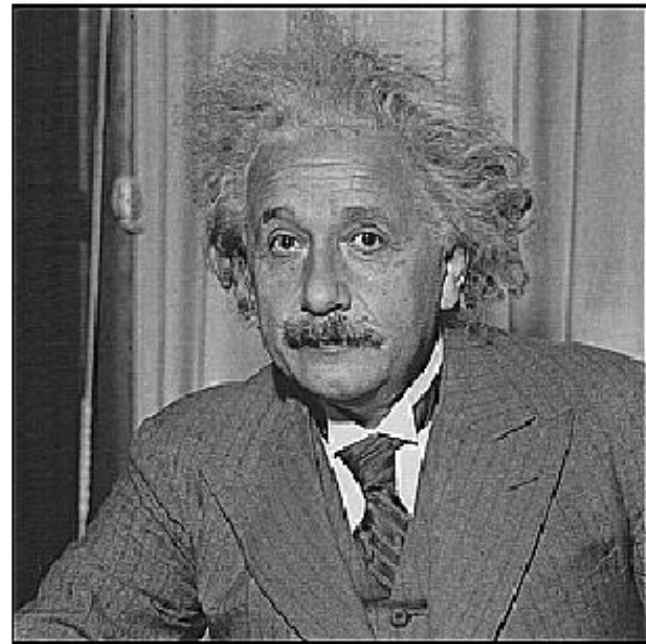
Sharpening filter

- Accentuates differences with local average

Sharpening



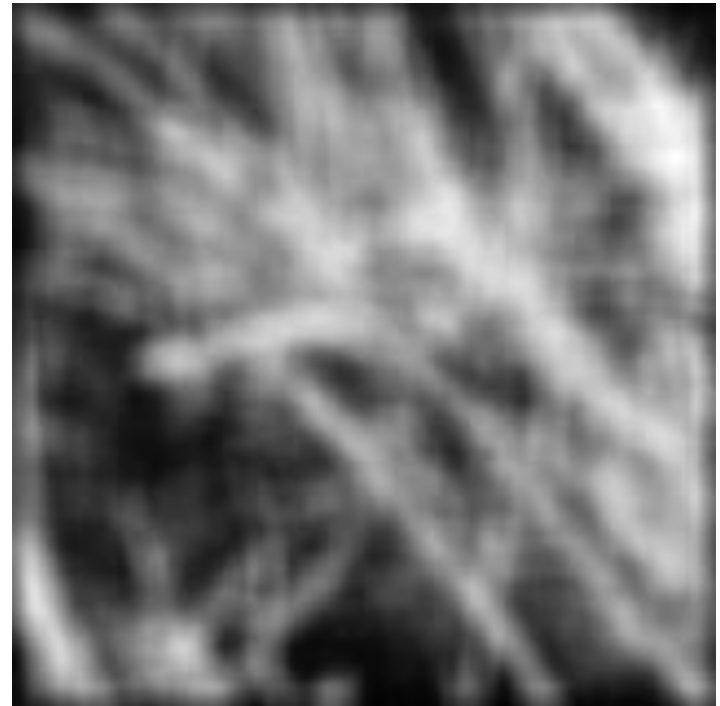
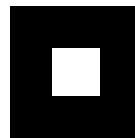
before



after

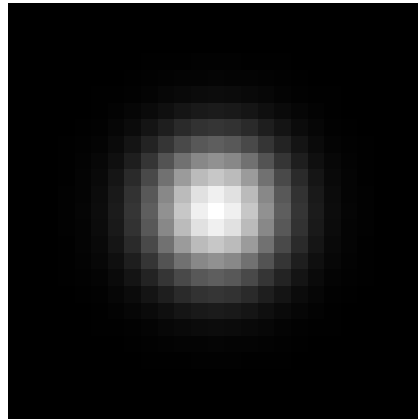
Smoothing with box filter revisited

- Smoothing with an average actually doesn't compare at all well with a defocused lens
- Most obvious difference is that a single point of light viewed in a defocused lens looks like a fuzzy blob; but the averaging process would give a little square



Smoothing with box filter revisited

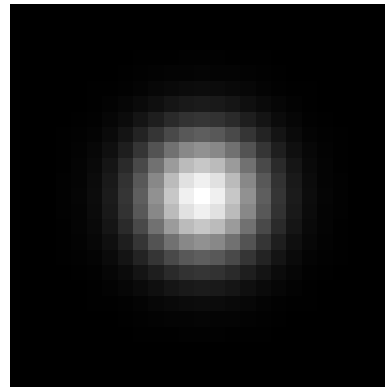
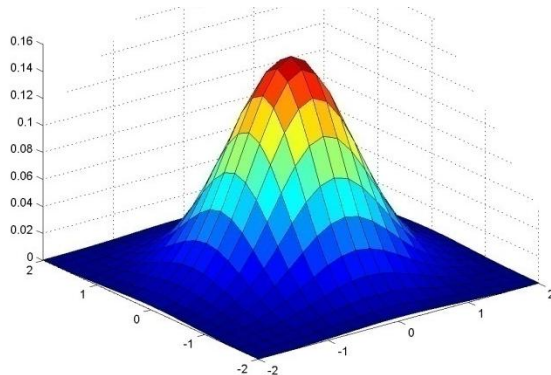
- Smoothing with an average actually doesn't compare at all well with a defocused lens
- Most obvious difference is that a single point of light viewed in a defocused lens looks like a fuzzy blob; but the averaging process would give a little square
- Better idea: to eliminate edge effects, weight contribution of neighborhood pixels according to their closeness to the center, like so:



“fuzzy blob”

Gaussian Kernel

$$G_{\sigma} = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



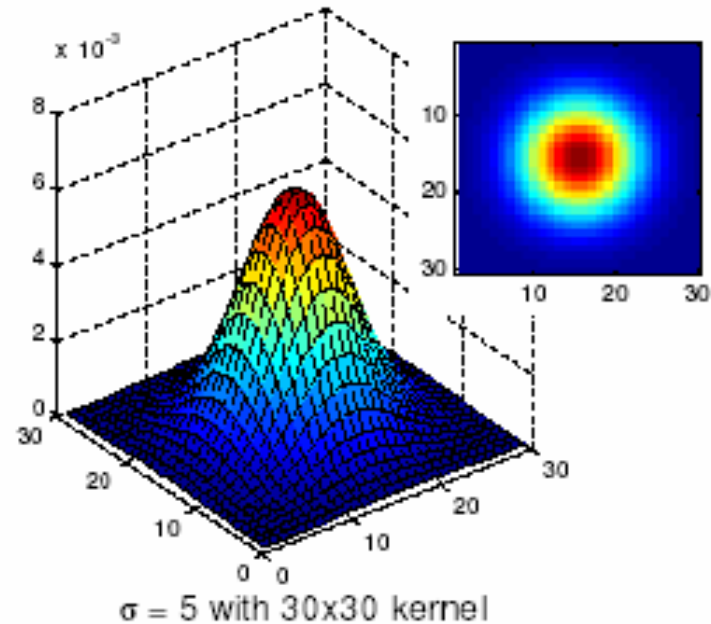
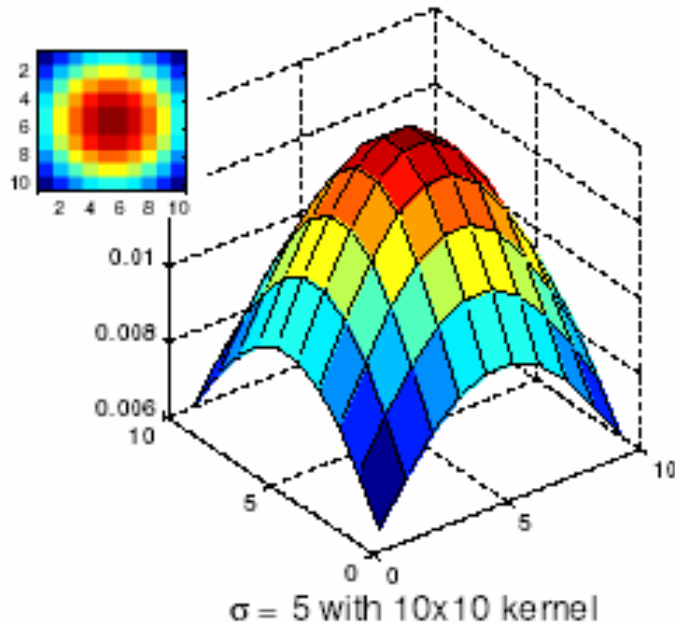
0.003	0.013	0.022	0.013	0.003
0.013	0.059	0.097	0.059	0.013
0.022	0.097	0.159	0.097	0.022
0.013	0.059	0.097	0.059	0.013
0.003	0.013	0.022	0.013	0.003

5 x 5, $\sigma = 1$

- Constant factor at front makes volume sum to 1 (can be ignored, as we should re-normalize weights to sum to 1 in any case)

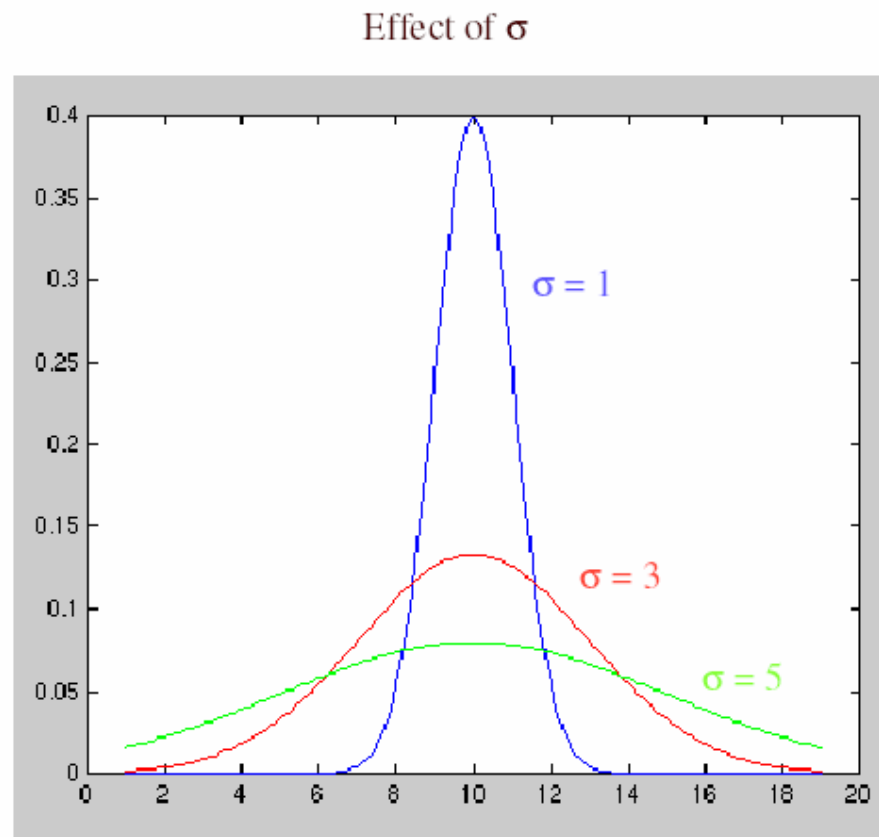
Choosing kernel width

- Gaussian filters have infinite support, but discrete filters use finite kernels

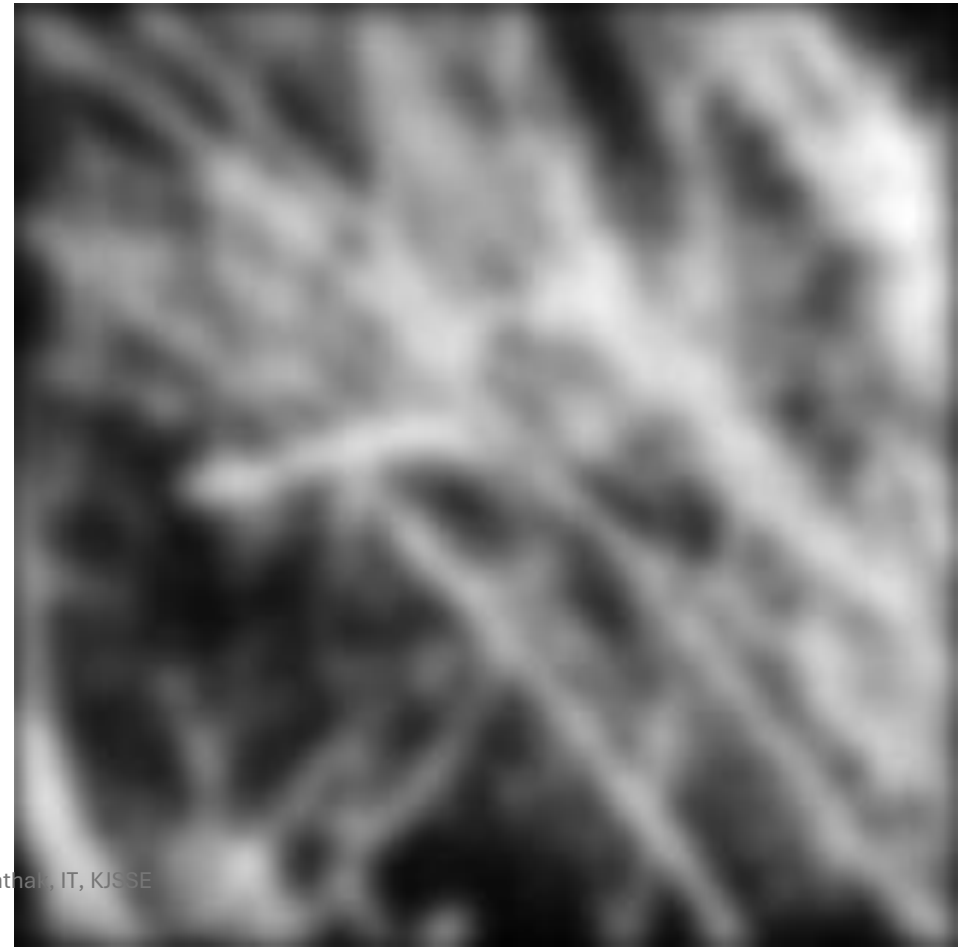
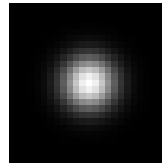


Choosing kernel width

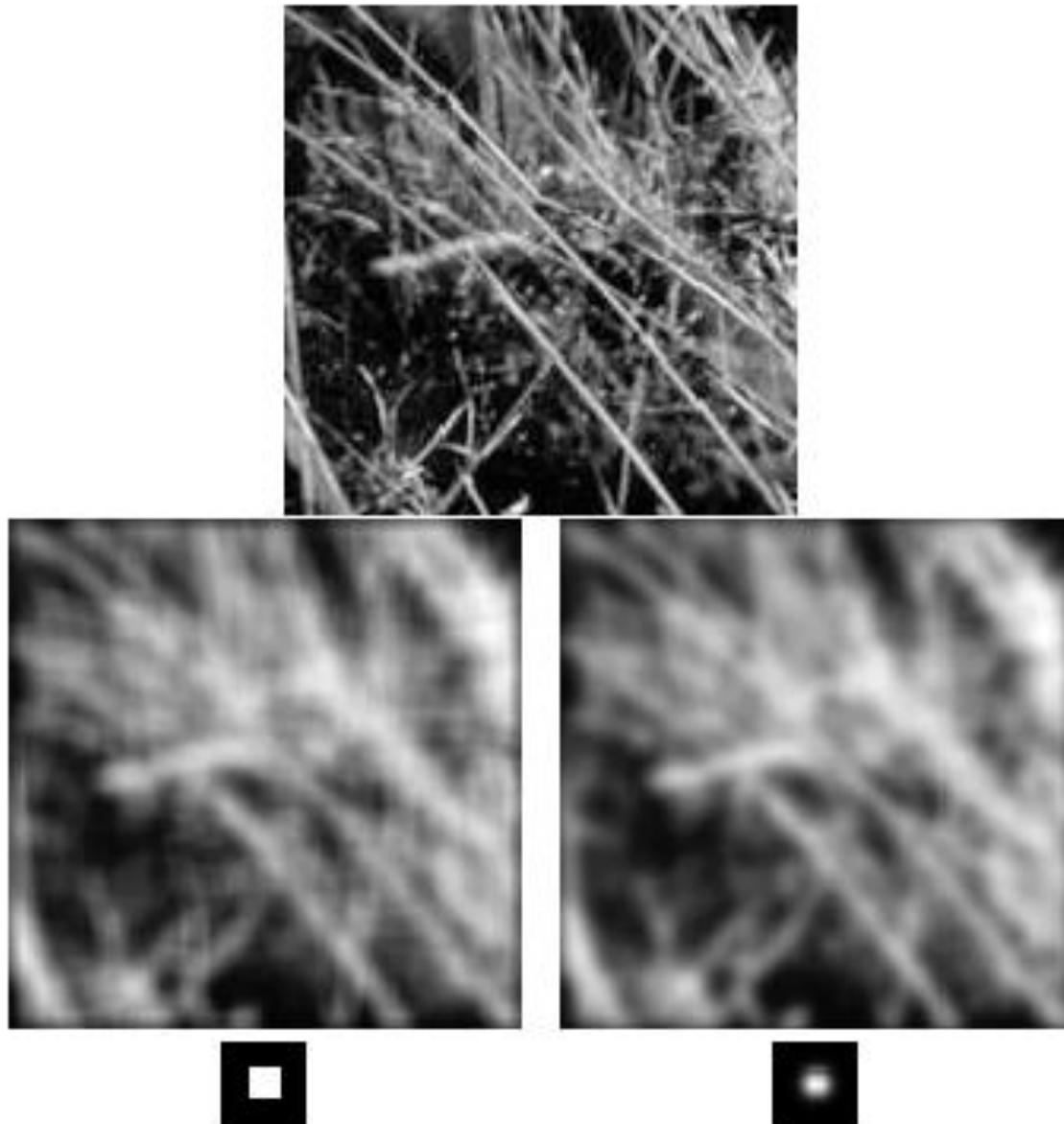
- Rule of thumb: set filter half-width to about 3σ



Example: Smoothing with a Gaussian



Mean vs. Gaussian filtering



Gaussian filters

- Remove “high-frequency” components from the image (low-pass filter)
- Convolution with self is another Gaussian
 - So can smooth with small-width kernel, repeat, and get same result as larger-width kernel would have
 - Convoluting two times with Gaussian kernel of width σ is same as convoluting once with kernel of width $\sigma\sqrt{2}$
- *Separable* kernel
 - Factors into product of two 1D Gaussians

Separability of the Gaussian filter

$$\begin{aligned} G_{\sigma}(x, y) &= \frac{1}{2\pi\sigma^2} \exp^{-\frac{x^2 + y^2}{2\sigma^2}} \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma} \exp^{-\frac{x^2}{2\sigma^2}} \right) \left(\frac{1}{\sqrt{2\pi}\sigma} \exp^{-\frac{y^2}{2\sigma^2}} \right) \end{aligned}$$

The 2D Gaussian can be expressed as the product of two functions, one a function of x and the other a function of y

In this case, the two functions are the (identical) 1D Gaussian

Separability example

2D convolution
(center location only)

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 3 & 3 \\ 3 & 5 & 5 \\ 4 & 4 & 6 \end{bmatrix}$$

The filter factors
into a product of 1D
filters:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

Perform convolution
along rows:

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 3 & 3 \\ 3 & 5 & 5 \\ 4 & 4 & 6 \end{bmatrix} = \begin{bmatrix} & 11 & \\ & 18 & \\ & 18 & \end{bmatrix}$$

Followed by convolution
along the remaining column:

$$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} * \begin{bmatrix} & 11 & \\ & 18 & \\ & 18 & \end{bmatrix} = \begin{bmatrix} & & \\ & 65 & \\ & & \end{bmatrix}$$

Separability

- Why is separability useful in practice?

Noise



Original



Salt and pepper noise



Impulse noise

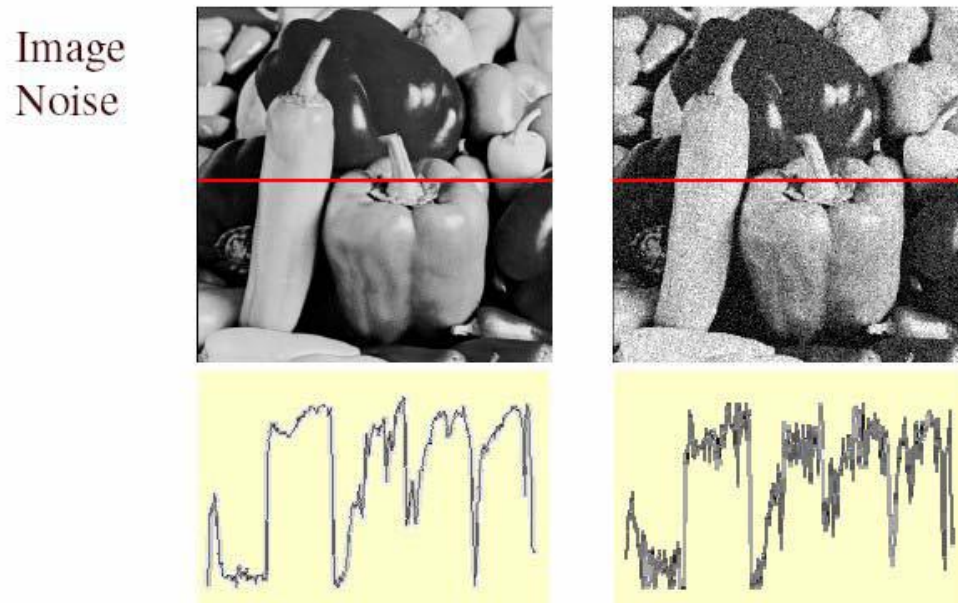


Gaussian noise

- **Salt and pepper noise:** contains random occurrences of black and white pixels
- **Impulse noise:** contains random occurrences of white pixels
- **Gaussian noise:** variations in intensity drawn from a Gaussian normal distribution

Gaussian noise

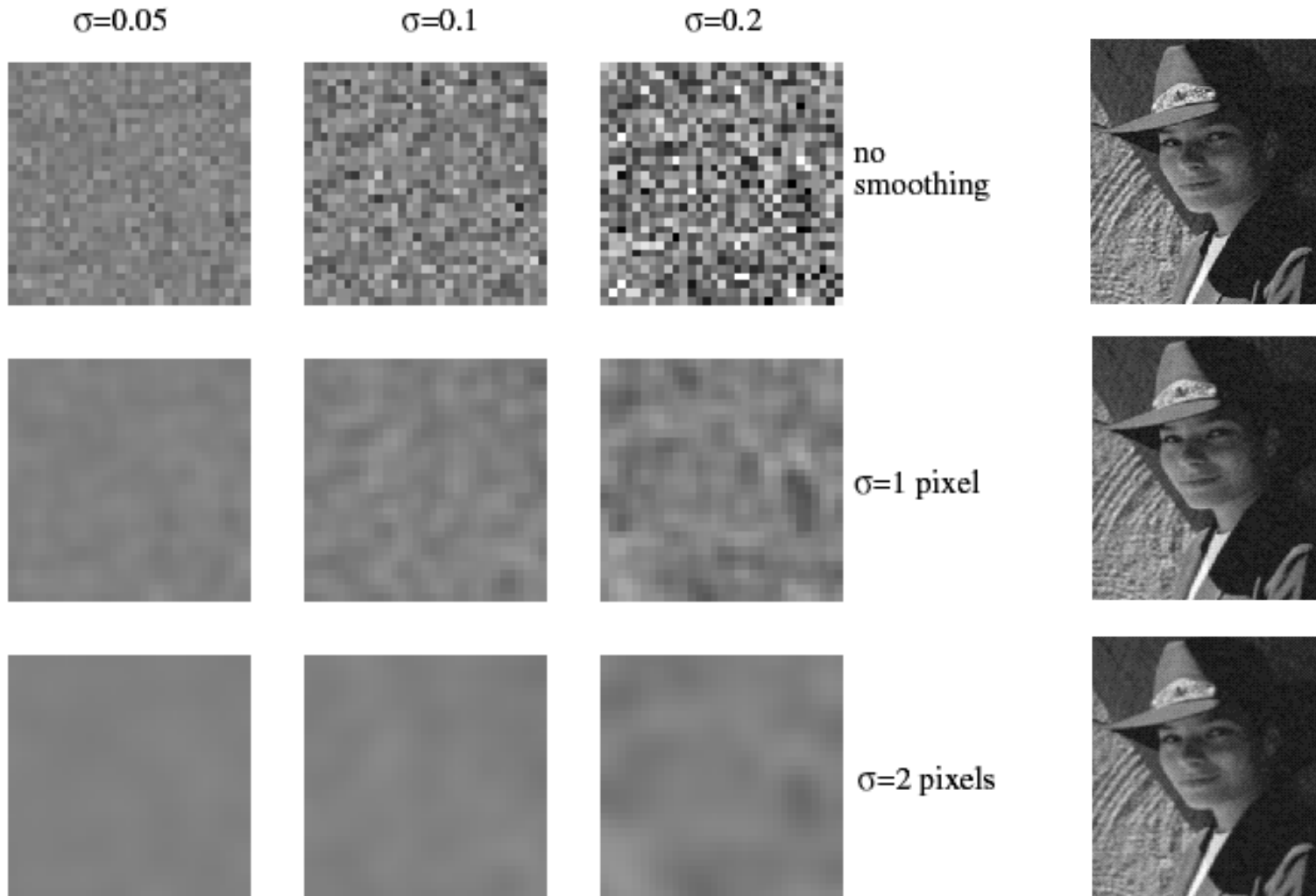
- Mathematical model: sum of many independent factors
- Good for small standard deviations
- Assumption: independent, zero-mean noise



$$f(x, y) = \overbrace{\hat{f}(x, y)}^{\text{Ideal Image}} + \overbrace{\eta(x, y)}^{\text{Noise process}}$$

Gaussian i.i.d. ("white") noise:
 $\eta(x, y) \sim \mathcal{N}(\mu, \sigma)$

Reducing Gaussian noise



Smoothing with larger standard deviations suppresses noise, but also blurs the image

Reducing salt-and-pepper noise

3x3



5x5



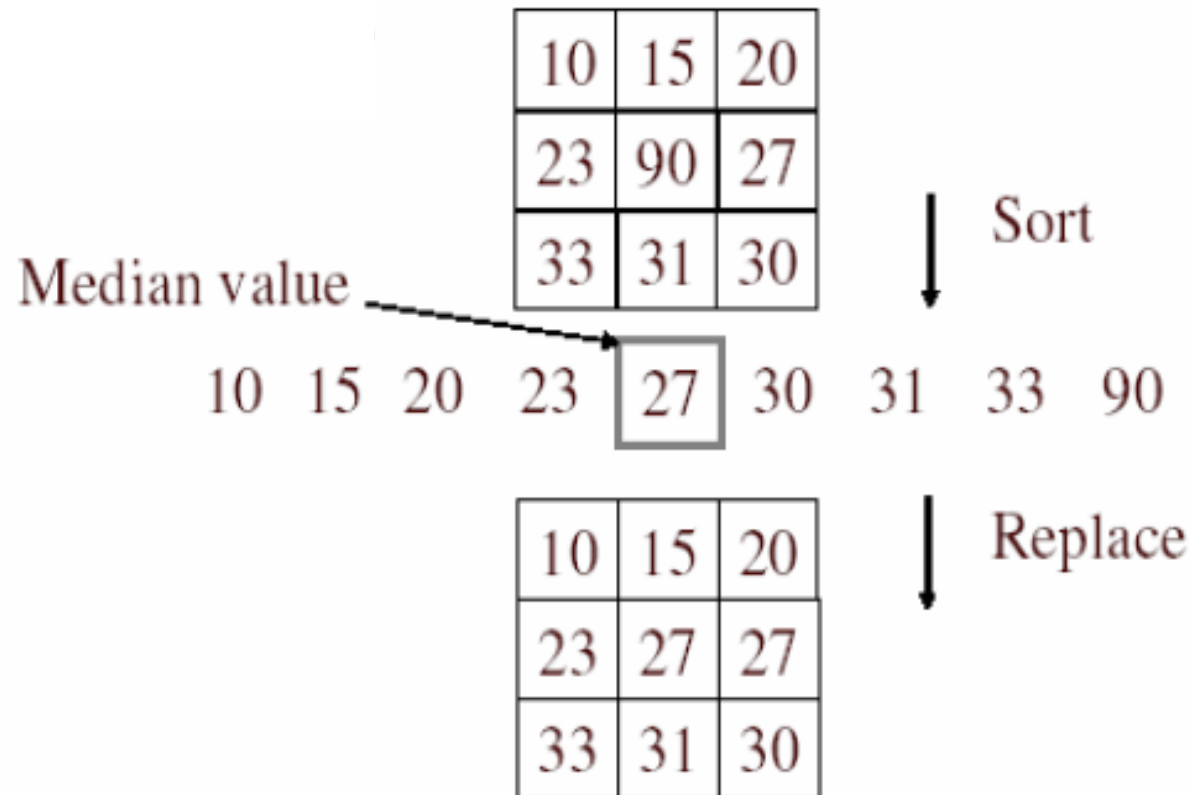
7x7



- What's wrong with the results?

Alternative idea: Median filtering

- A **median filter** operates over a window by selecting the median intensity in the window

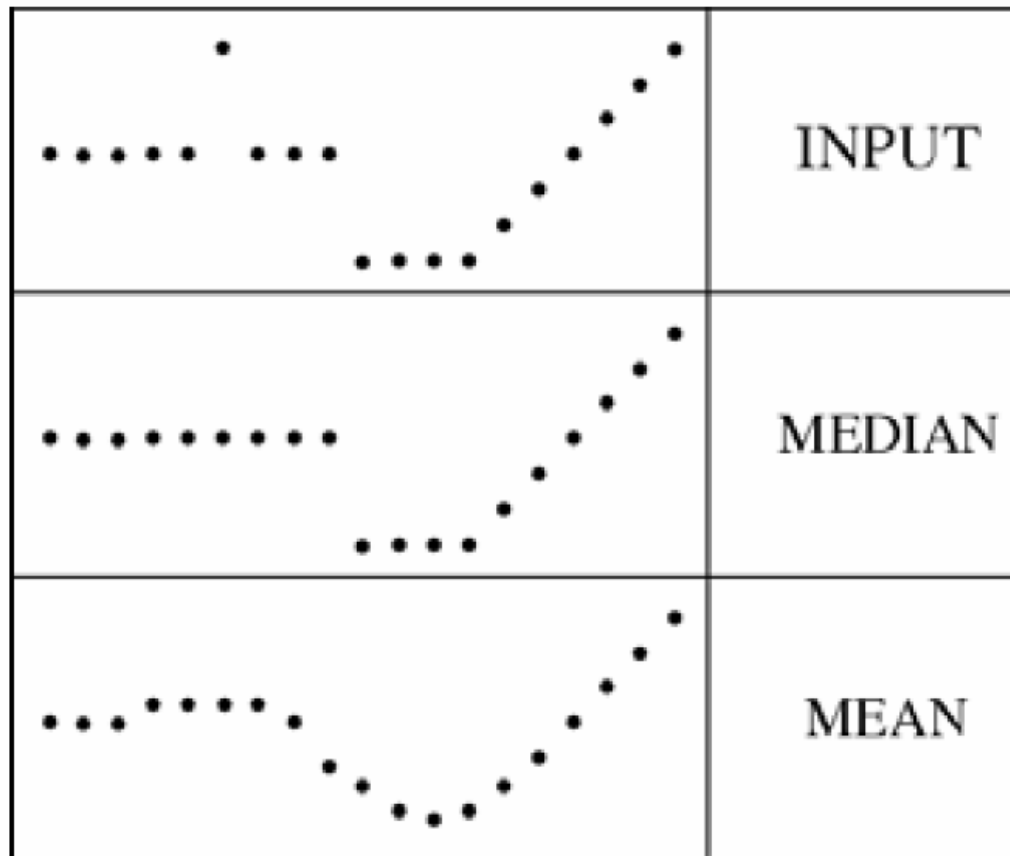


- Is median filtering linear?

Median filter

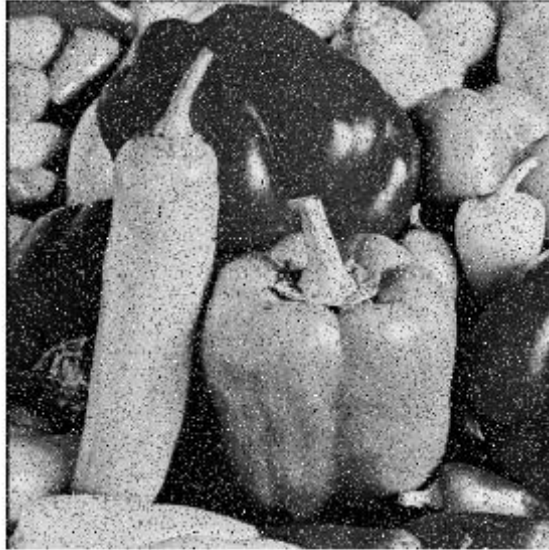
- What advantage does median filtering have over Gaussian filtering?
 - Robustness to outliers

filters have width 5 :

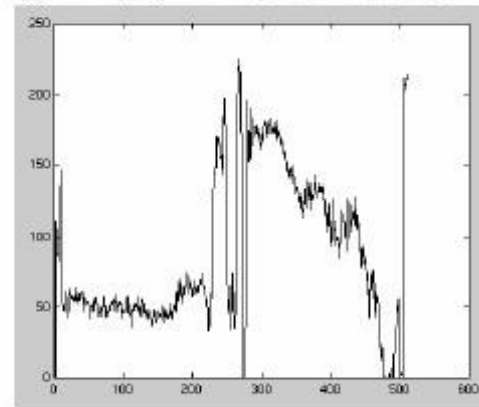
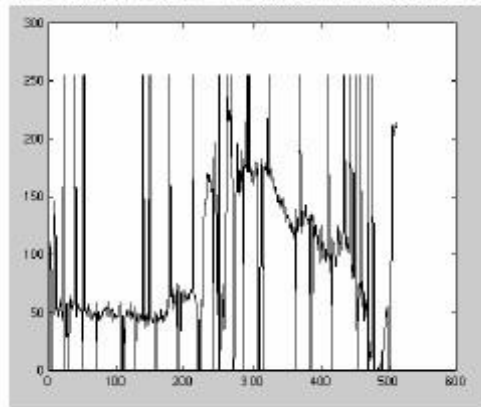


Median filter

Salt-and-pepper noise



Median filtered



- MATLAB: `medfilt2(image, [h w])`

Median vs. Gaussian filtering

3x3

5x5

7x7

Gaussian



Median



Image Resolution

Image resolution

- **Resolution** is a measure of the smallest thing we can discern in an image
 - Resolution affects directly to the quality of the image and determines the object that might be correctly imaged.
- Resolution types
 - Radiometric resolution
 - Pixel resolution
 - Spatial resolution
 - Temporal resolution
 - Spectral resolution
-  **NOTE:** The specific definition and units of resolution depends on the type of resolution

Radiometric resolution

- **Radiometric resolution** refers to the ability to discriminate differences in energy magnitude or levels of brightness.
- The finer the radiometric resolution of a sensor, the more sensitive it is to detecting small differences in reflected or emitted energy.
- The radiometric resolution is encoded on the **depth** of the pixel (i.e. the number of bits used to encode the intensity).

Radiometric resolution

- The radiometric resolution is closely related to the **dynamic range**.
- The dynamic range is the ratio between the largest and smallest possible values of a changeable quantity.
- In medical imaging large dynamic ranges are necessary to image single cell metabolism (which can be seen in almost absolute darkness)



Pixel resolution

- **Pixel resolution** refers to the total number of pixels in an digital image.
 - For example. If an image has M rows and N columns, then its resolution can be defined as $M \times N$.

Pixel resolution

- Pixel resolution affects the image size
 - Visual inspection becomes harder
 - This may become important for computational reasons;
 - time necessary to process the image,
 - memory requirements,
 - etc.

Pixel resolution

- Often, the higher pixel resolution the higher the quality of the image...
 - ...however “quality” of the image is a very ambiguous term.

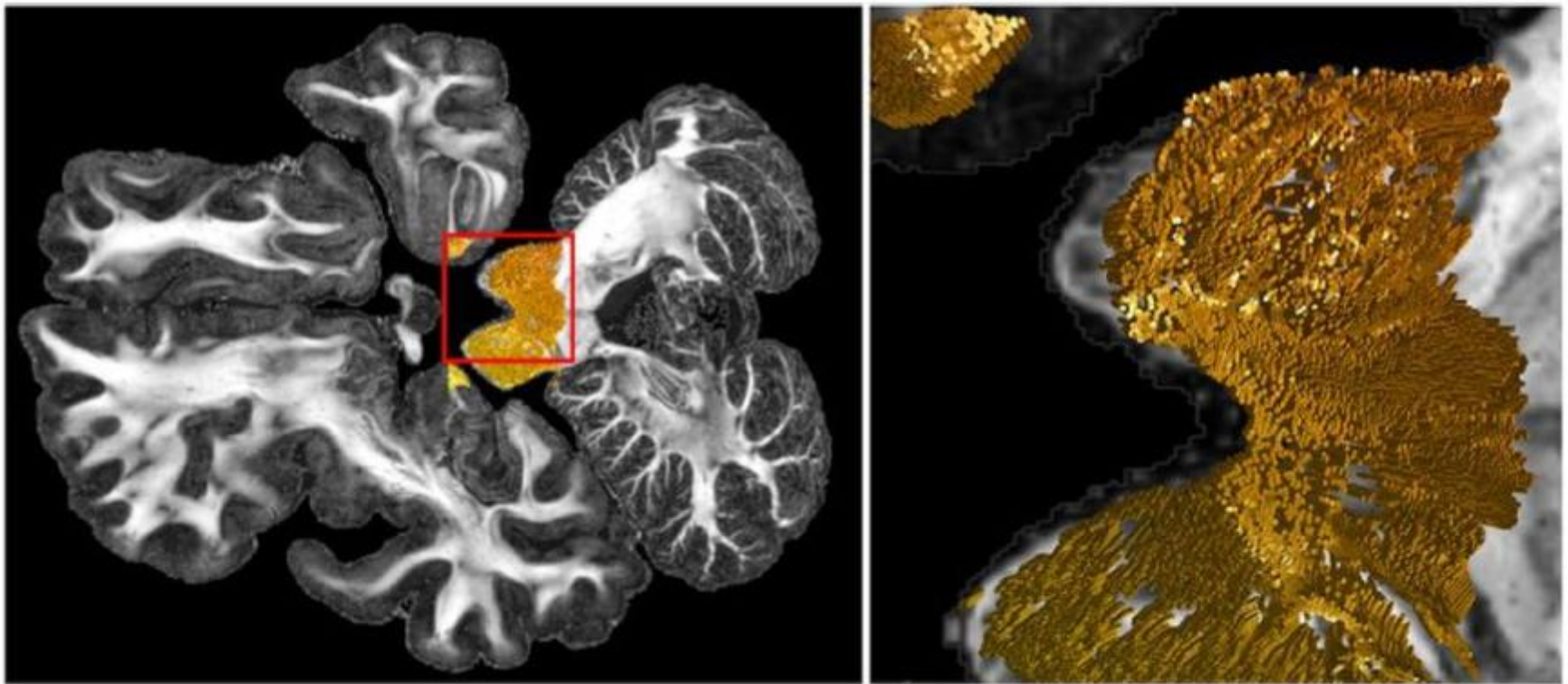


Fig. 7 (Color online) 3D visualization of 20816 reconstructed nerve fibers (15 MB) from 36 PLI slides of a human brain. The region with size of 27.39 mm × 22.72 mm × 3.20 mm that is marked in red on the

left hand side is zoomed on the right. The color of the nerve fibers represents their direction. In the small area shown, all fibers have nearly the same direction

Pixel resolution

Table 1 Examples of large medical image data (fps = frames per second; fpd = frames per day)

Data	Resolution	No of images	No of slices	No of pixel	No of bits	Total memory
Whole body MRI scan	8 mm	1	250	256 × 256	8	16 MB
Screening mammography	50 μm	4	1	5 000 × 6 000	16	230 MB
Whole body CT scan	1 mm	1	2 000	512 × 512	12	750 MB
4D sequence of a beating heart		20 fps	240	512 × 512	12	per sec: 1.75 GB
PLI human brain scan	20 μm	1	3 200	8 000 × 8 000	16	200 GB
IRMA mammography database	50 μm	10 517	1	5 000 × 6 000	16	590 GB
Fluorescence microscopy		2 fpd				per year: 1 TB
Microscopic human brain scan	1.5 μm	1	7 200	106 667 × 106 667	16	66 TB
LIFE full body MRI cohort	8 mm	200 000	250	256 × 256	8	3 PB

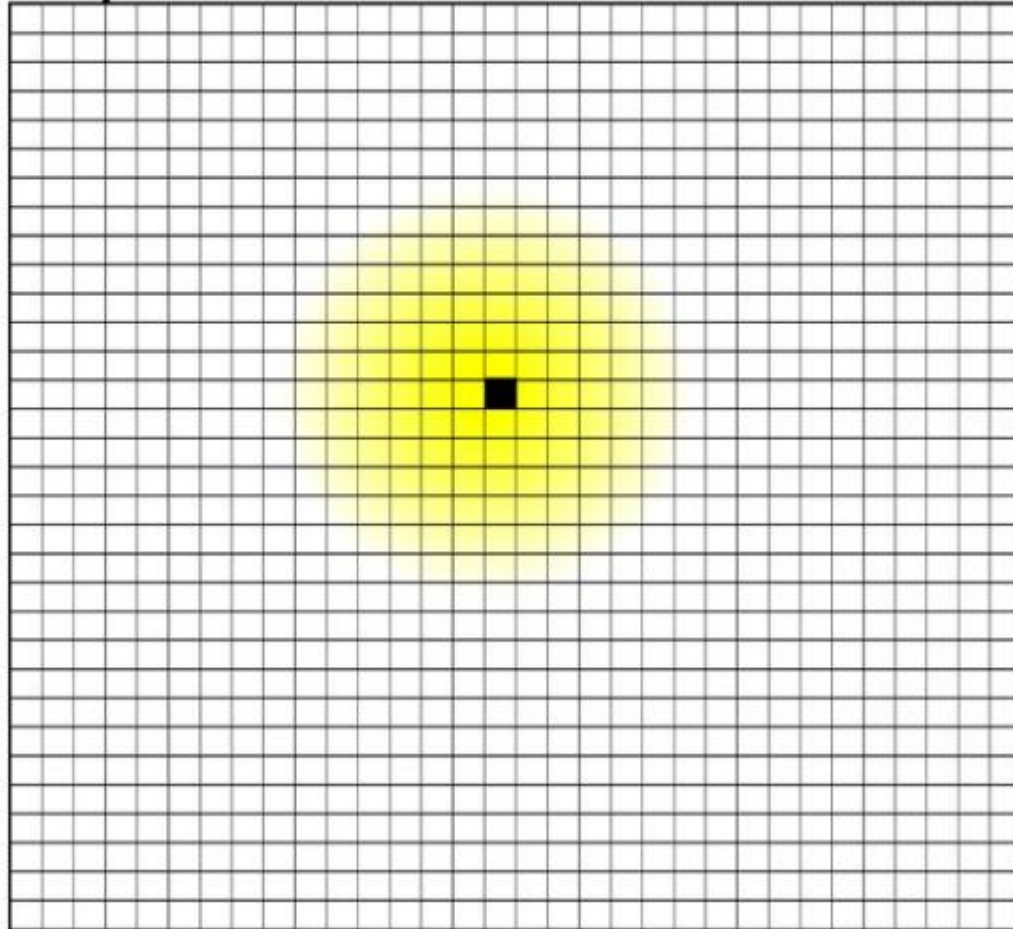
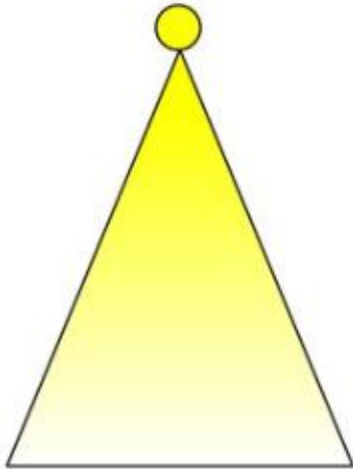
Table from: [Scholl et al (2011) Comput Sci Res Dev 26:5-13]

Spatial resolution

- There are several definitions for **spatial resolution**:
 - Distance covered by a pixel
 - The smallest discernible object
 - Distance that can be reached by a sensor (Remote sensing)
 - Pixel size in the real world
- In general they all mean the same:
 - The relation between the physical distance and the pixel size

Spatial resolution

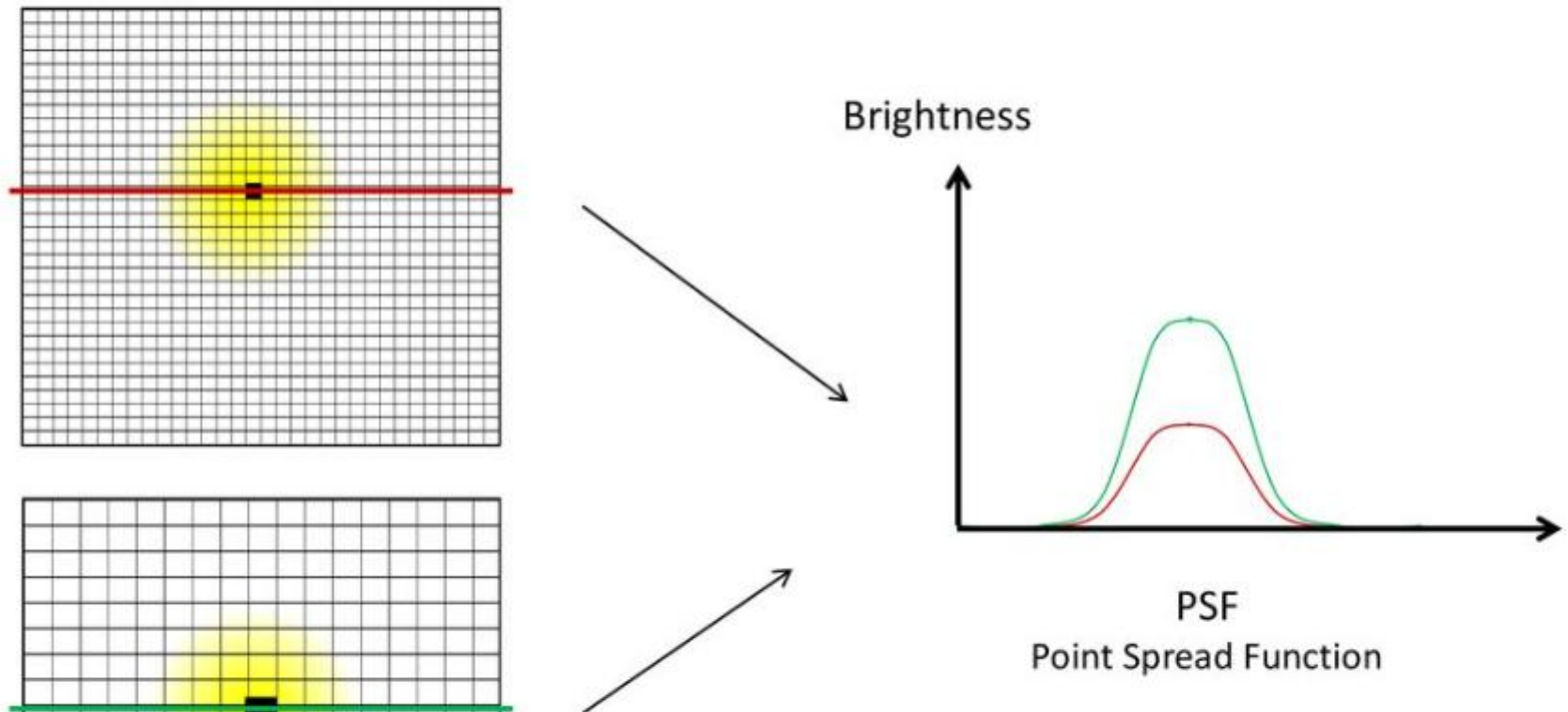
- Spatial resolution is limited by the PSF but also by the pixel resolution.



[Modified from Prof. Claridge]



Spatial resolution



PSF
Point Spread Function

[Modified from Prof. Claridge]

Spatial Resolution

- Spatial resolution (depending on the imaging modality) may also be affected by:
 - Depth at which the imaging system is aiming to sample
 - In optical systems, resolution degrades rapidly with increasing depth
 - Geometric arrangements of sources and detectors
 - e.g. slice size in MRI
 - Also, settings based on transmission have better resolution than their equals based on reflectance (remember, biological tissues are forward scatterers).

Temporal Resolution

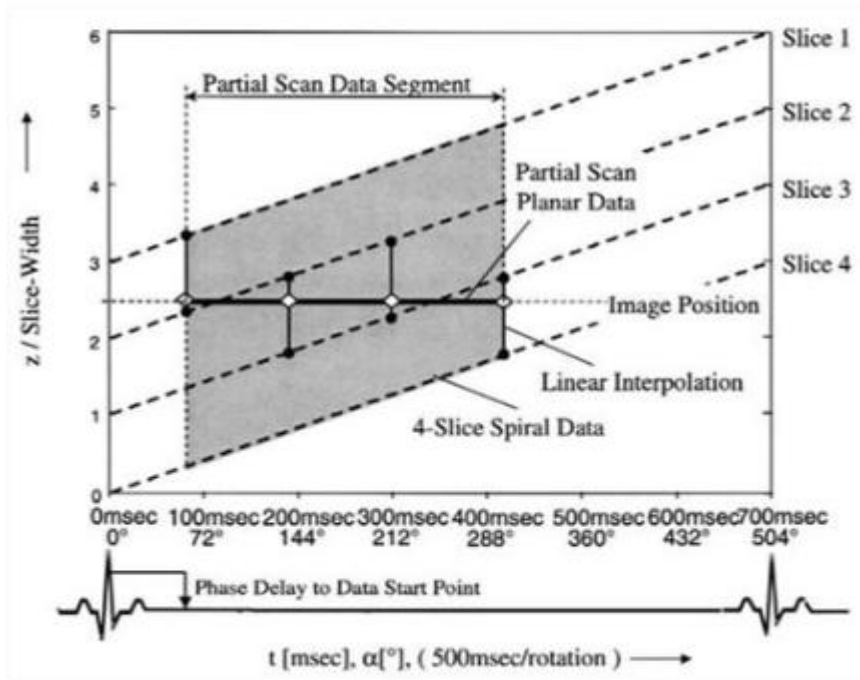
- **Temporal resolution** relates to the duration of time for acquisition of a single frame of a dynamic process. [Radiopedia: <http://radiopaedia.org/articles/temporal-resolution>]
- It determines the precision of a measurement with respect to time; and in this sense it is related to **sampling rate**.

Temporal Resolution

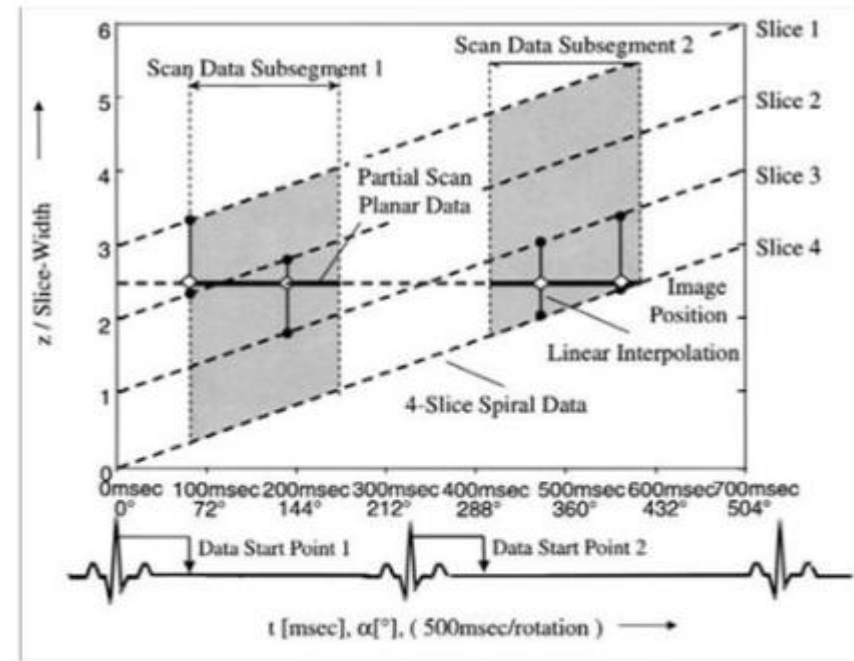
- In biomedical imaging it relates to the ability to observe the evolution of a physiological process –we can't observe things faster than our temporal resolution-,
- ...as well as putting a practical limit on the benefit of sampling ever faster –there's no point to sample faster, if the physiological process is inherently slow-.

Temporal Resolution

- Example: Electrocardiogram [Flohr and Ohnesorge (2001) Journal of Computer Assisted Tomography 25(6):907-923]



Low temporal resolution;
high longitudinal resolution



High temporal resolution;
low longitudinal resolution



Spectral Resolution

- **Spectral resolution** is defined by the wavelength interval size and number intervals that the sensor is measuring [Lalwani et al (2012) IJETAE, 2(5):315-317].
 - Or in other words; the ability of the sensor to define wavelengths intervals
- The finer the spectral resolution, the narrower the wavelength range for a particular channel [Canada Center for Remote Sensing; Sect 2.3 Spatial Resolution, Pixel Size, and Scale Free]

Super-resolution

- **Super-resolution** is the image processing that extracts a high-resolution image from multiple low resolution images of the same scene [Sandar et al (2014) JIKREC, 3(1):1006-1009]
 - The low resolution may contain aliasing artifacts. [Robinson et al; New Applications of Super-resolution in Medical Imaging]

Super-resolution

- More than just interpolation

Interpolation



Interpolation => 1 input image -> 1 output image

Super-resolution



Superresolution => K input images -> 1 output image



Super-resolution

An example of
a super
resolution
algorithm for
MRI

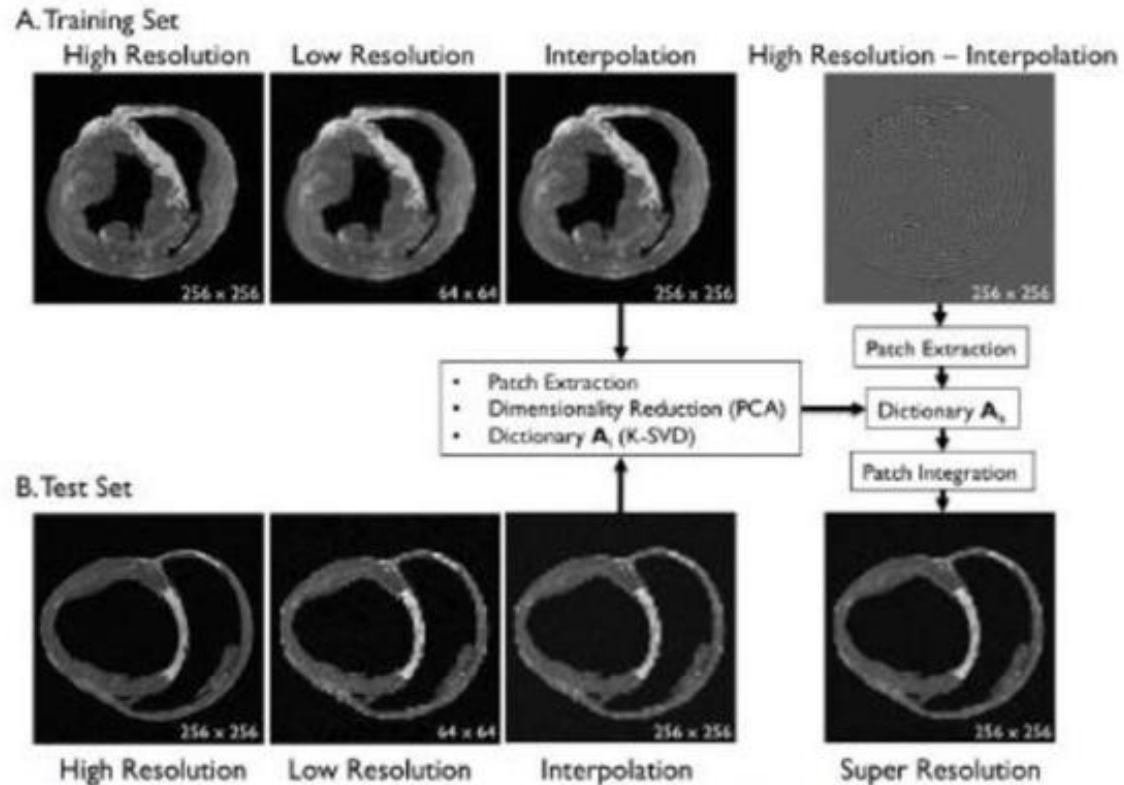


FIGURE 1. Algorithm of single-image super-resolution using sparse representation. The details of the algorithm are described in Zeile et al. [6]. **A.** Training set ($n=14$ hearts, total 2,100 images). In the original high-resolution late gadolinium enhancement (LGE) image (a 256×256 matrix), the region of high signal intensity (SI) (i.e. bright pixels) indicates myocardial infarction (MI). The high-resolution image was scaled down by a factor of 4 to generate a low-resolution image (a 64×64 matrix). The low-resolution image was then scaled up by a factor of 4 to the original size (a 256×256 matrix) by interpolation. The methods of scale-up and scale-down are shown in Fig. 2. Both the high-resolution and interpolated images were divided into small overlapping patches, and the pairs of matching patches were extracted to form the training dictionary. Each of these patch-pairs underwent a pre-processing stage that removes the low-frequency components from high-resolution patches and extracts features from low-resolution patches. Dimensionality reduction using principal component analysis (PCA) was also applied on the features of the low-resolution patches, making the dictionary training step much faster. A low-resolution dictionary A_1 was trained for the low-resolution patches using the K-SVD algorithm [7], such that they could be represented sparsely. A corresponding high-resolution dictionary A_2 was constructed for the high-resolution patches, such that it matched the low-resolution dictionary A_1 . **B.** Test set ($n=10$ hearts, total 1,500 images). As in the training set, low-resolution images (a 64×64 matrix) were also constructed from the high-resolution images (a 256×256 matrix) by scale-down by a factor of 4, and the low-resolution images (a 64×64 matrix) were scaled up to the destination size (a 256×256 matrix) by interpolation. Pre-processed low-resolution patches were extracted from each location, and then sparse-coded using the trained low-resolution dictionary A_1 . The representations found in the low-resolution dictionary A_1 were then used to recover the high-resolution patches by multiplying them with the high-resolution dictionary A_2 . The recovered high-resolution patches were merged by averaging in the overlap area to create the resulting image (a 256×256 matrix).

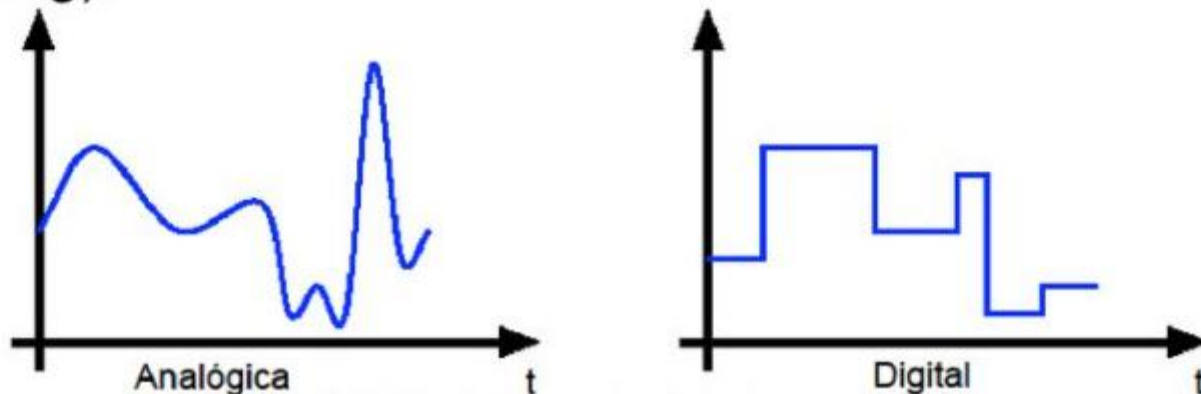
Image Quality and Uncertainties in Image Formation

Uncertainties of image formation

- Many stages of the process of (digital) image formation are characterised by uncertainties
- It is useful to be able to estimate magnitude of potential errors
- Some of the techniques
 - Spatial / spectral calibration
 - Signal to noise ratio (SNR)

Digitization

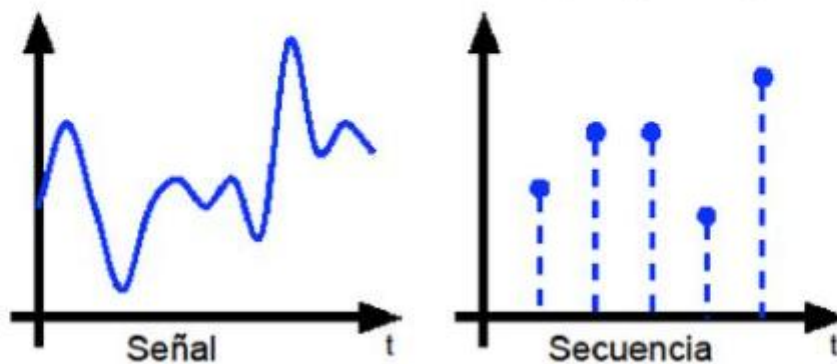
- **Digitization** is the process of transforming an analog signal (infinite values) into a digital signal (discrete number of values).
- In medical imaging, digitization often means converting the analog output of the sensor into a discrete value somehow proportional to recorded radiation intensity
 - Sensors' output most times are a continuous voltage (analog).



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Digitization

- Digitization takes two stages:
 - **Sampling** – Take samples convolving the sampled signal (i.e. intensity) with the sampling signal (e.g. Dirac's delta)
 - Sampling converts a **continuous signal** into a **discrete sequence**, but the result remains analog.
 - **Quantization** – Converts the acquired samples (analog) to discrete values (digital).



Digitization

- Factors affecting the size of the pixel include:
 - Altering the area covered by the pixel e.g. tilting the sensor
 - Enlarging the radiation collection area of the sensor
 - Changing the diameter of the scanning beam
 - Inter-linear space

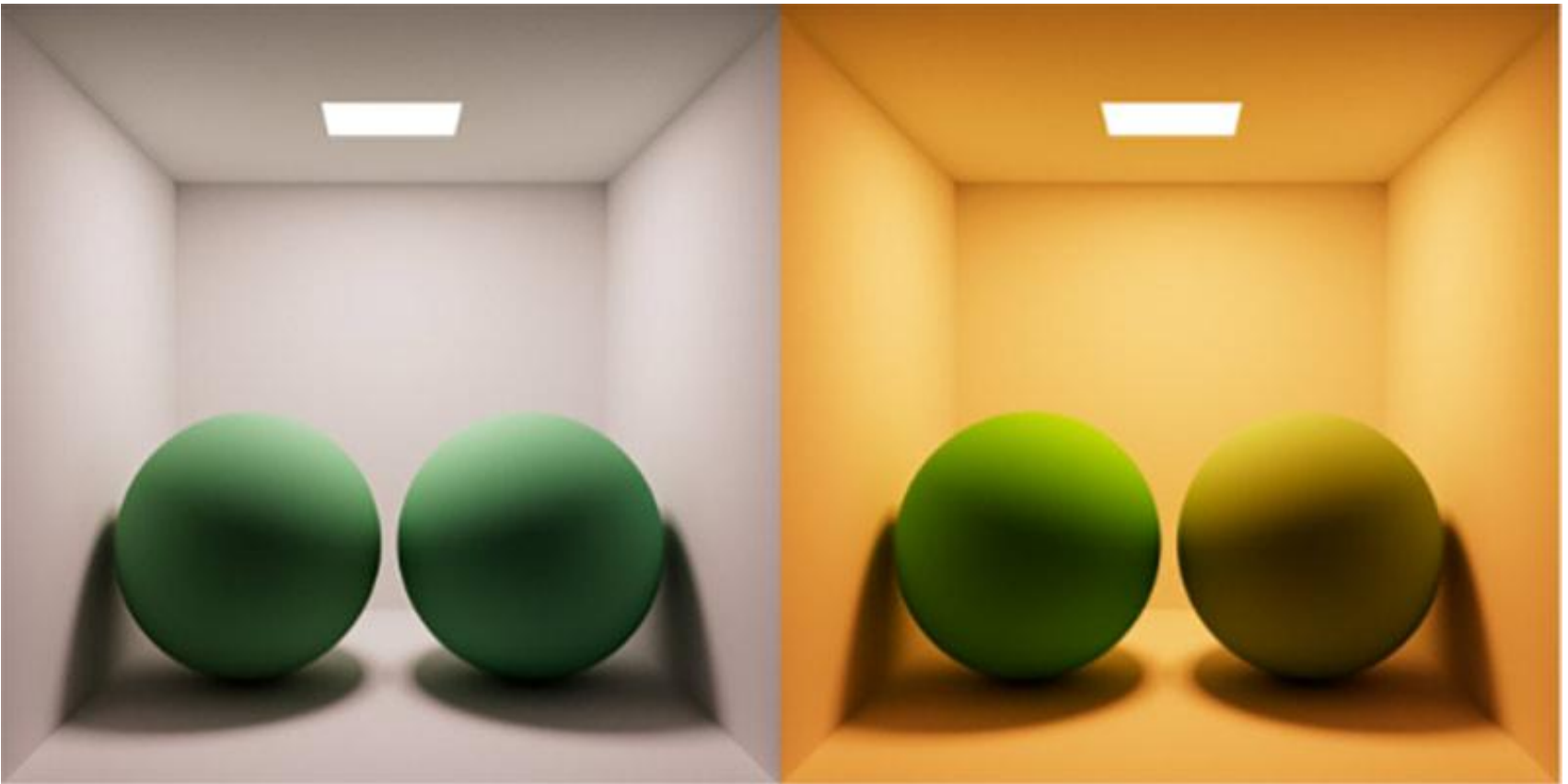
Quantum efficiency

- **Quantum efficiency** (QE) of a photodetector is the ratio between photons excited by the incoming radiation or exposition to the **valence band*** (n_e) and the average number of photons that reach the sensor ($\overline{n_p(\lambda)}$)

$$QE(\lambda) = \frac{n_e(\lambda)}{n_p(\lambda)} < 1$$

- A perfect photodetector (noiseless) will have a QE=1.
- QE depends on the wavelength

Metamerism



Metamerism



Metamerism

- **Metamerism** is the perception of the same “colour” (i.e. sensed spectral emission) from different originating objects e.g. different histophysiological configurations [self definition]
 - It destroys monotonic behaviour and affects reconstruction
- Metamerism is mainly a problem of a many-to-one projection to a lower dimensional space. [Preece and Claride (2004) IEEE TPAMI 26(7):913-922]

Metamerism

- Metamerism is more common with neutral colors like grays, whites, and dark colors.
- Metamerism is less common with lighter or more saturated colors.
- Color metamerism occurs when colors, having different spectral compositions, appear the same to a human eye.
 - Spectral composition is the distribution of light energy at each wavelength that is emitted, transmitted, or reflected by a color sample.

Metamerism

- The phenomenon where two colors appear identical under certain lighting conditions but differ under others.
- Importance in Image Processing:
 - Critical in color reproduction, color matching, and accurate image rendering.
- Two different spectral power distributions producing the same color sensation.
- Example: Matching colors using different pigments or light sources.

Metamerism

- Color metamerism is caused by three basic factors:
 - Light source:
 - Materials:
 - Human color vision:

Metamerism

- Types of Metamerism
 - **Illuminant Metamerism:** Color match under one light source but not another.
 - **Observer Metamerism:** Different observers perceive colors differently due to variations in vision.
 - **Device Metamerism:** Differences in color reproduction between devices.

Metamerism

- **Metamerism in Image Processing Applications**
 - Color calibration
 - Color management systems
 - Digital imaging and printing
 - Display technology

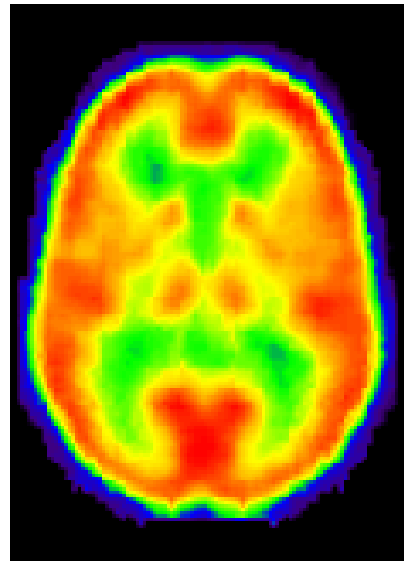
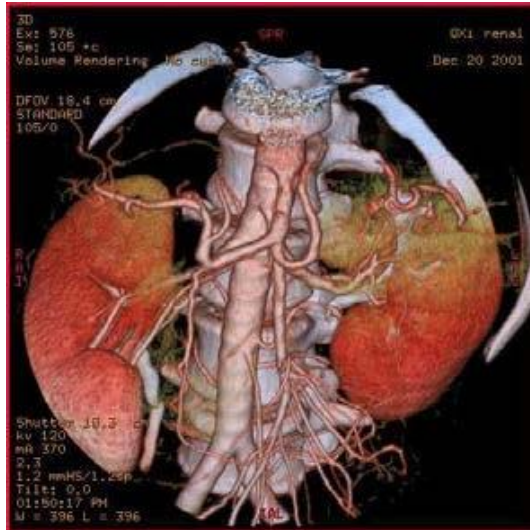
Metamerism

- Challenges Due to Metamerism-
 - Maintaining color consistency across devices and lighting conditions
 - Impact on quality control in imaging workflows
- Methods to Handle Metamerism
 - Using standardized light sources for viewing
 - Spectral imaging and multispectral analysis
 - Color profiling and device characterization

Metamerism

- Metamerism depends on:
 - The imaging system (including the illuminant and the filters)
 - ...specially the number of filters
 - Rule of thumb: The more filters, the more difficult it is that metamerism happens.
 - Corollary: hyperspectral imaging virtually eliminates metamerism
 - The optical properties of the biological sample
 - The acquisition setup (i.e. transmittance or reflectance)
 - Two objects may be metameric under one setup e.g. reflectance, but distinguishable under another e.g. transmittance.

Medical Imaging Modalities



Magnetic Resonance Imaging

- Non-invasive imaging technology that produces three dimensional detailed anatomical images.
- Used for disease detection, diagnosis, and treatment monitoring.
- It is based on sophisticated technology that excites and detects the change in the direction of the rotational axis of protons found in the water that makes up living tissues.



Magnetic Resonance Imaging

- MRIs employ powerful magnets which produce a strong magnetic field that forces protons in the body to align with that field. When a radiofrequency current is then pulsed through the patient, the protons are stimulated, and spin out of equilibrium, straining against the pull of the magnetic field. When the radiofrequency field is turned off, the MRI **sensors** are able to detect the energy released as the protons realign with the magnetic field. The time it takes for the protons to realign with the magnetic field, as well as the amount of energy released, changes depending on the environment and the chemical nature of the molecules. Physicians are able to tell the difference between various types of tissues based on these magnetic properties.

<https://youtu.be/1CGzk-nV06g>

Magnetic Resonance Imaging

- To obtain an MRI image, a patient is placed inside a large magnet and must remain very still during the imaging process in order not to blur the image.
- Contrast agents (often containing the element Gadolinium) may be given to a patient intravenously before or during the MRI to increase the speed at which protons realign with the magnetic field.
- The faster the protons realign, the brighter the image.

Magnetic Resonance Imaging

What is MRI used for?

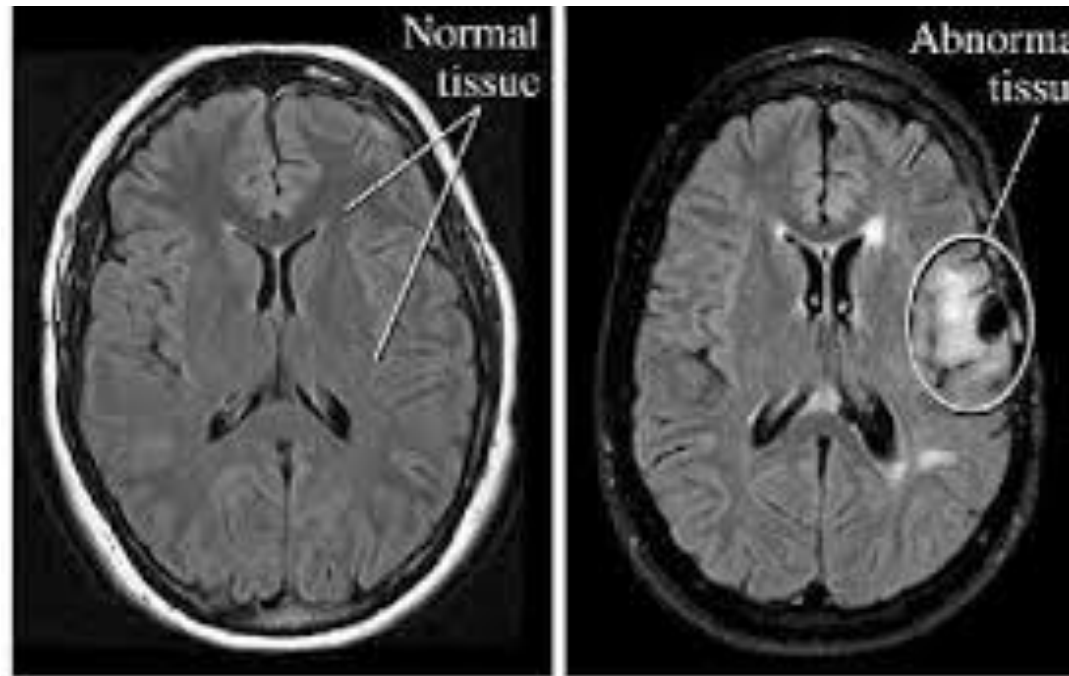
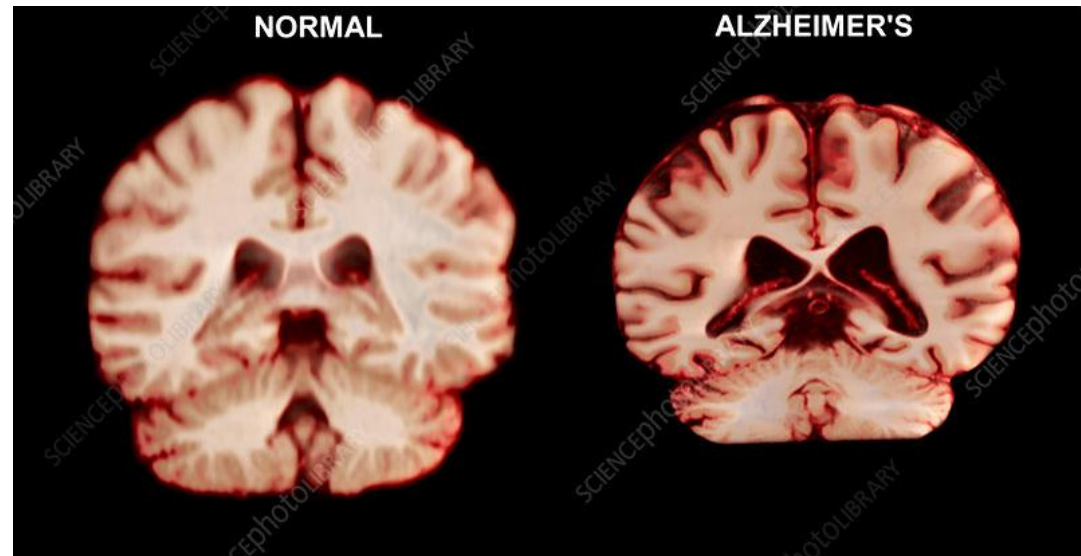
- MRI scanners are particularly well suited to image the non-bony parts or soft tissues of the body.
- They differ from **computed tomography** (CT), in that they do not use the damaging **ionizing radiation** of **x-rays**.
- The brain, spinal cord and nerves, as well as muscles, ligaments, and tendons are seen much more clearly with MRI than with regular x-rays and CT
- for this reason MRI is often used to image knee and shoulder injuries.

Magnetic Resonance Imaging



Magnetic Resonance Imaging

- In the brain, MRI can differentiate between white matter and grey matter
- Can also be used to diagnose aneurysms and tumors.
- Because MRI does not use x-rays or other **radiation**, it is the imaging modality of choice when frequent imaging is required for diagnosis or therapy, especially in the brain.
- However, MRI is more expensive than x-ray imaging or CT scanning.



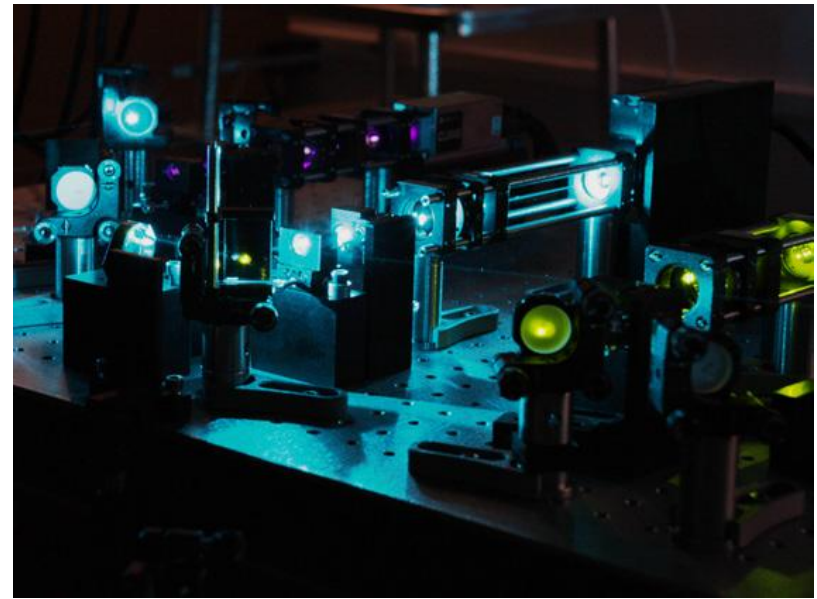
Magnetic Resonance Imaging

- One kind of specialized MRI is functional Magnetic Resonance Imaging (fMRI.)
- Used to observe brain structures and determine which areas of the brain “activate” (consume more oxygen) during various cognitive tasks.
- Used to advance the understanding of brain organization and offers a potential new standard for assessing neurological status and neurosurgical risk.

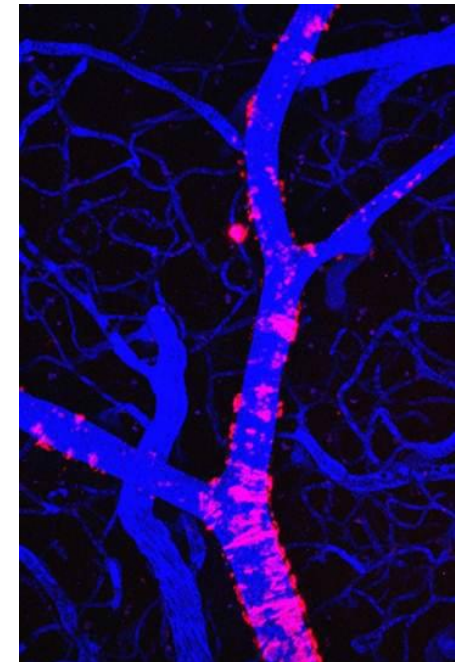
Optical imaging

Optical imaging

- Technique for non-invasively looking inside the body, as is done with x-rays. Unlike x-rays, which use ionizing radiation, optical imaging uses visible light and the special properties of photons to obtain detailed images of organs and tissues as well as smaller structures including cells and even molecules.
- These images are used by scientists for research and by clinicians for disease diagnosis and treatment



Laser set-up in high resolution optical imaging laboratory



Multiphoton microscopy of amyloid deposits in mouse model of Alzheimer's Disease.

Optical imaging

Advantages of optical imaging?

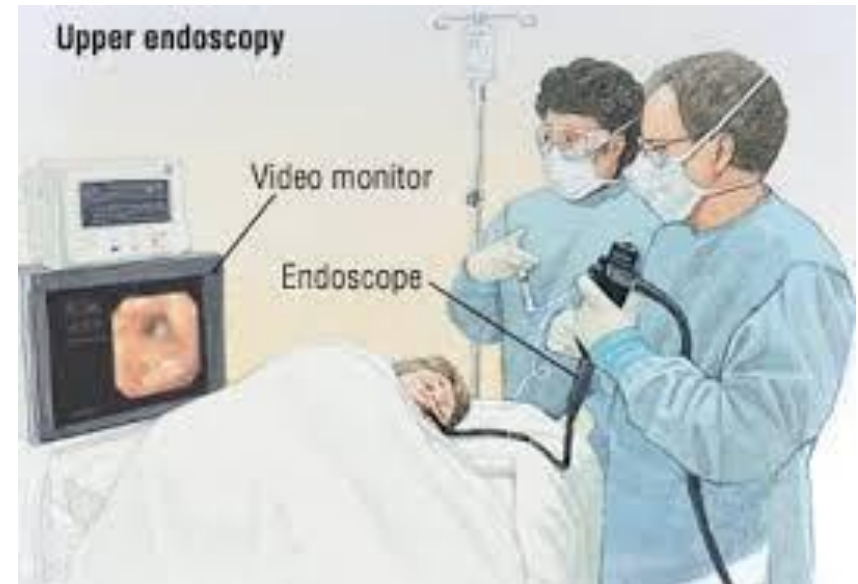
- Reduces patient exposure to harmful radiation by using non-ionizing radiation, which includes visible, ultraviolet, and infrared light.
- Much safer for patients, and significantly faster, optical imaging can be used for lengthy and repeated procedures over time to monitor the progression of disease or the results of treatment.
- Useful for visualizing soft tissues. Soft tissues can be easily distinguished from one another due to the wide variety of ways different tissues absorb and scatter light.
- Because it can obtain images of structures across a wide range of sizes and types, optical imaging can be combined with other imaging techniques, such as MRI or x-rays, to provide enhanced information for doctors monitoring complex diseases or researchers working on intricate experiments.
- Optical imaging takes advantage of the various colors of light in order to see and measure many different properties of an organ or tissue at the same time. Other imaging techniques are limited to just one or two measurements

Optical imaging

Types of optical imaging-

1. Endoscopy:

- The simplest and most widely recognized type of optical imaging is endoscopy.
- An endoscope consists of a flexible tube with a system to deliver light to illuminate an organ or tissue.
- For example, a physician can insert an endoscope through a patient's mouth to see the digestive cavity to find the cause of symptoms such as abdominal pain, difficulty swallowing, or gastrointestinal bleeding.
- Endoscopes are also used for minimally invasive robotic surgery to allow a surgeon to see inside the patient's body while remotely manipulating the thin robotic arms that perform the procedure



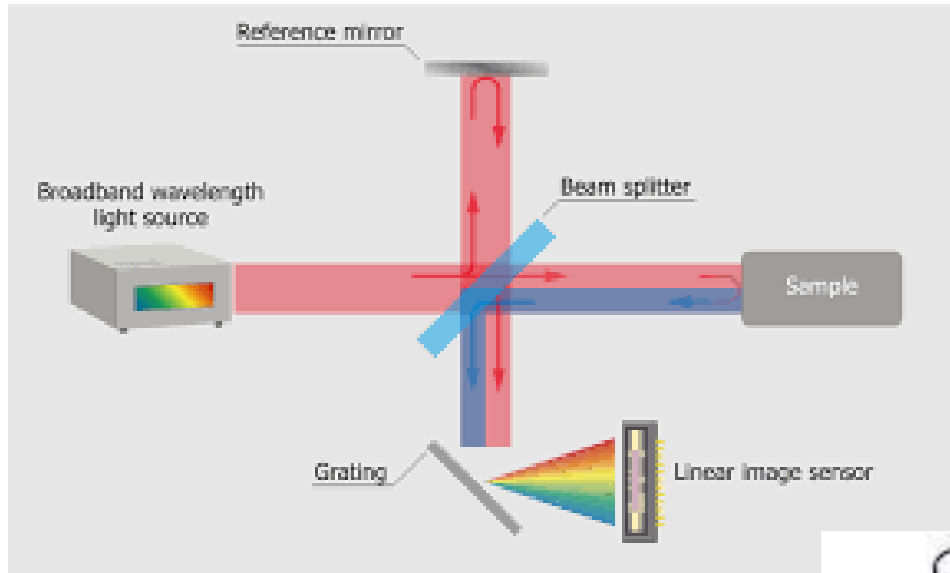
Optical imaging

- **Optical Coherence Tomography (OCT):**

- A technique for obtaining sub-surface images such as diseased tissue just below the skin.
- OCT is a well-developed technology with commercially available systems now in use in a variety of applications, including art conservation and diagnostic medicine.
- For example, ophthalmologists use OCT to obtain detailed images from within the retina.
- Cardiologists also use it to help diagnose coronary artery disease.

Optical imaging

Optical Coherence Tomography (OCT):



Source: <https://www.hamamatsu.com/eu/en/applications/medical-imaging/oct.html>



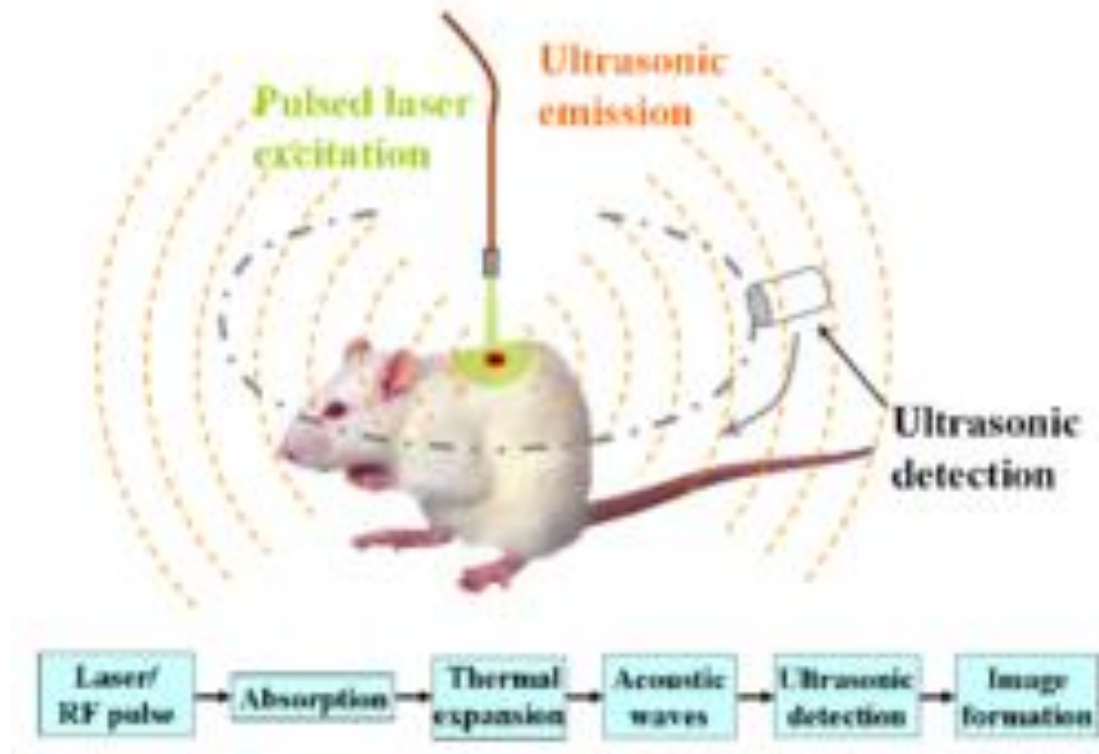
OCT Eye Test - Optical Coherence Tomography

www.drswatisatheeyeclinic.com

Optical imaging

- **Photoacoustic Imaging:**

- During photoacoustic imaging, laser pulses are delivered to a patient's tissues; the pulses generate heat, expanding the tissues and enabling their structure to be imaged.
- The technique can be used for a number of clinical applications including monitoring blood vessel growth in tumors, detecting skin melanomas, and tracking blood oxygenation in tissues.



Source: https://en.wikipedia.org/wiki/Photoacoustic_imaging

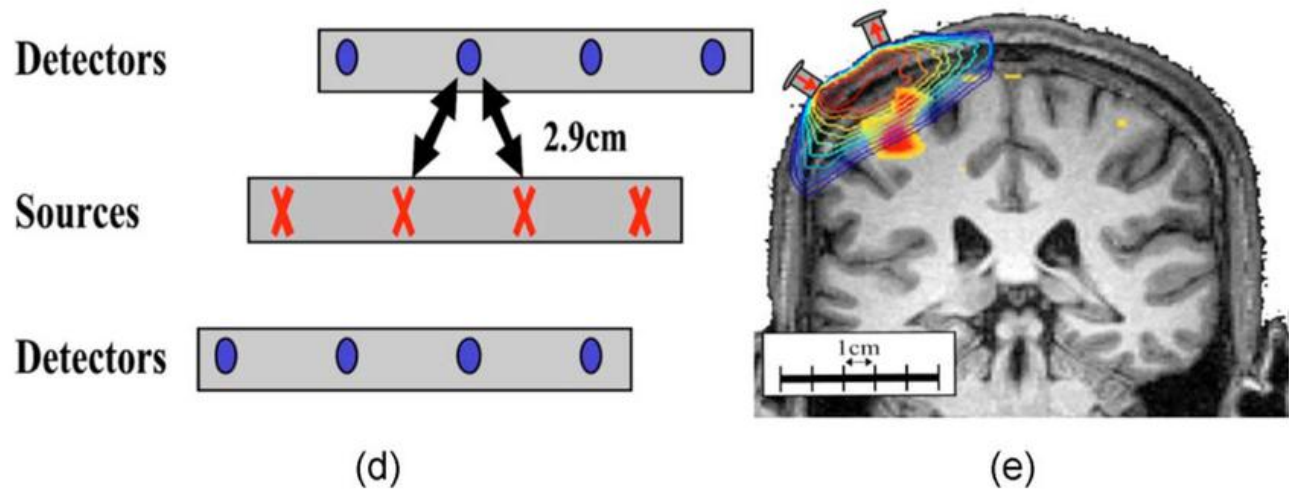
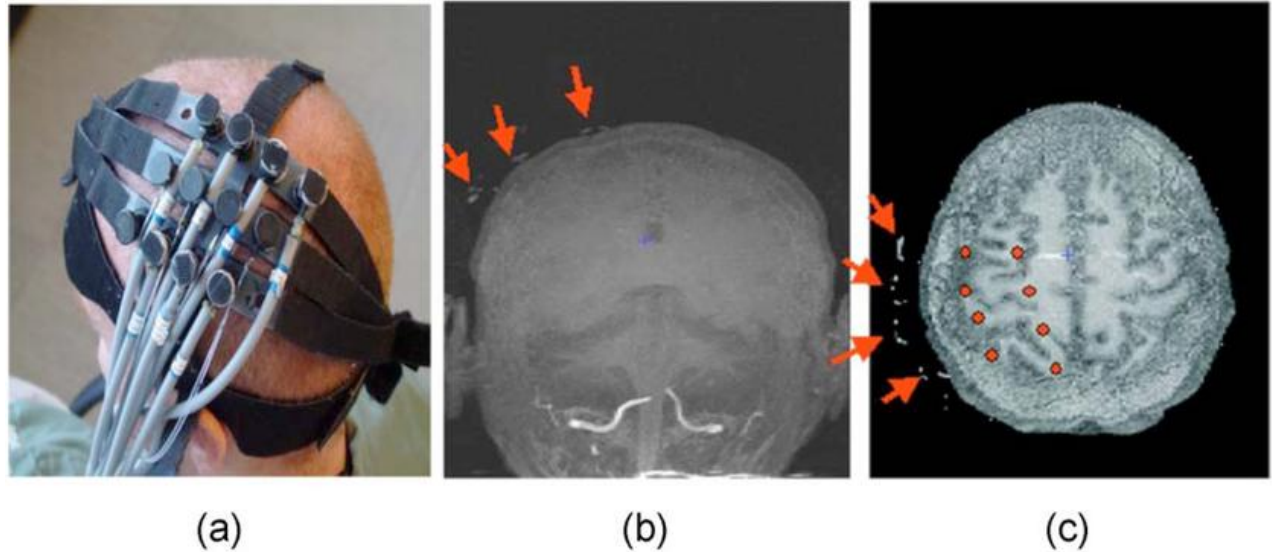
Optical imaging

Diffuse Optical Tomography (DOT):

- Used to obtain information about brain activity.
- A laser that uses near-infrared light is positioned on the scalp.
- The light goes through the scalp and harmlessly traverses the brain.
- The absorption of light reveals information about chemical concentrations in the brain.
- The scattering of the light reflects physiological characteristics such as the swelling of a neuron upon activation to pass on a neural signal.

Optical imaging

Diffuse Optical Tomography (DOT):



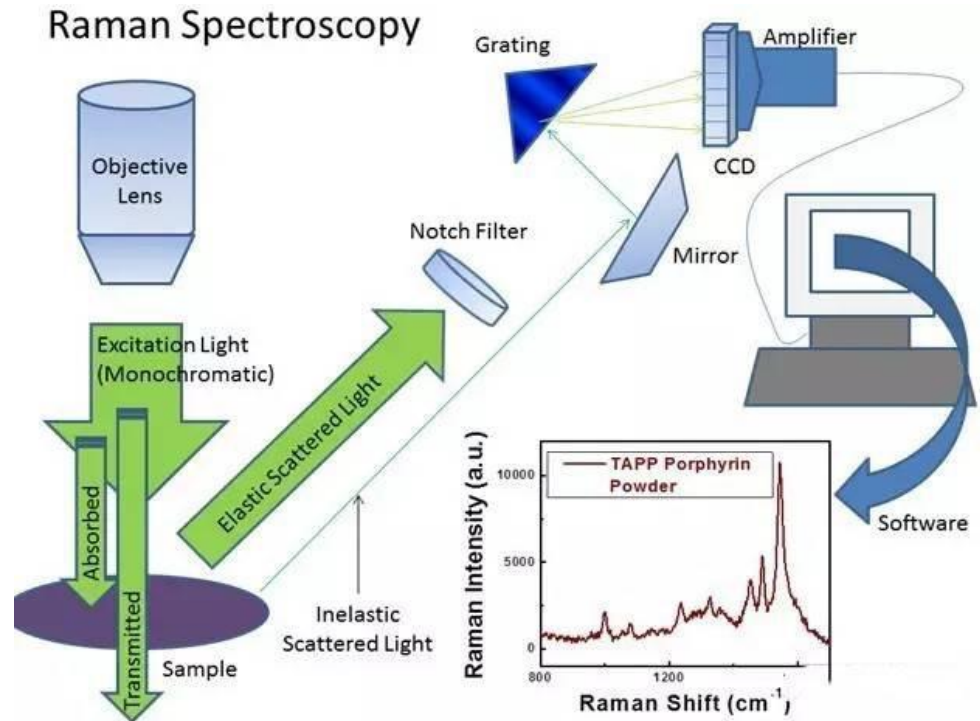
Source:

https://www.researchgate.net/publication/23485456_Huppert_TJ_Diamond_SG_Boas_DADirect_estimation_of_evoked_hemoglobin_changes_by_multimodality_fusion_imaging_J_Biomed_Opt_13054031

Optical imaging

Raman Spectroscopy:

- This technique relies on what is known as Raman scattering of visible, near-infrared, or near-ultraviolet light that is delivered by a laser.
- The laser light interacts with molecular vibrations in the material being examined, and shifts in energy are measured that reveal information about the properties of the material.
- The technique has a wide variety of applications including identifying chemical compounds and characterizing the structure of materials and crystals.
- In medicine, Raman gas analyzers are used to monitor anesthetic gas mixtures during surgery.



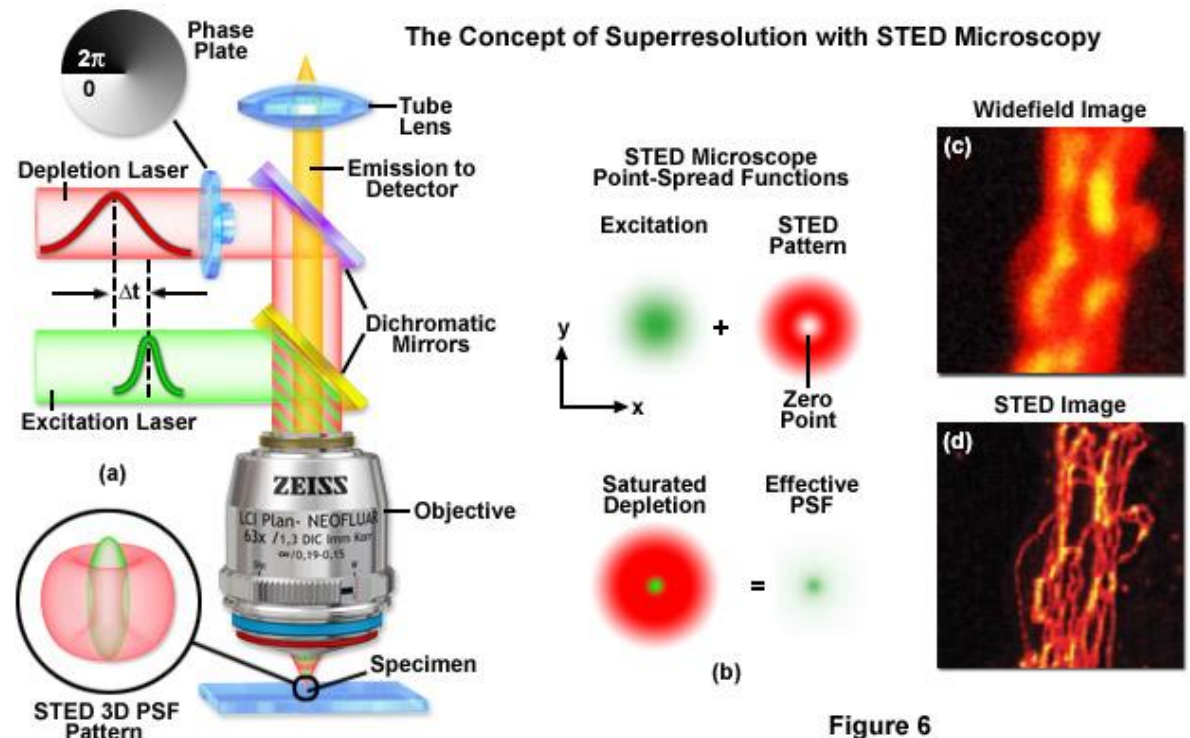
Source:

https://universallab.org/blog/blog/introduction_to_raman_spectroscopy_fundamentals/

Optical imaging

Super-resolution Microscopy:

- This form of light microscopy encompasses a number of techniques used in research to obtain very high resolution images of individual cells, at a level of detail not feasible using normal microscopy.
- One example is a technique called photoactivated localization microscopy (PALM), which uses fluorescent markers to pinpoint single molecules.
- PALM can be performed sequentially to create a super-resolution image from the series of molecules isolated in the sample tissue

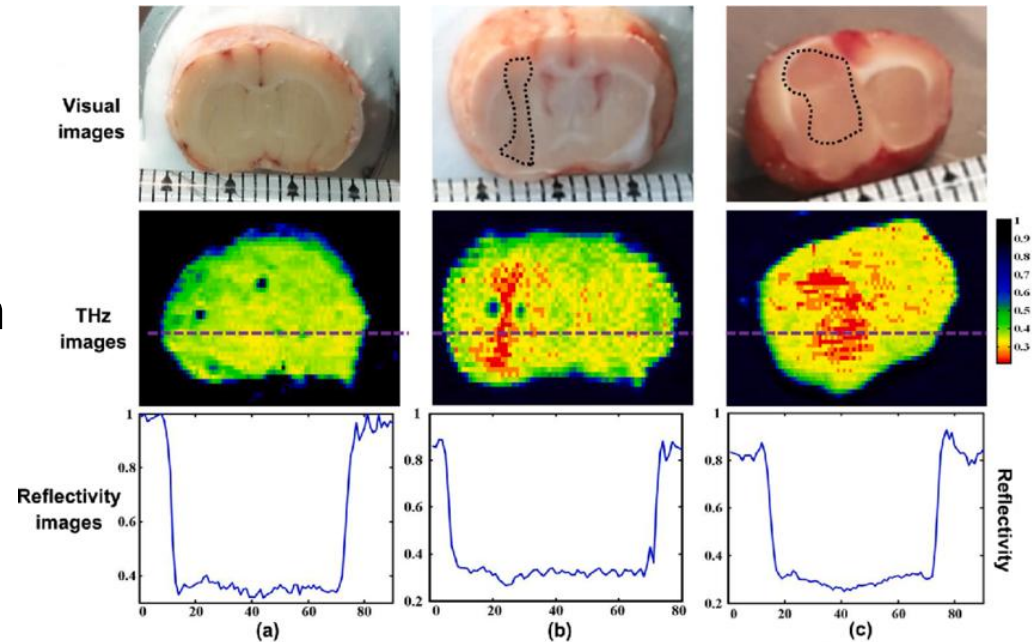


Source: <https://zeiss-campus.magnet.fsu.edu/articles/superresolution/introduction.html>

Optical imaging

Terahertz Tomography:

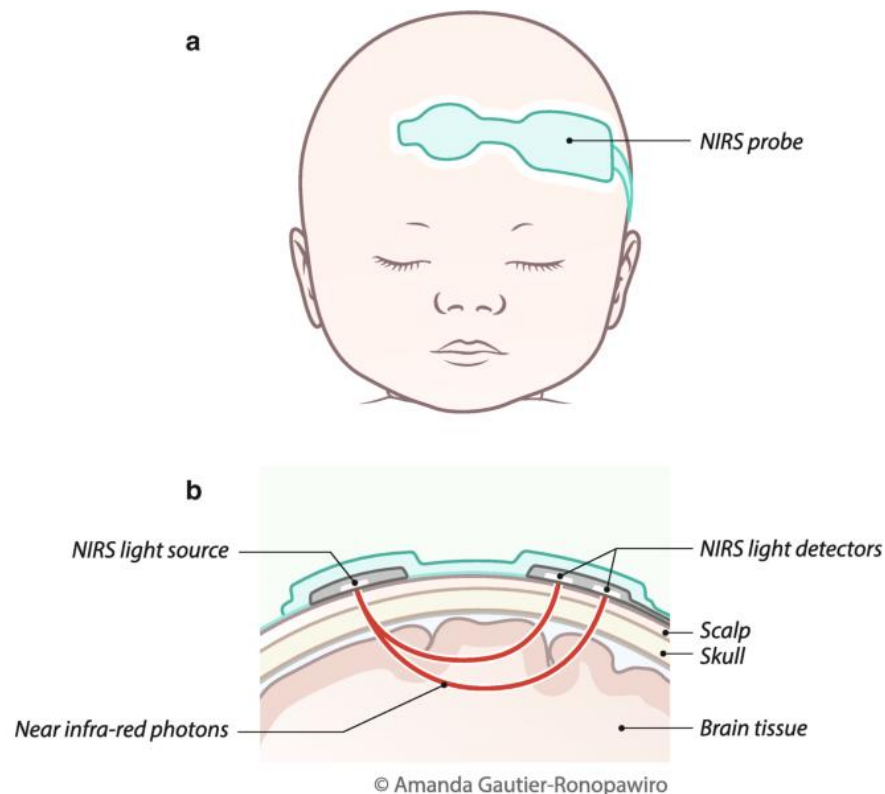
- This relatively new, experimental technique involves sectional imaging using terahertz radiation.
- Terahertz radiation consists of electromagnetic waves, which are found on the spectrum between microwaves and infrared light waves.
- Terahertz radiation can “see” what visible and infrared light cannot, and holds great promise for detecting unique information unavailable via other optical imaging methods.



Source: https://www.researchgate.net/figure/The-results-of-terahertz-THz-radiation-test-Visual-imaging-THz-imaging-and-the_fig1_339431212

Near-infrared spectroscopy (NIRS)

- Non-invasive technique that can measure tissue oxygen saturation in organs such as the brain, kidney, and intestine.
- By monitoring changes in the attenuation of near-infrared light passing through the brain, NIRS can provide cerebral regional oxygen saturation measurements (CrSO_2).
- NIRS has been used in neonatal intensive care units (NICUs) for various indications, including monitoring extremely premature infants and neonates with encephalopathy, congenital heart disease (CHD), anemia, respiratory support, and CNS injuries.
- Factors such as device type, sensor position, head position, and care procedures can affect NIRS measurements.



An illustration of a cerebral NIRS probe attached to the forehead of an infant and an illustration of the basic technology used in NIRS. (© Amanda Gautier-Ronopawiro)

Source: https://link.springer.com/chapter/10.1007/978-3-031-55972-3_17

Near-infrared spectroscopy (NIRS)

- Technique that measures the amount of oxygen inside the brain tissues.
- The device has a thin cable attached to a sensor/probe—a small, soft patch on the baby's side of the forehead.
- The probe uses near-infrared light, which is very safe.
- The light goes a few centimeters into the brain and measures the color of the red blood cells (the cells that carry oxygen around the body) as it changes according to the amount of oxygen they carry

Microscopy

- Microscopes are of three basic types: optical, electron (or ion), and scanning probe.
- **Optical microscopy** (light microscopy) is a very common tool in biological research which employs visible light to magnify tiny objects.
- Light sources commonly used in optical microscopes include arc-discharge lamps, incandescent tungsten-halogen bulbs, LEDs, and lasers. Leica Microsystems, imaging equipment maker, offers automated light microscopes with choice of illumination between halogen and LED.
- Illumination techniques-bright field, dark field, cross-polarized light, and phase contrast illumination. These techniques provide increased contrast while viewing the sample.

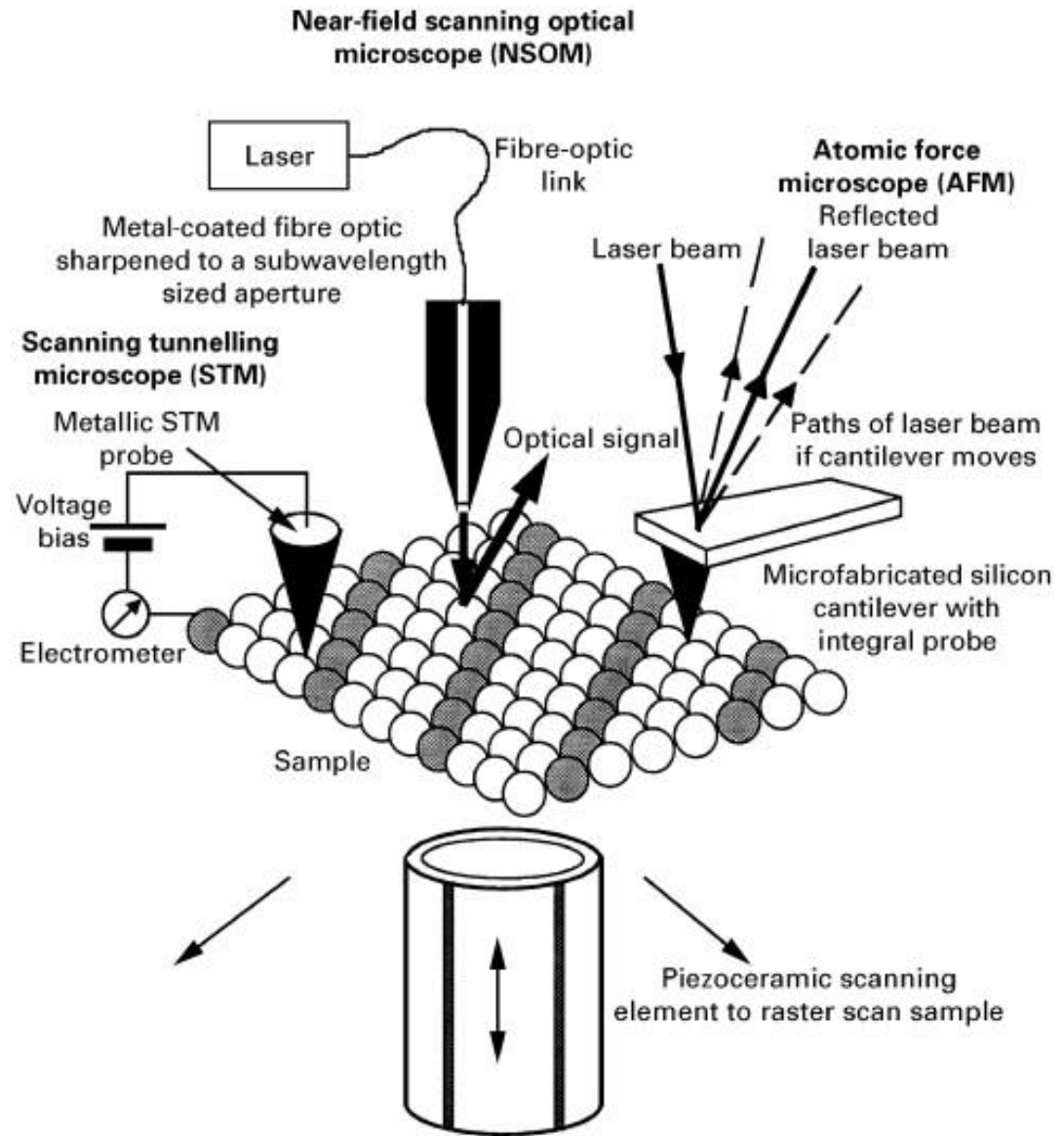


<https://www.news-medical.net/life-sciences/Optical-Electron-and-Scanning-Probe-Microscopy.aspx>

Microscopy

Scanning probe microscopy (SPM)-

- Use a range of tools to produce images of surfaces and structures at the nanoscale level.
- A sharp physical probe scans the sample surface and sends the data gathered to a computer that generates a high-resolution image of the sample surface, which can be visualized by the user.
- SPM users don't see the sample surface directly - they see an image that represents the surface of the sample.
- Different types of SPMs are atomic force microscopes, magnetic force microscopes, and scanning tunneling microscopes.



Source: <https://www.sciencedirect.com/topics/nursing-and-health-professions/scanning-probe-microscope>



Microscopy

Electron microscopy

- Uses an electron beam to form an image of the sample.
- Greater resolving power compared to the light microscope which allows the viewing of finer details of tiny objects.
- Uses electromagnetic or electrostatic lenses to control the path of electrons which are sensitive to magnetic fields.
- The resolving power of an electron microscope is inversely proportional to the irradiation wavelength.

Confocal imaging

- Confocal microscopy offers several advantages over conventional widefield optical microscopy, including the ability to control depth of field, elimination or reduction of background information away from the focal plane (that leads to image degradation), and the capability to collect serial **optical sections** from thick specimens.
- The basic key - use of spatial filtering techniques to eliminate out-of-focus light or glare in specimens whose thickness exceeds the immediate plane of focus.

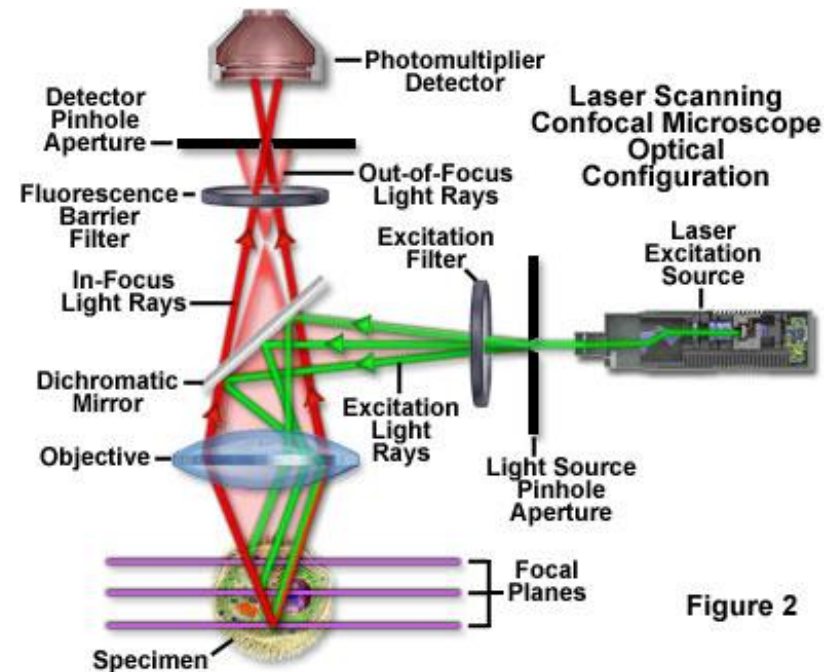


Figure 2

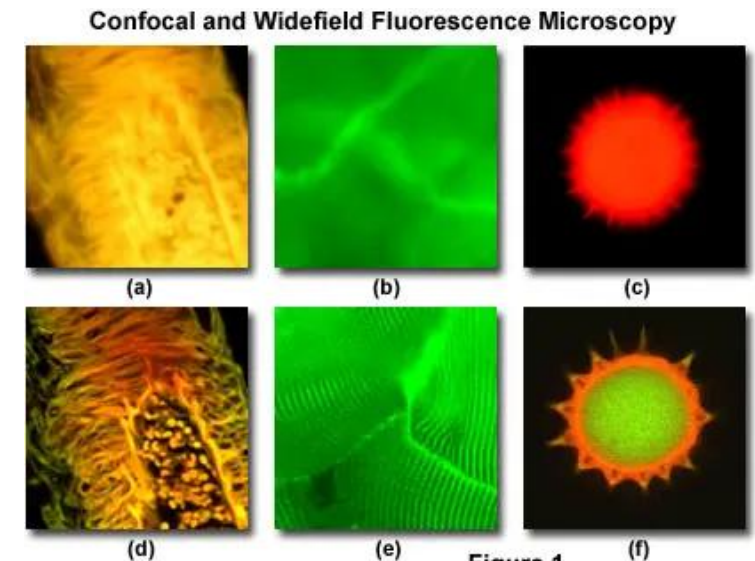


Figure 1

One and two photon imaging

- <https://blog.biodock.ai/one-vs-two-photon-microscopy/>

Magnetic Resonance Imaging (MRI)

- Introduction
- The Components
- The Technology
- Physics behind MR



Most slides taken from
[http://](http://www.slideworld.org/viewslides.aspx/Magnetic-Resonance-Imaging-%28MRI%29-MR-Imaging-Physics-ppt-2086416)
[www.slideworld.org/](http://www.slideworld.org/viewslides.aspx/Magnetic-Resonance-Imaging-%28MRI%29-MR-Imaging-Physics-ppt-2086416)
[viewslides.aspx/Magnetic-Resonance-Imaging-](http://www.slideworld.org/viewslides.aspx/Magnetic-Resonance-Imaging-%28MRI%29-MR-Imaging-Physics-ppt-2086416)
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[Physics-ppt-2086416](http://www.slideworld.org/viewslides.aspx/Magnetic-Resonance-Imaging-%28MRI%29-MR-Imaging-Physics-ppt-2086416)



Introduction

■ What is MRI?

- Magnetic resonance imaging (MRI) is a spectroscopic imaging technique used in medical settings to produce images of the inside of the human body.
- MRI is based on the principles of nuclear magnetic resonance (NMR), which is a spectroscopic technique used to obtain microscopic chemical and physical data about molecules
- In 1977 the first MRI exam was performed on a human being. It took 5 hours to produce one image.

Introduction

- How Does it Work?
 - The magnetic resonance imaging is accomplished through the absorption and emission of energy of the radio frequency (RF) range of the electromagnetic spectrum.

MRI Image Aquisition

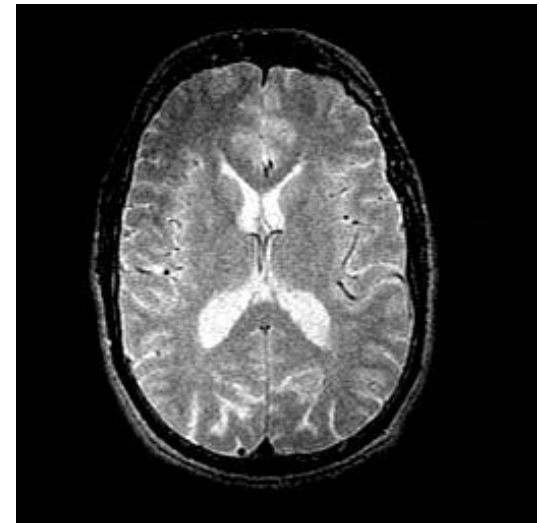
- Pulse sequences are characterized by a repetition time (TR) and a receiver delay time (TE)
- TR and TE are used to control signal levels from different tissues
- Receiver signals computer analyzed to form an image
- Mathematical algorithms based on the Fourier Transform
- Tomographic and volume images can be aquired

MRI Image Contrast

- contrast in MRI produced by variations in proton density and by variations in relaxation times in different tissues
- paramagnetic contrast agents can be employed to enhance contrast
- MRI offers anatomical diagnostic imaging features similar to CT, but, with "tuneable" contrast
- however, MRI does not image bone well because of low water content
- magnetic and rf fields levels used in MRI have no known biological risks



Spin-Echo Sequence
TR = 250 ms
TE = 20 ms



Spin-Echo Sequence
TR = 2000 ms
TE = 80 ms

MRI Scanner

- solenoid magnet field scanner looks much like a CT scanner
- rf transmitter/reciever coil typical in scanner cowling, however, special coil assemblies for head and extremities imaging used
- scanner situated in a rf shielded room
- ferromagnetic metals, pacemakers are not compatible with MRI exams
- MRI is noisy (knocking noises) due rapid application of imaging field gradients



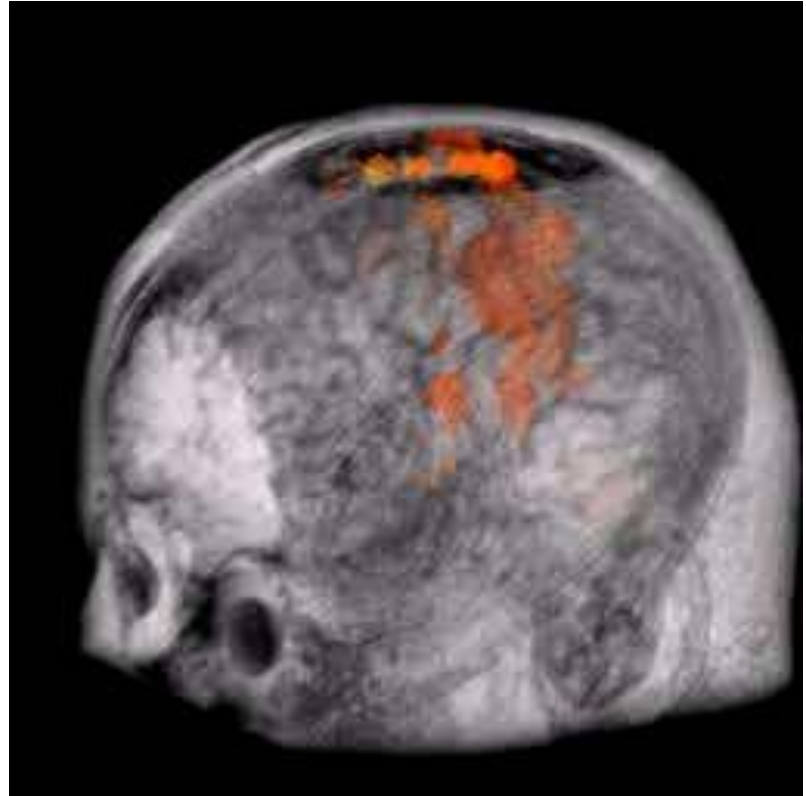
MRI Scanner



**"bird cage" coil assembly
for head imaging**

BOLD Imaging (fMRI)

- MRI can measure physiological processes (i.e., functional MRI)
- oxy- to deoxy-hemoglobin results in changes to local relaxation processes thus affecting signal level
- technique called blood oxygen level-dependant (BOLD) MRI
- hemodynamic changes can be imaged and correlated to neuronal activity
- subjects given specific challenges during imaging to isolate associated activity areas in the brain



motor strip localization
(co-registered 3D image)

Contrast

- imaging relies on contrast differences
- in diagnostic imaging, contrast must distinguish anatomy, and/or physiological processes
- different imaging modalities produce contrast through differing physical processes
- various modalities offer advantages and disadvantages

X-Ray Modalities

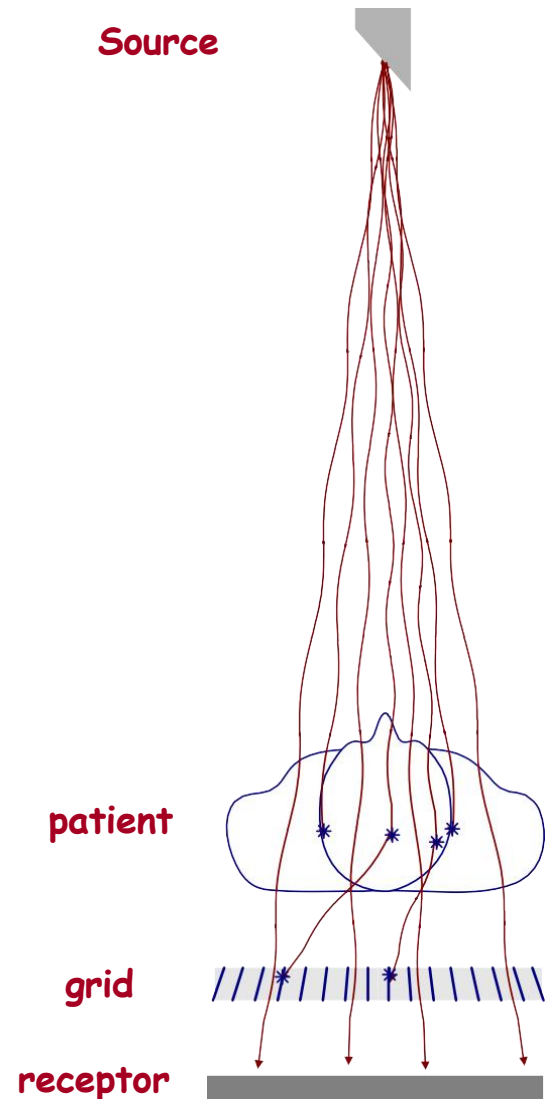
- x-ray modalities are the most common imaging modalities in medical diagnostic imaging
- modalities include:
 - Radiography
 - Fluoroscopy
 - Computed Tomography

X-Ray Contrast

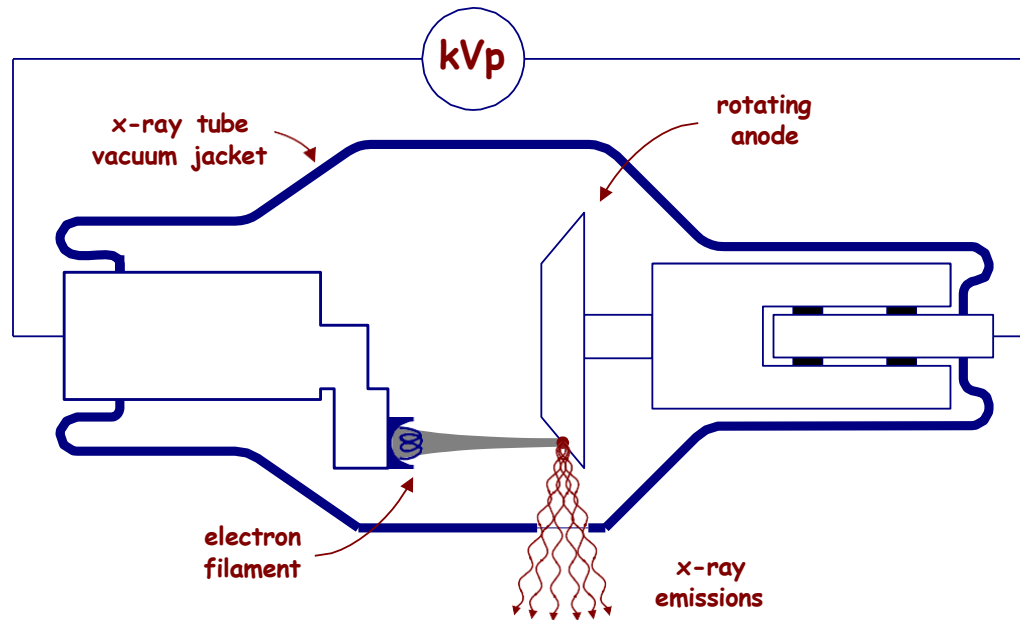
- low energy x-rays produce contrast through absorption in tissue
- relative absorption depends on tissue density and atomic composition
- down-side: absorption and scattering results in ionization (radiation dose) and potential biological damage, however, **benefit outweighs risk**

X-Ray Imaging Basics

- source produces collimated beam of x-rays
- x-rays absorbed, scattered or transmitted through patient
- if imaged, scattered x-rays reduce contrast, typically removed by a grid
- receptor captures an image of the transmitted x-rays



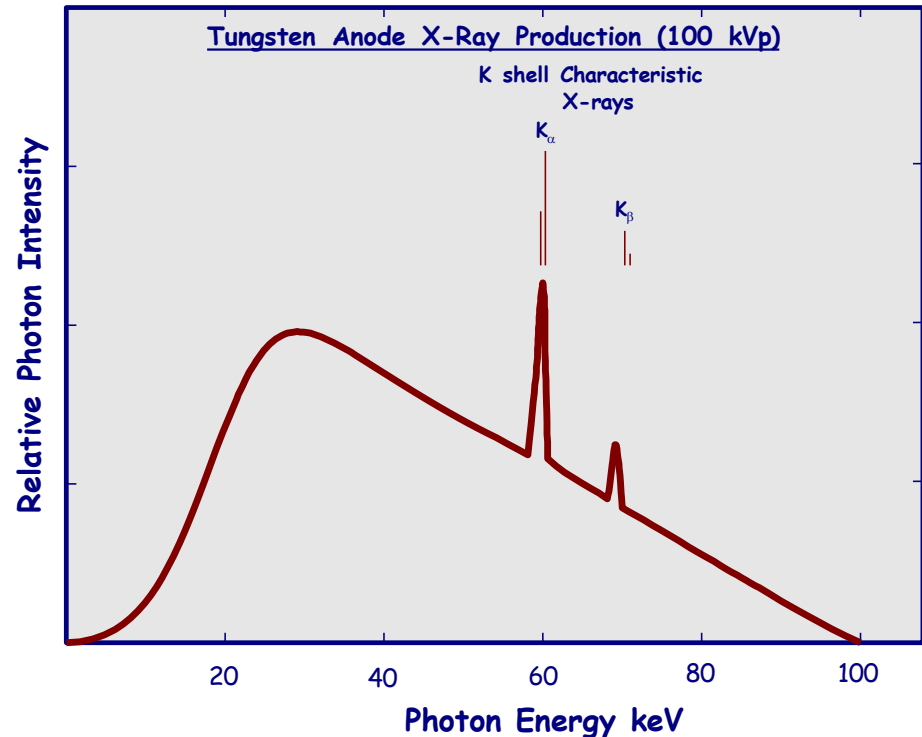
X-ray Production



- X-ray vacuum tube: apply an DC voltage (kVp) between a cathode (electrode filament) and anode
- high energy electrons striking the anode produce
 - heat (typically > 99% of electron energy),
 - bremsstrahlung radiations, and
 - characteristic x-ray radiations

Anode Target X-Ray Spectrum

- polyenergetic bremsstrahlung (i.e., braking radiation) spectrum, and
- monoenergetic characteristic (fluorescent) spectral lines
- upper energy limit set by generator kVp (typical diagnostic energies 50 - 120 kVp)
- in practice, lower energy x-ray spectrum preferentially attenuated (filtered, hardened) by inherent and added filtration
- attenuation desirable since low energy x-rays otherwise totally absorbed in patient, and contribute disproportionately to patient dose



Radiography



Wall Stand



Table

- radiography or plain x-rays, the most common x-ray imaging modality
- in radiography, static anatomy images produced, typically on film
- film not very sensitive to x-rays, fluorescent "screen" used to convert x-rays to visible light and expose film
- typical radiography suite comprises a gantry mounted tube, a table, and a wall stand

Radiograph Example

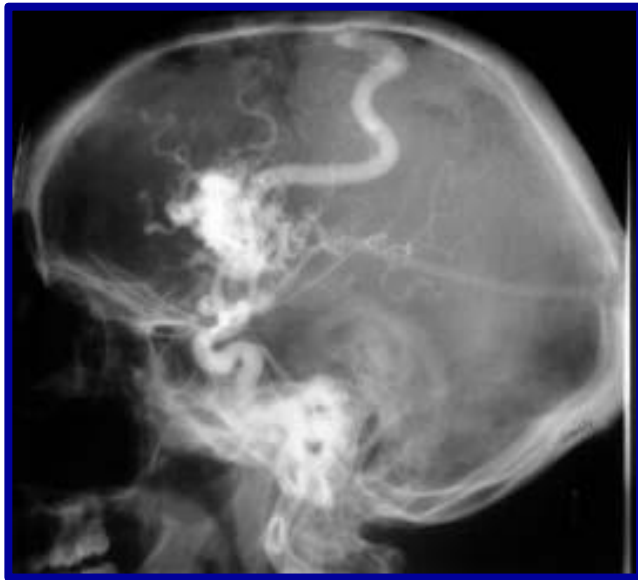
- plain x-rays used to image most aspects of anatomy
- chest x-ray a common radiographic procedure
- negative image produced for reading by radiologist
- dark image regions correspond to high x-ray transmission
- image visualizes lung field and silhouette of mediastinum
- used to diagnose lung and mediastinal pathologies (e.g., pneumonia, and cardiomegaly)



Pnuemocystis

Contrast Enhancement

- contrast agents (dyes) can be injected into the blood vessels (angiograms) and cavities to improve visibility
- for example: iodine and barium absorbs more x-rays than tissue



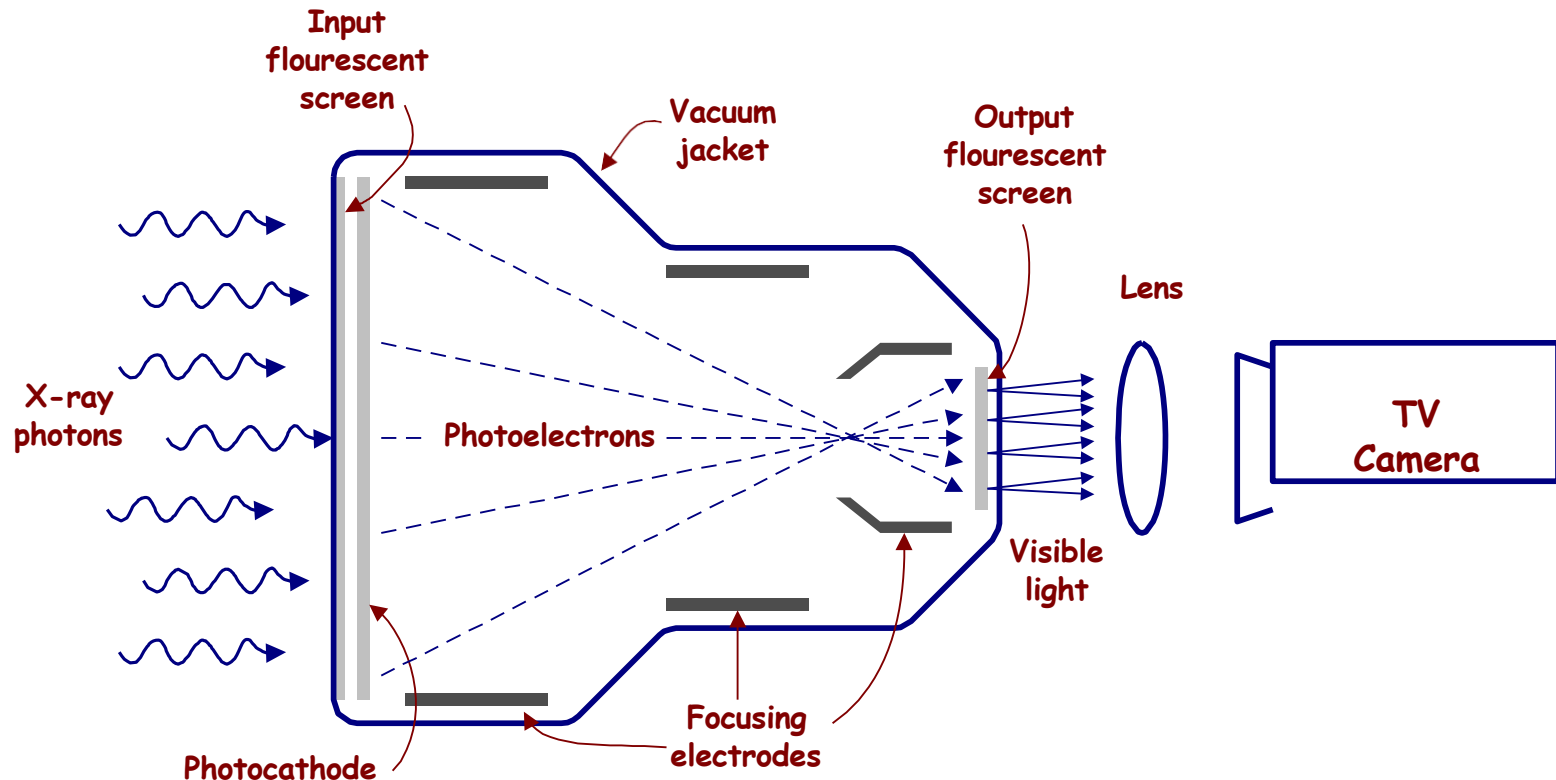
Cerebral Arteries



Air-Contrast Barium Enema

Fluoroscopy

- fluoroscopy used to obtain real time x-ray images
- image receptor converts x-ray image into a TV signal
- video images can also be recorded (film, video-tape)



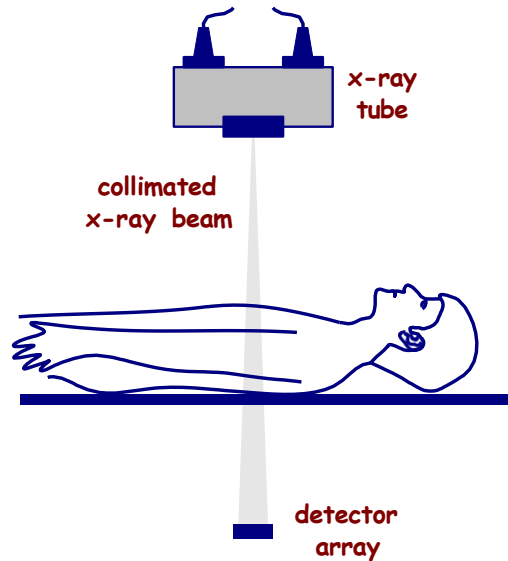
Fluoroscopy Suites



- table and c-arm arrangements available
- fluoroscopy typically used for observing the digestive tract, catheter guiding, and cardiac angiography

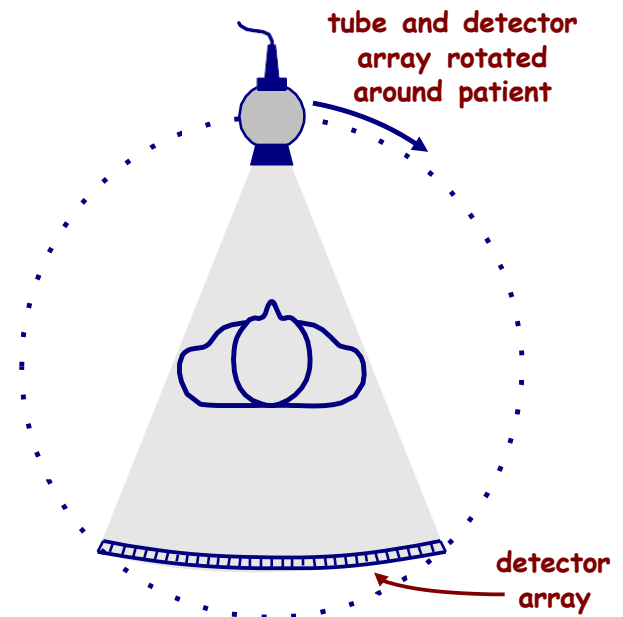


X-ray Computed Tomography (CT)



- conventional x-rays are projection images, and overlying structures can obscure anatomical details
- in CT slice projections (profiles) through patient measured by a detector array

- by rotating the tube and detector array, profiles are taken at multiple angles
- a computer then processes the profiles using a mathematical algorithm (convolution) to create a cross-sectional image on a video screen



CT Scanner



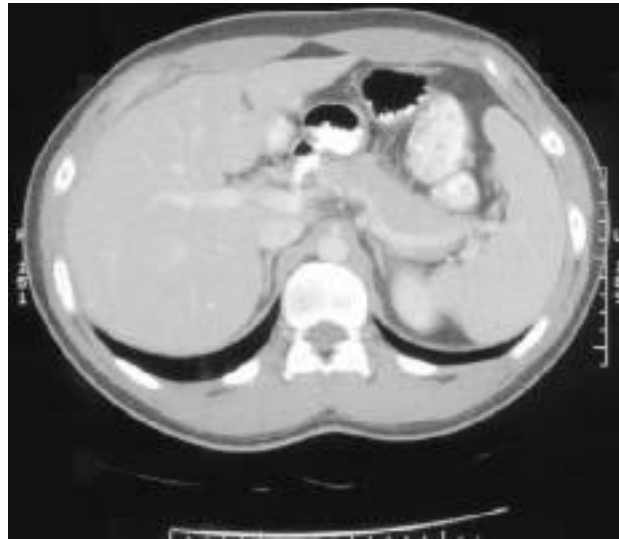
- cabling covers rotating tube and detector electronics
- central port and table for patient
- computer console for control and image viewing

CT Slice Images



abdominal scan at
kidney level

abdominal scan
spleen/liver level



head scan showing
ventricles

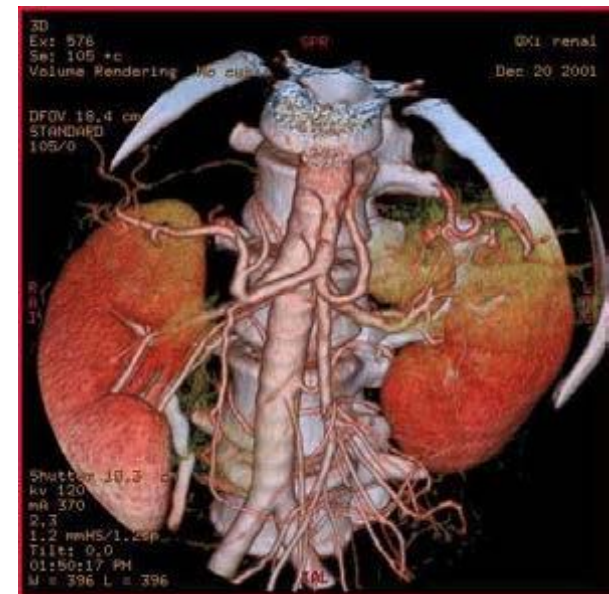
- CT eliminates the shadow overlap problem of conventional X-rays
- contrast agents commonly used in CT

Helical CT

- modern CT scanners use continuous tube rotations and table translation
- with respect to patient, the tube follows a helical path
- results in faster scans (e.g., a single breath hold lung scan)
- helical scan profiles are interpolated to form slice images
- modern computer reconstruction can reformat data to view slices at arbitrary angles
- three-dimensional rendered images of complex blood vessels like the renal arteries or aorta are also possible



Simulated helical x-ray beam path for a scan of the of the abdomen. The highlighted area is a man's stomach (man is lying on his back with his arms over his head).



3D rendering of kidneys

3D Rendered CT Images



Heart



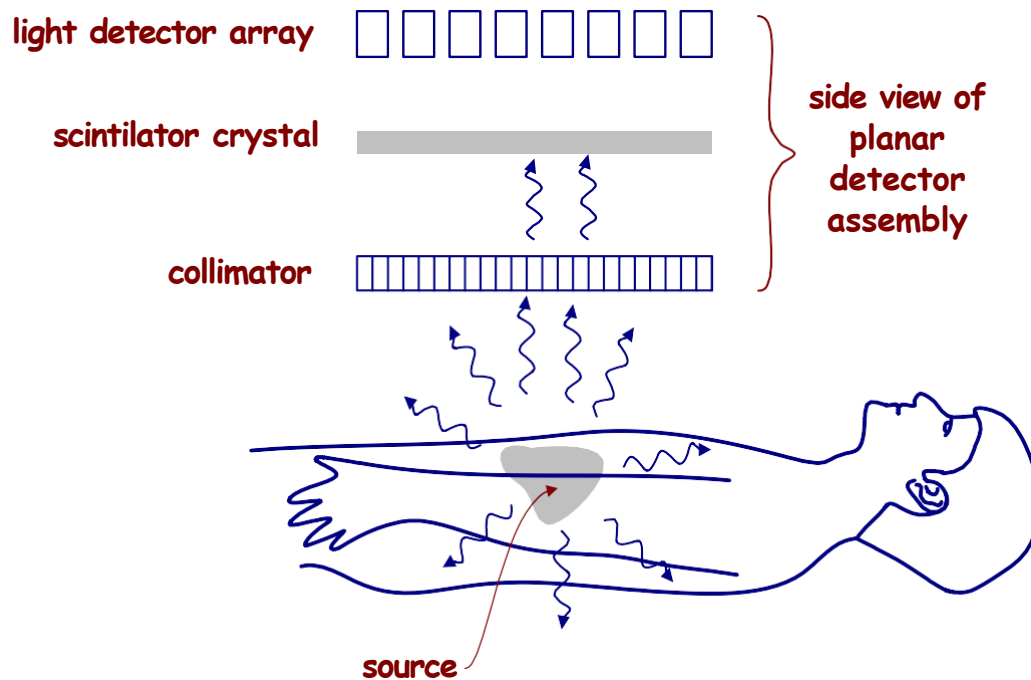
Colon Fly Through

Nuclear Medicine Imaging

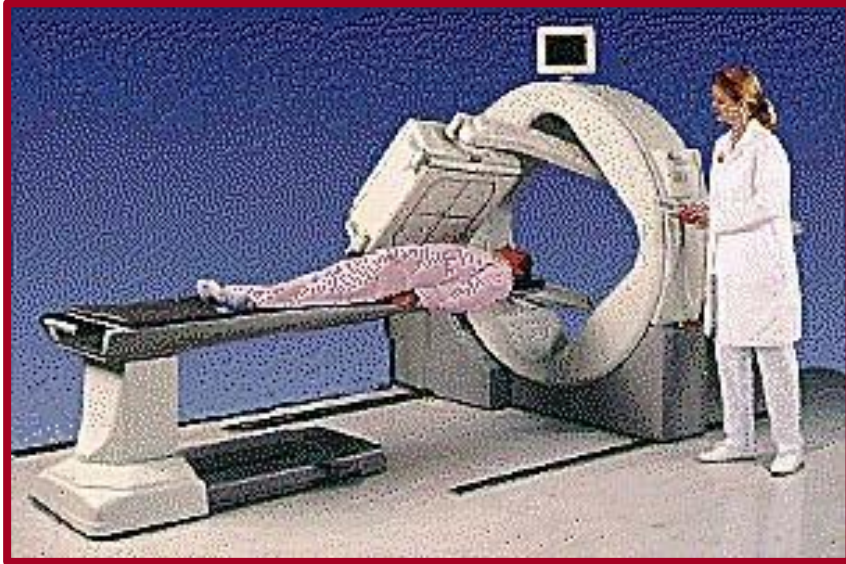
- radio-isotopes are natural and artificially produced unstable isotopes that decay through gamma-ray and/or particulate emissions (e.g., positrons)
- ideal imaging isotopes feature low dose to the patient (e.g., short physical and/or biological half lives)
- medical isotopes produced in nuclear reactors and by particle accelerators
- nuclear medicine images visualize radioisotope concentrations
- by “tagging” radio-isotopes to biological molecules, physiological processes can be measured
- **nuclear imaging is functional, not anatomic**

Planar and SPECT Cameras

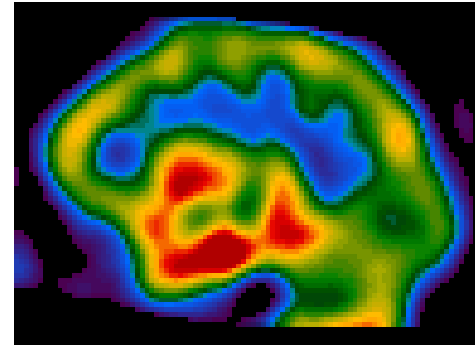
- relies on isotopes that emit γ -rays (e.g., ^{99m}Tc)
- planar camera comprises a collimator, scintillator crystal (e.g., NaI) and a light detector array
- by rotating a planar camera, data for tomographic images acquired
- SPECT an acronym for single photon emission computed tomography



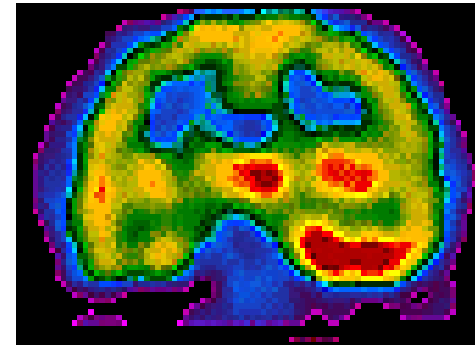
SPECT Camera & Images



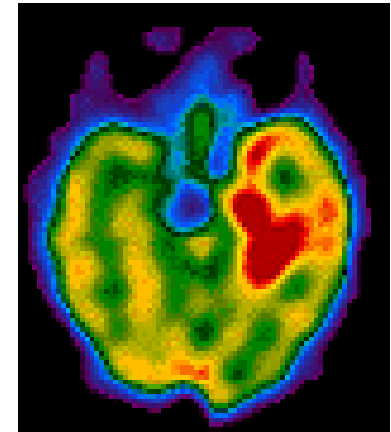
rotating planar SPECT camera



sagittal



coronal

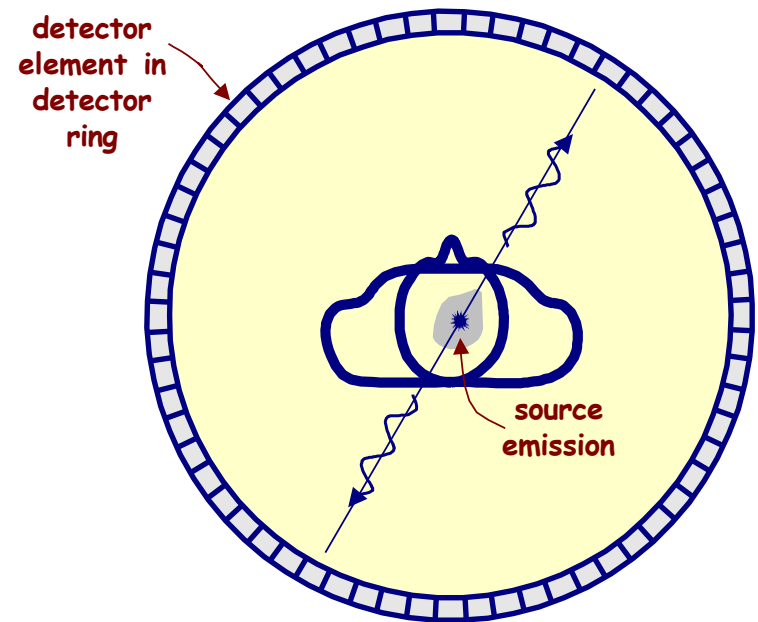


transaxial

Tc-99m HMPAO SPECT perfusion images, showing decreased blood perfusion to posterior frontal and anterior temporal lobes

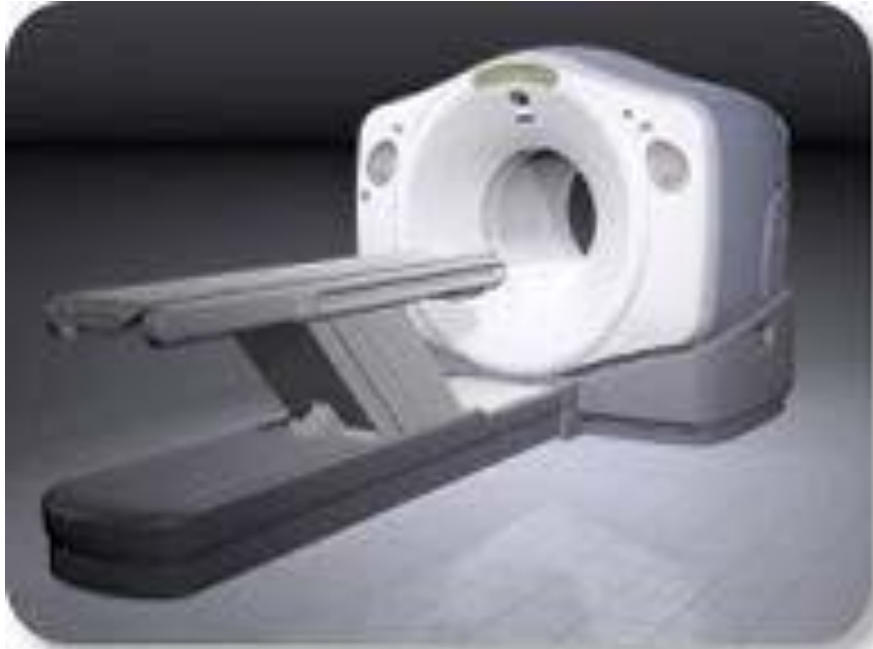
PET Imaging

- some radio isotopes decay with the emission of a positron (e.g., ^{18}F)
- positrons annihilate with electrons shortly after emission, resulting in emission of two coincident 511 keV photons traveling in opposite directions
- positron emission tomography (PET) camera detects coincident photon emissions to form tomographic data sets for computer image reconstruction
- PET has higher sensitivity and resolution than SPECT
- ^{18}F FDG commonly used in PET to detect increased cellular metabolism (e.g., detecting and staging cancer)



PET camera

State of the Art PET-CT Scanner



- PET-CT systems generate PET functional and CT anatomy images of a patient in a single study

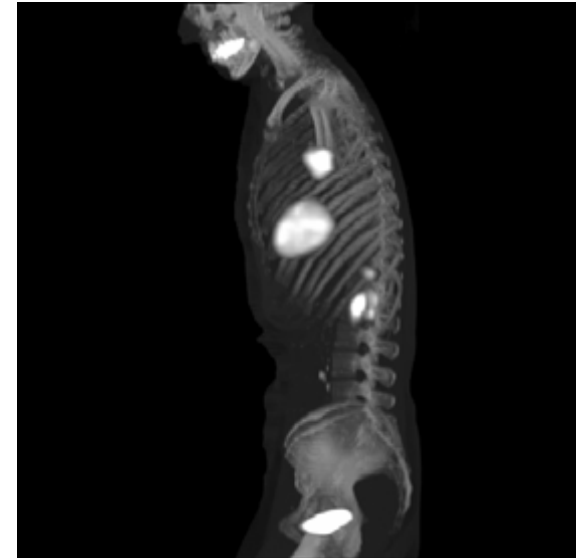
3D Image Co-registration



PET Uptake



CT Anatomy



Co-Registered

- functional and anatomical images registered and fused to form a single image

EEG, ECG, EMG

Slide ref :Mitesh Shrestha

What is Signal ?

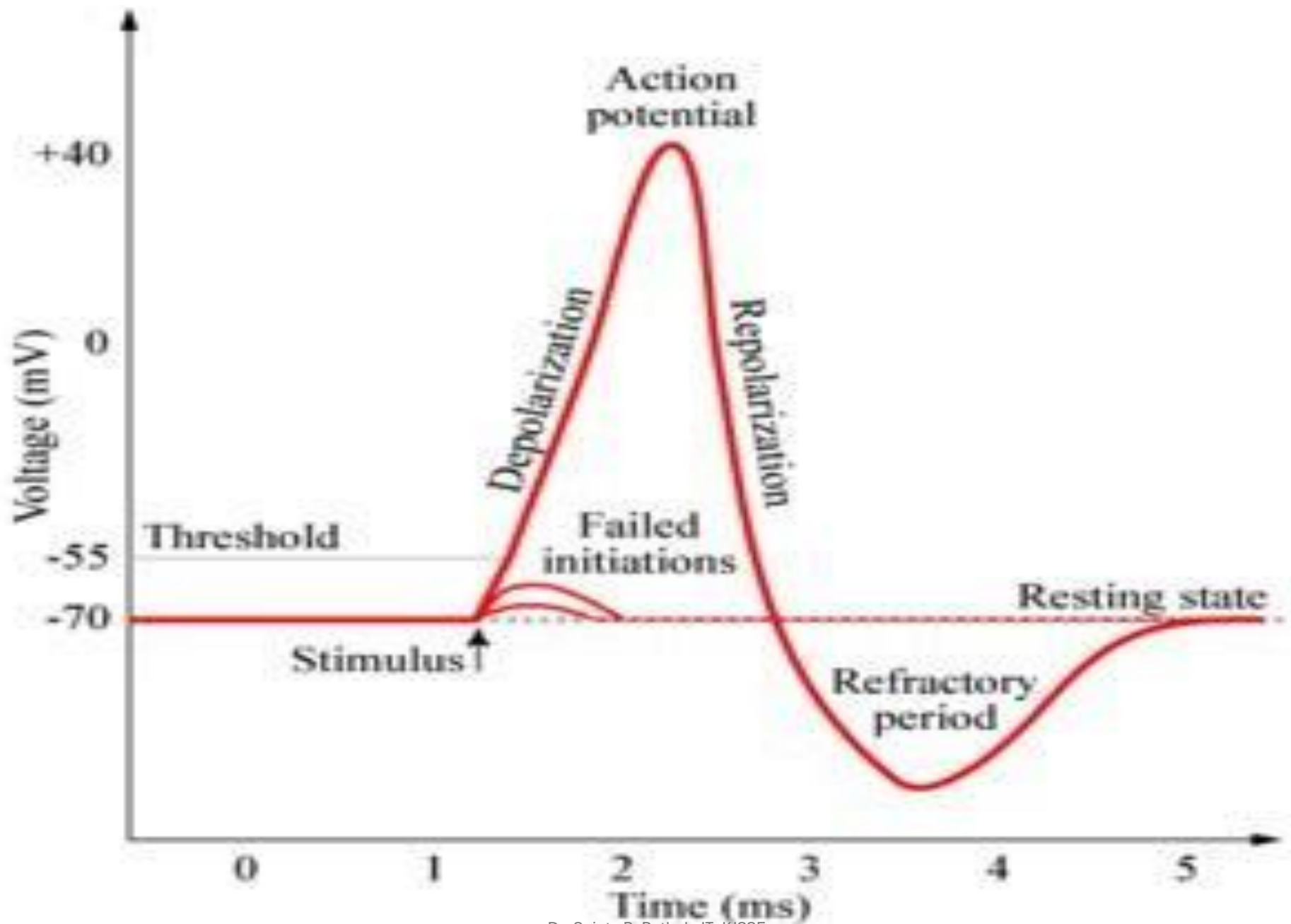
- A signal is defined as a fluctuating quantity or impulse whose variations represent information. The amplitude or frequency of voltage, current, electric field strength, light, and sound can be varied as signals representing information.
- A signal can be simply defined as a function that conveys information.
- Signals are represented mathematically as functions of one or more independent variables.
- Examples: voltage, current, electric field strength, light, sound, etc.

Biomedical signals

- Biomedical signals means the bio-signals which are generated in biological systems only.
- Biomedical signals are observations of physiological activities of organisms, ranging from gene and protein sequences, to neural and cardiac rhythms, to tissue and organ images.
- Examples of biomedical signals:
 - ECG (Electrocardiogram) signal,
 - EEG (Electroencephalogram) signal, etc.

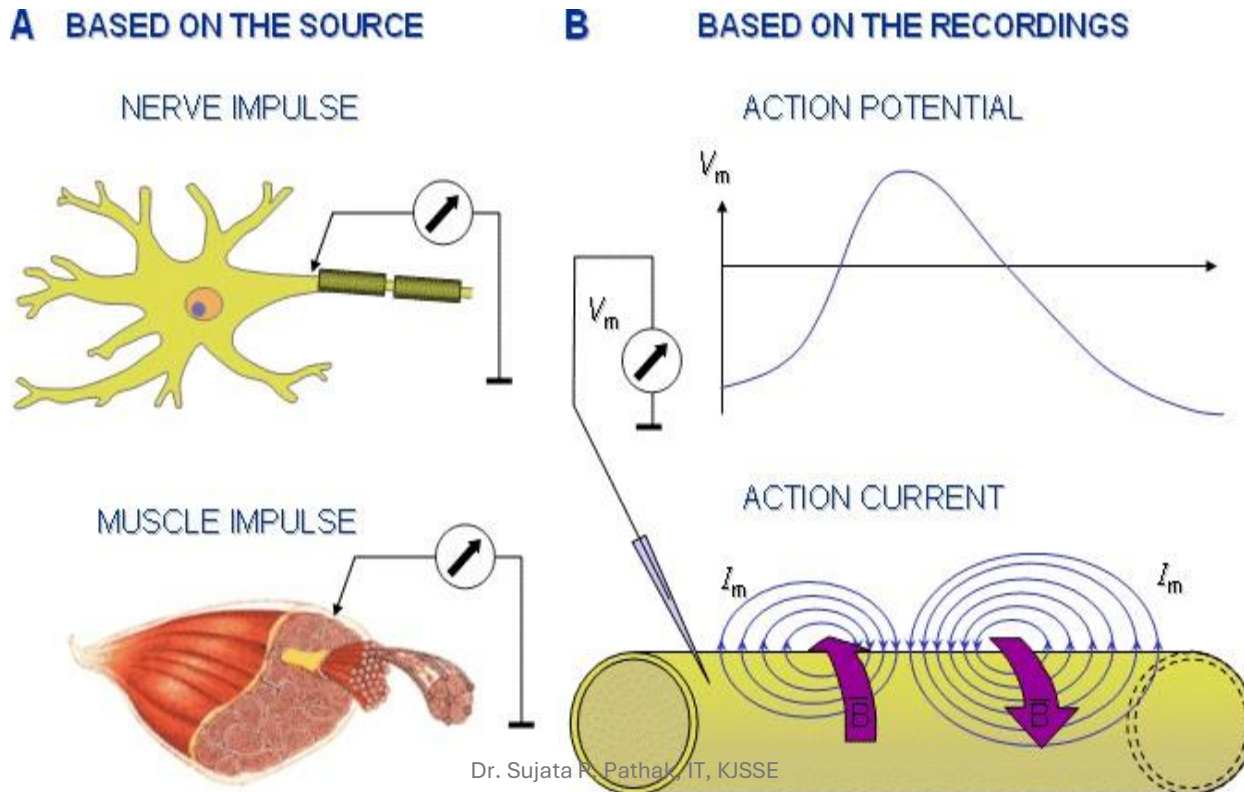
How biomedical signals are generated ?

- Biomedical signals are electrical or magnetic signals generated by some biological activity in the human body.
- Human body is composed of living tissues that can be considered as a power station.
- Action of living tissues in terms of bioelectric potentials generate multiple electric signals from two internal sources- muscles and nerves system.



What are biopotentials

- *An electric potential that is measured between points in living cells, tissues, and organisms, and which accompanies all biochemical processes.*
- Also describes the transfer of information between and within cells



Classification of Biomedical Signals

- Bioelectric Signal
- Bioacoustics Signal
- Biomechanical Signal
- Biochemical Signal
- Bio-magnetic Signal
- Bio- optical signal

Bioelectric signal

- The electrical signals which we can measure mainly on the surface of the body is known as bioelectric signal.
- It is generated by muscle cells and nerve cells.
- Basic source is the cell membrane potential.
- Examples: ECG, EEG, EMG, EOG

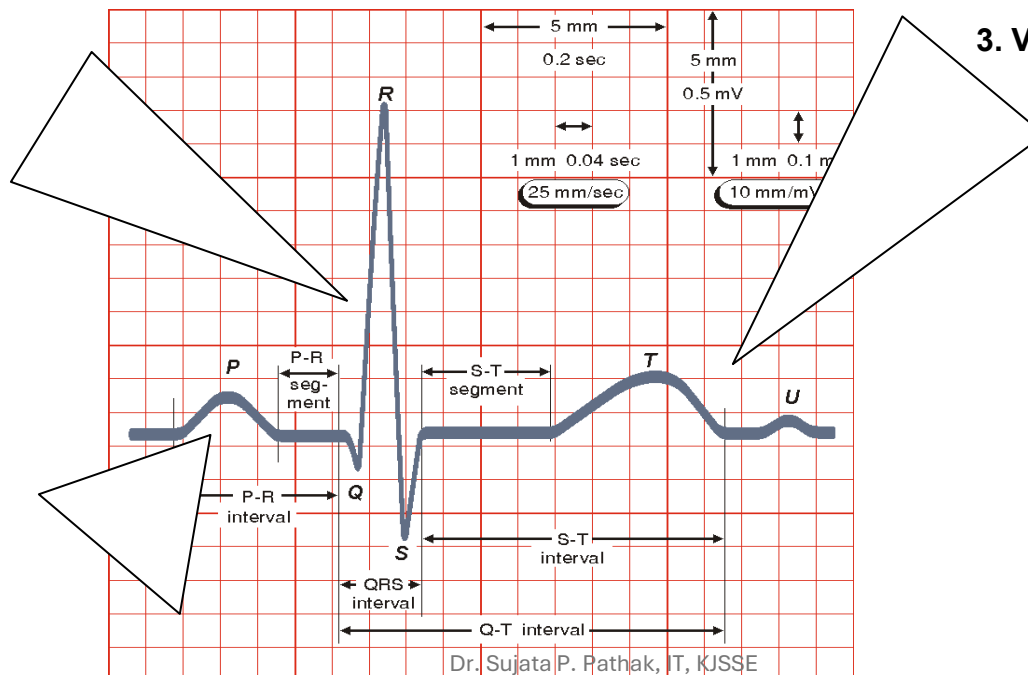
Electrocardiography (ECG)

- Measures galvanically the electric activity of the heart
- Well known and traditional, first measurements by Augustus Waller using capillary electrometer (year 1887)
- Very widely used method in clinical environment
- Very high diagnostic value

2. Ventricular depolarization

3. Ventricular repolarization

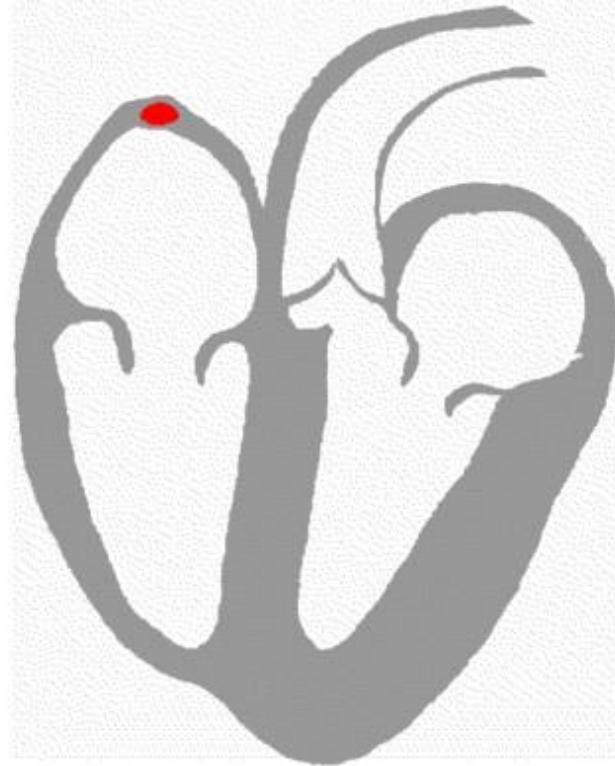
1. Atrial depolarization



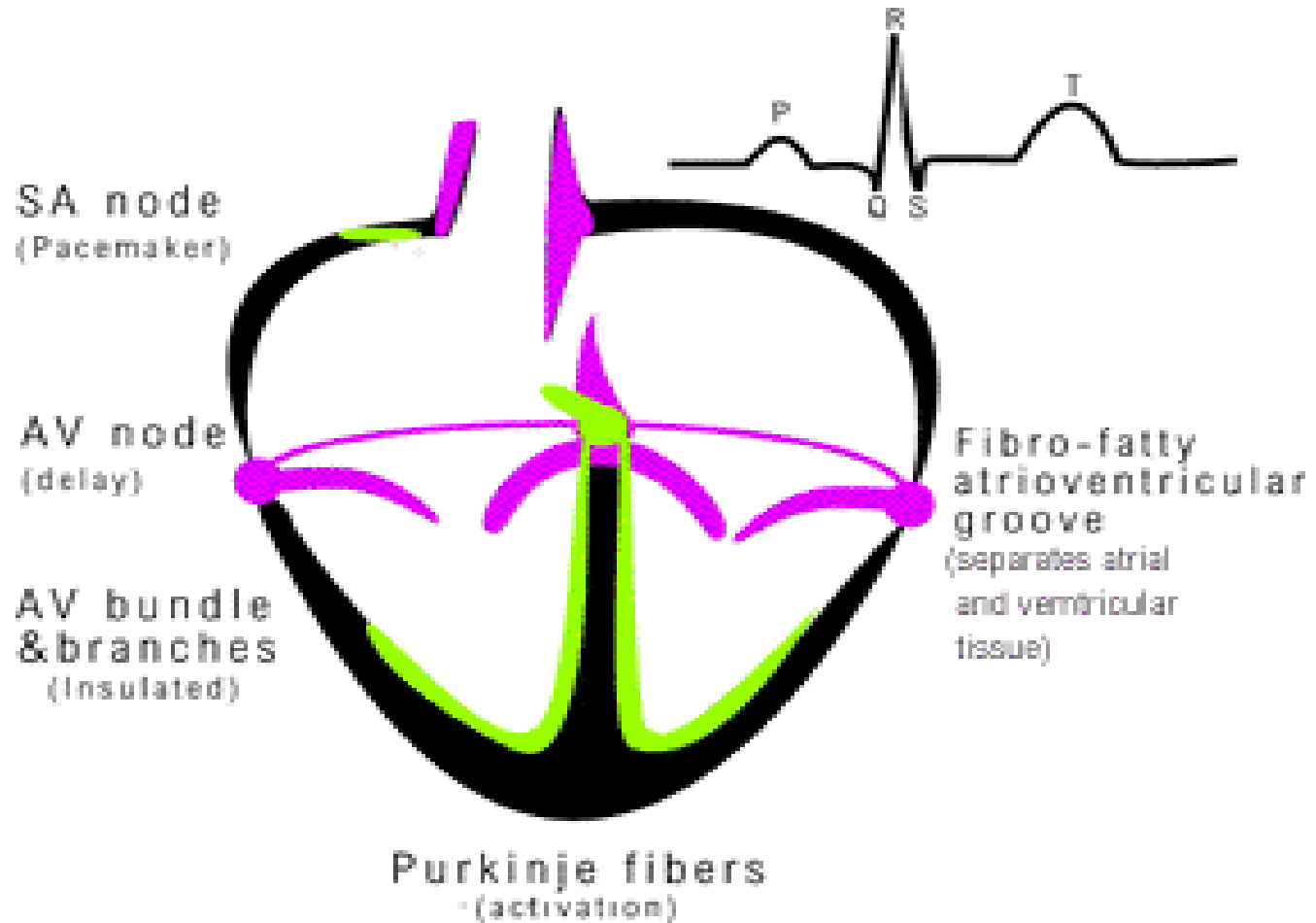
Electrocardiography

- The heart is an electrical organ, and its activity can be measured non-invasively
- Wealth of information related to:
 - The electrical patterns proper
 - The geometry of the heart tissue
 - The metabolic state of the heart
- Standard tool used in a wide-range of medical evaluations

ECG principle



Cardiac Electrical Activity



■ Contractile

■ Conductive

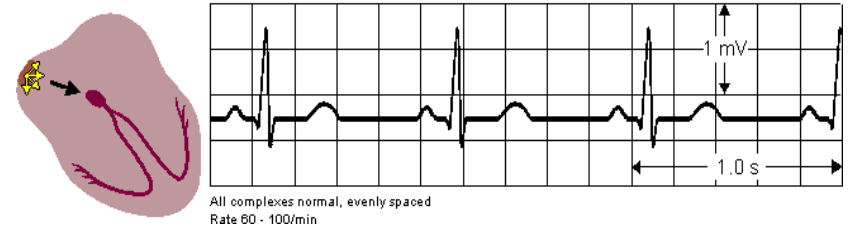
■ Nonconductive

ECG basics

- Amplitude: 1-5 mV
- Bandwidth: 0.05-100 Hz
- Largest measurement error sources:
 - Motion artifacts
 - 50/60 Hz powerline interference
- Typical applications:
 - Diagnosis of ischemia
 - Arrhythmia
 - Conduction defects

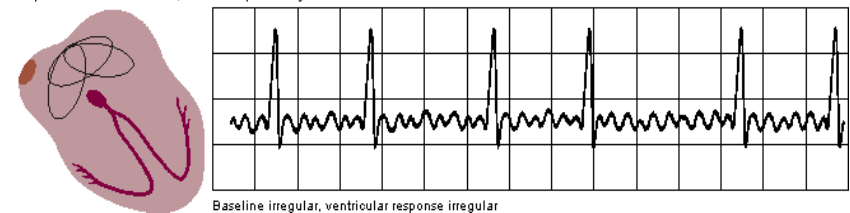
NORMAL SINUS RHYTHM

Impulses originate at S-A node at normal rate



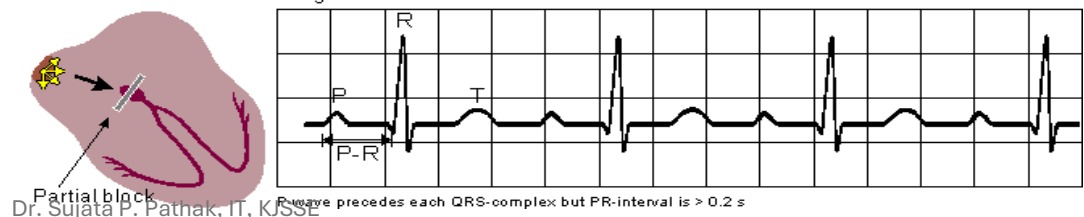
ATRIAL FIBRILLATION

Impulses have chaotic, random pathways in atria



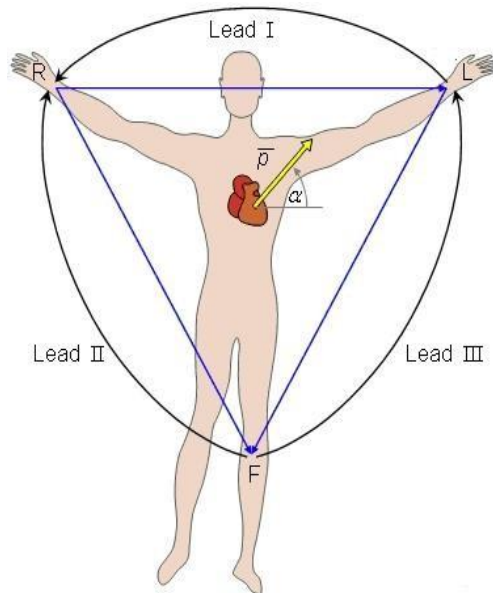
A-V BLOCK, FIRST DEGREE

Atrio-ventricular conduction lengthened



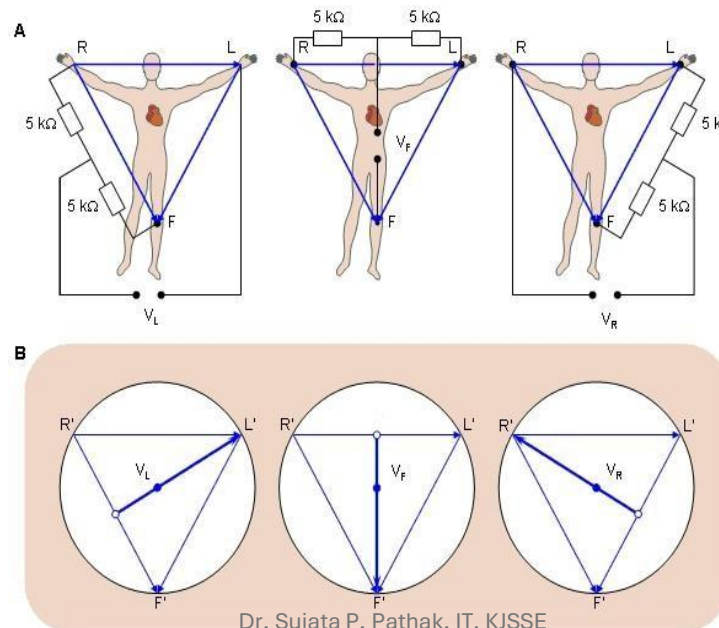
12-Lead ECG measurement

- Most widely used ECG measurement setup in clinical environment
- Signal is measured non-invasively with 9 electrodes
- Lots of measurement data and international reference databases
- Well-known measurement and diagnosis practices
- This particular method was adopted due to historical reasons, now it is already rather obsolete



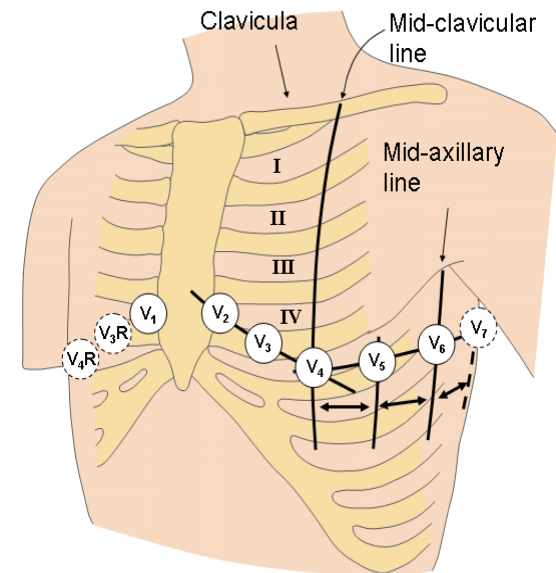
Einthoven leads: I, II & III

Goldberger augmented leads: V_R , V_L & V_F

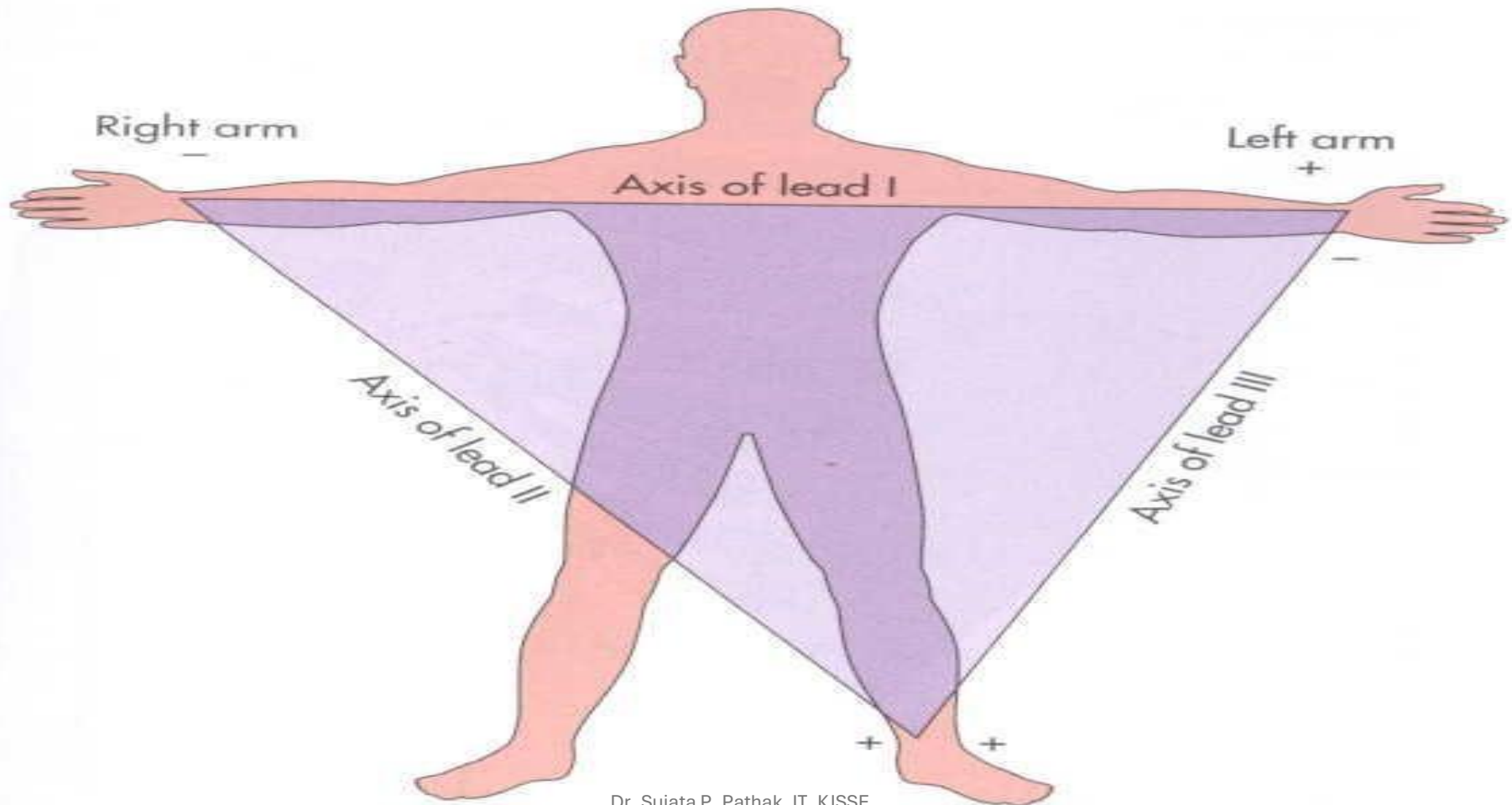


Dr. Sujata P. Pathak, IT, KJSSE

Precordial leads: V_1 - V_6

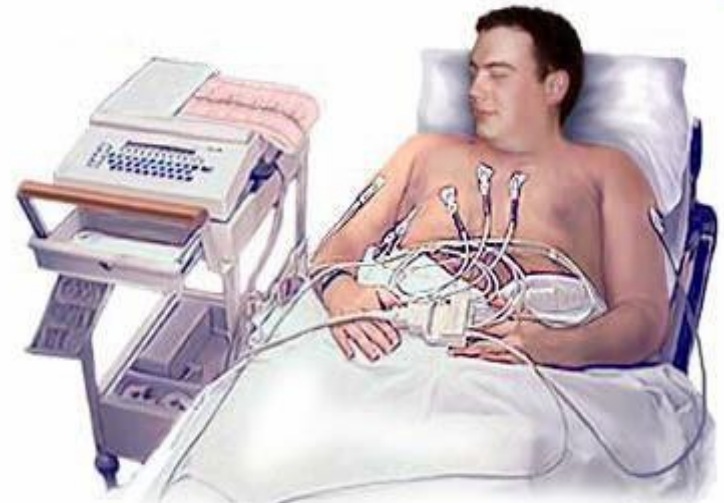
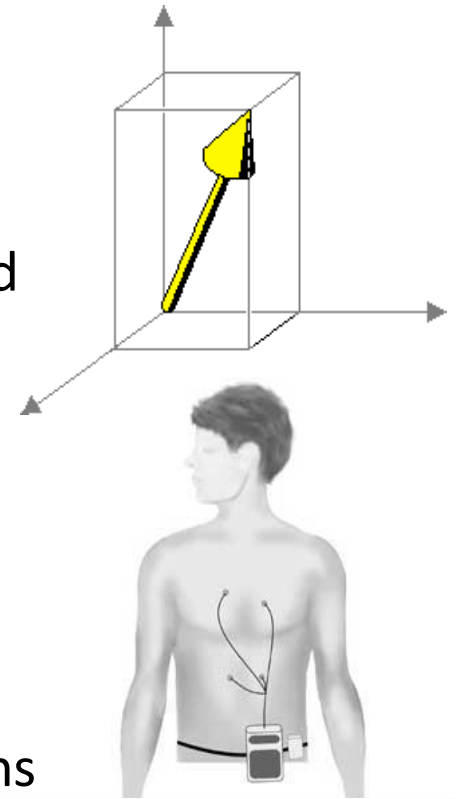


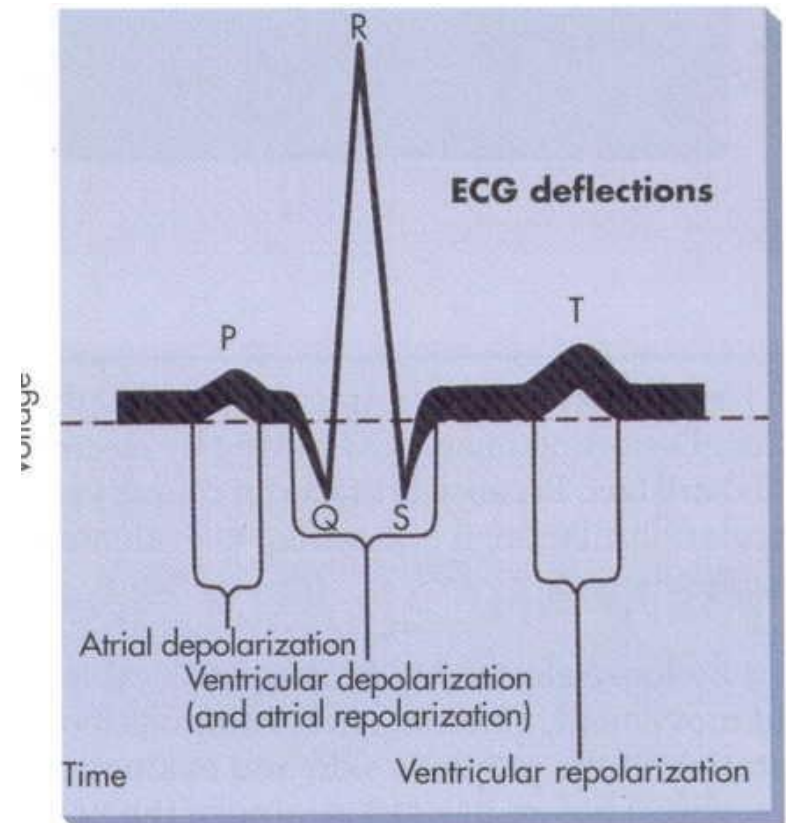
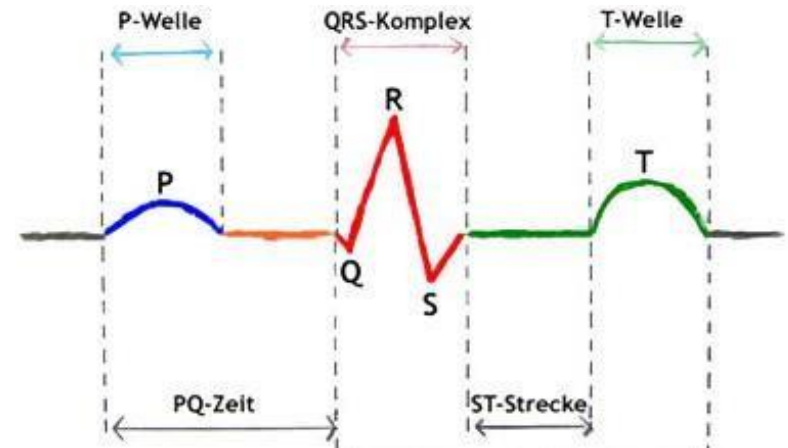
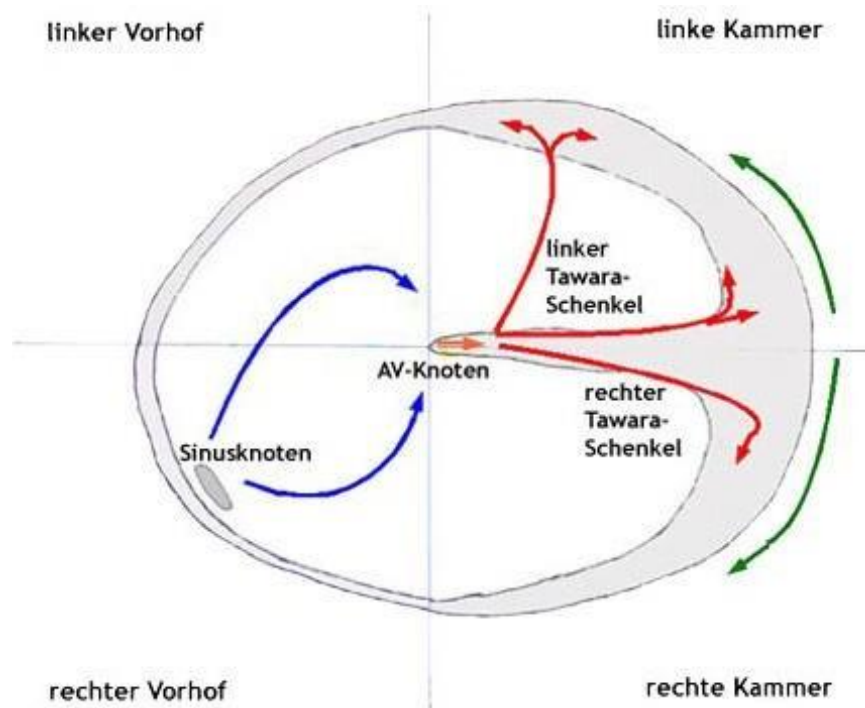
EINTHOVENS TRIANGLE



Why is 12-lead system obsolete?

- Over 90% of the heart's electric activity can be explained with a dipole source model
 - Only 3 orthogonal components need to be measured, which makes 9 of the leads redundant
- The remaining percentage, i.e. nondipolar components, may have some clinical value
- 12-lead system does, to some extent, enhance pattern recognition and gives the clinician a few more projections to choose from
- ...but....
- If there was no legacy problem with current systems, 12-lead system would've been discarded ages ago



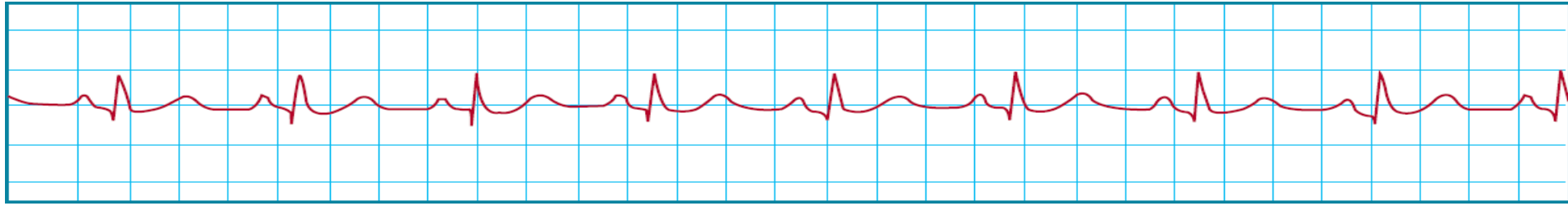


Origination of the QRS - Signal

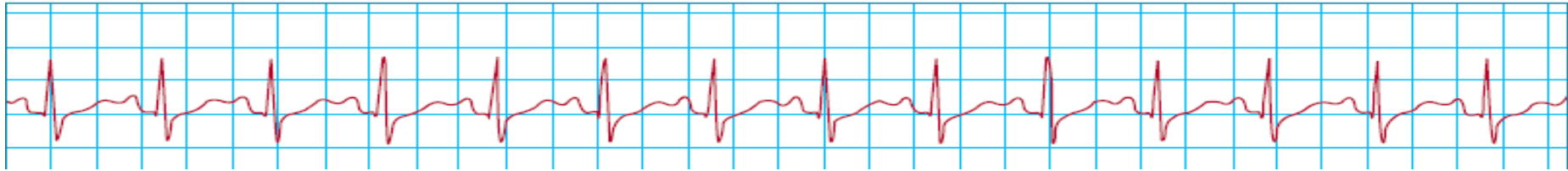
ECG - applications

- **Diagnostics**
- **Functional analysis**
- **Implants (pace maker)**
- **Biofeedback (Heart rate variability, HRV)**
- **Peak Performance Training, Monitoring**

Normal sinus rhythm



Sinus Tachycardia—Rate 122



Sinus Bradycardia—Rate 48



RATE

- P wave rate 60 - 100 bpm with <10% variation - ***Normal***
- Rate < 60 bpm = ***Sinus Bradycardia***
 - Results from:
 - Excessive vagal stimulation
 - SA nodal ischemia (Inferior MI)
- Rate > 100 bpm = ***Sinus Tachycardia***
 - Results from:
 - Pain / anxiety
 - CHF
 - Volume depletion
 - Pericarditis
 - Chronotropic Drugs (Dopamine)

Electroencephalography (EEG)

- An electrophysiological monitoring method to record electrical activity of the brain.
- Typically noninvasive, with the electrodes placed along the scalp, although invasive electrodes are sometimes used in specific applications.
- EEG measures voltage fluctuations resulting from ionic current within the neurons of the brain.
- In clinical contexts, EEG refers to the recording of the brain's spontaneous electrical activity over a period of time, as recorded from multiple electrodes placed on the scalp.
- Diagnostic applications generally focus on the spectral content of EEG, that is, the type of neural oscillations (popularly called "brain waves") that can be observed in EEG signals.

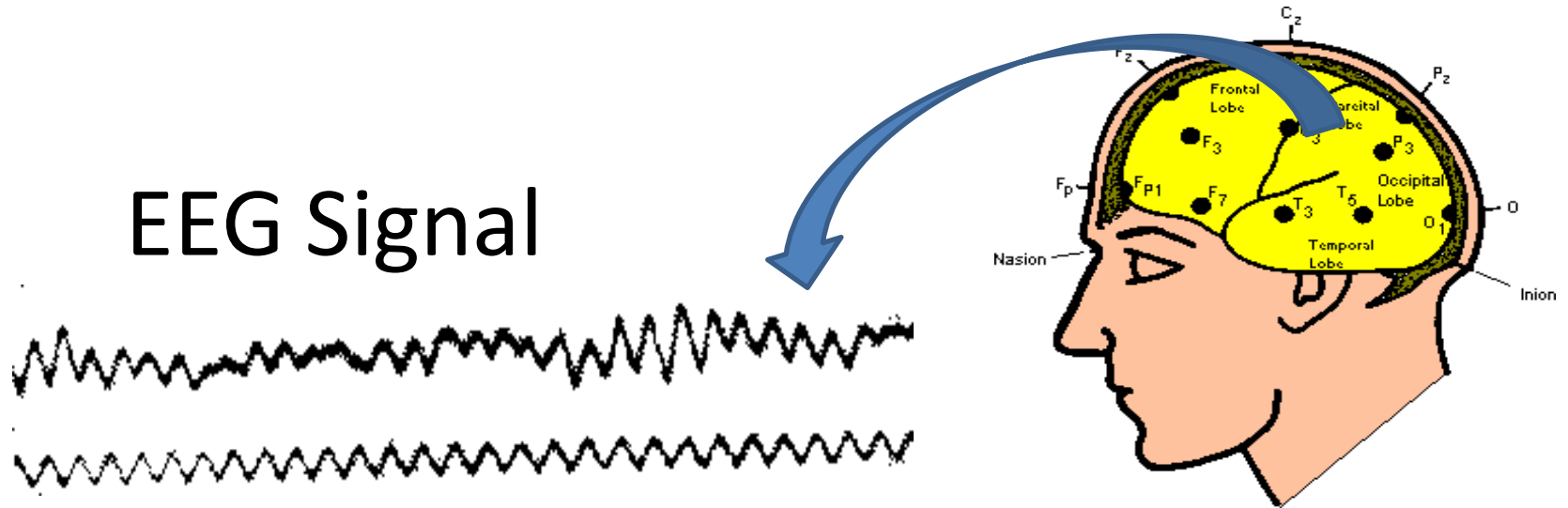
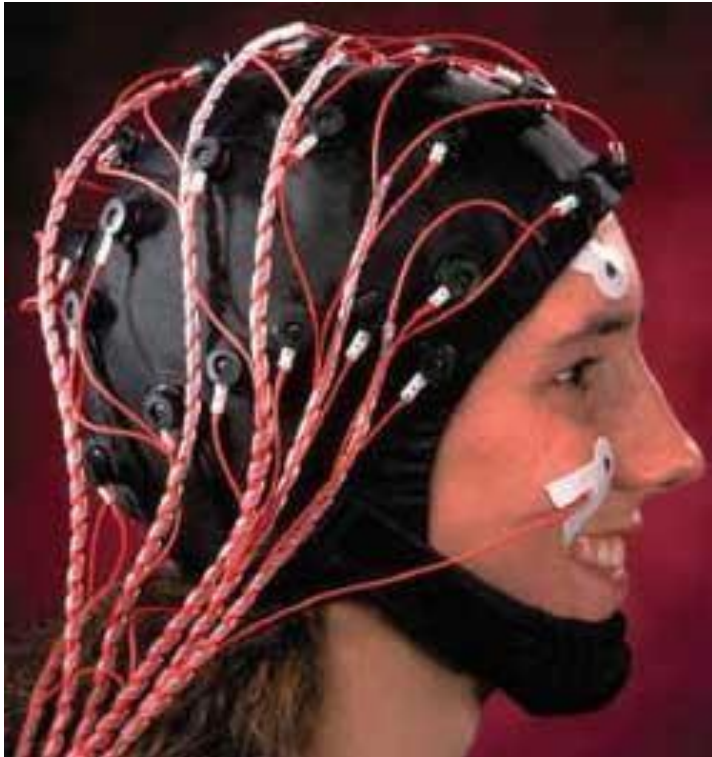
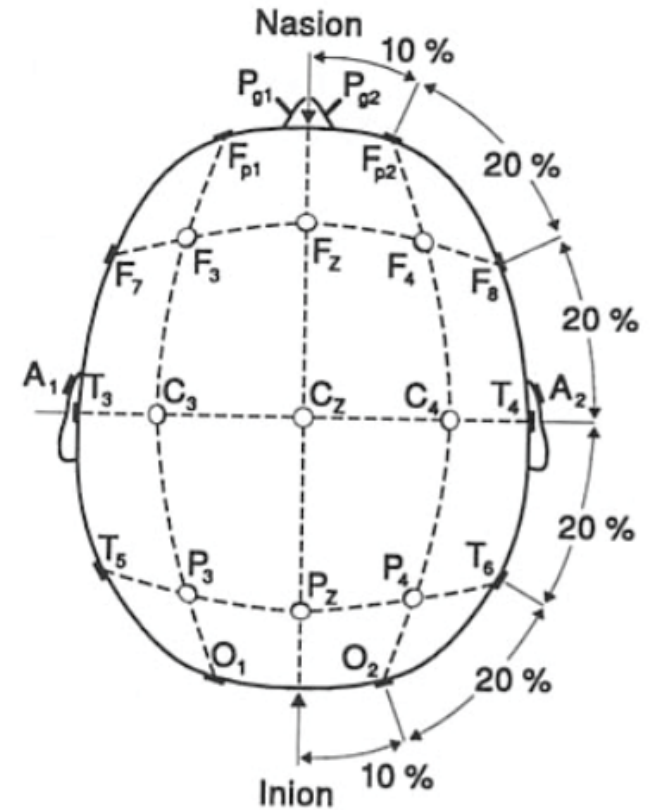


Fig.: EEG signal originating from the brain

- The Electroencephalogram (EEG) is a recording of electrical activity originating from the brain.
- It is recorded on the surface of the scalp using electrodes, thus the signal is retrievable non-invasively.
- Signal varies in terms of amplitude and frequency
- Normal frequency range: 0.5Hz to 50 Hz.



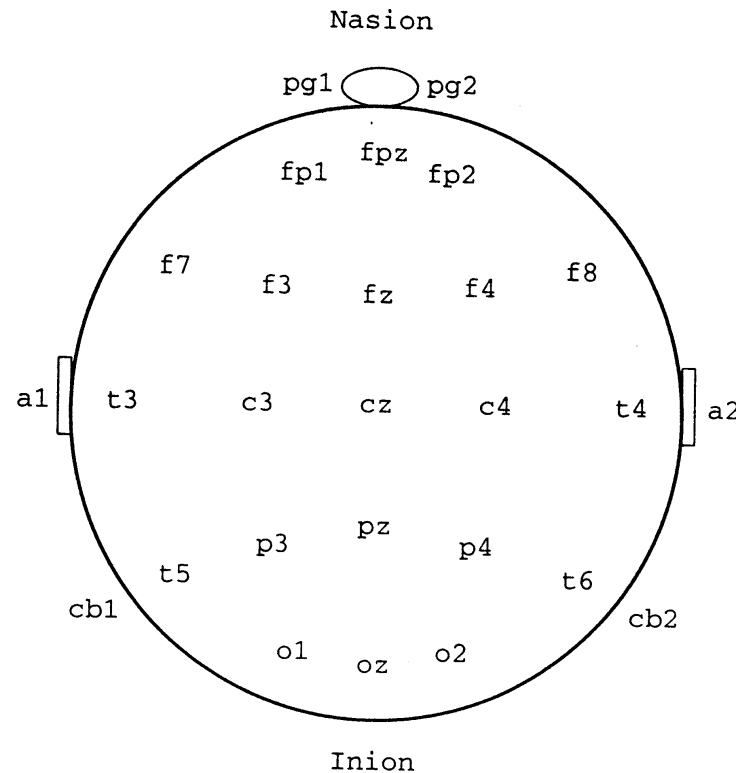
EEG Electrode – cap



**locations of
the 10/20 system**

Electroencephalogram (EEG)

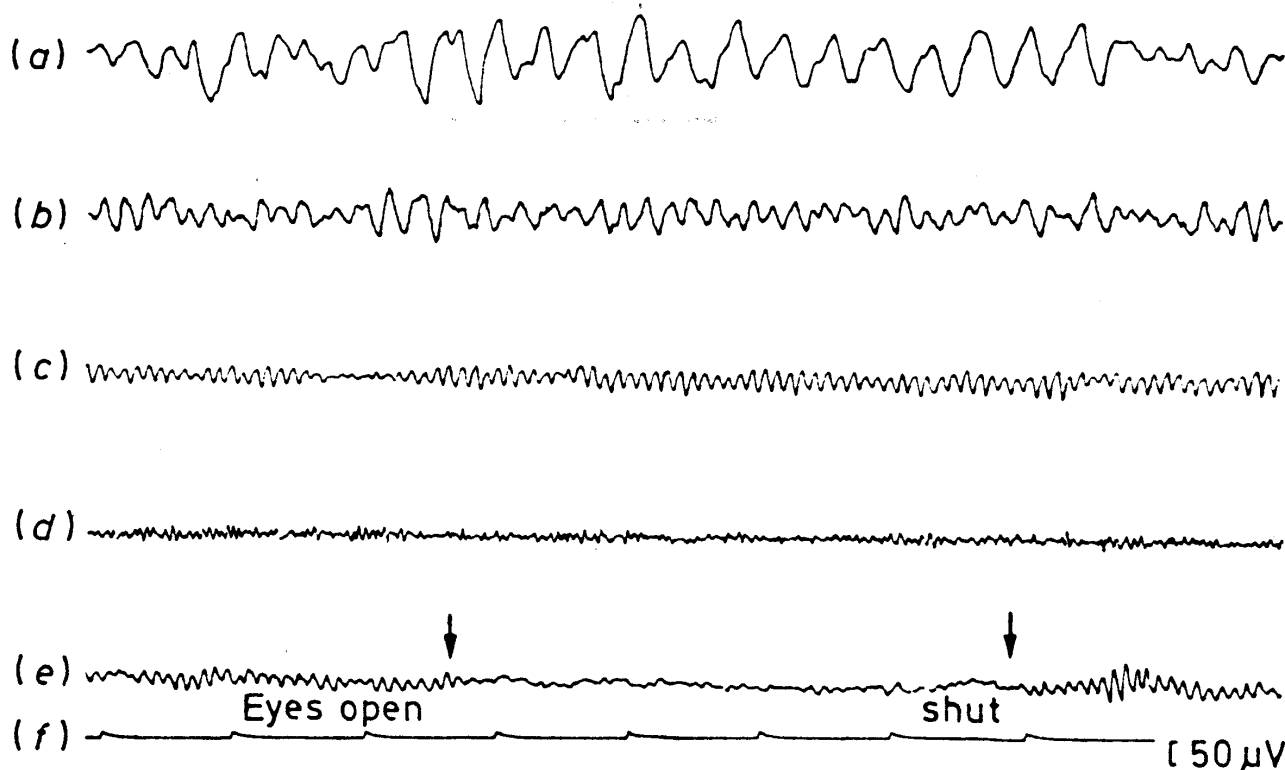
- The 10-20 system of electrode placement for EEG recording.



Electroencephalogram (EEG)

- The commonly used terms for EEG frequency range:
 - Delta (0.5-4 Hz): deep sleep
 - Theta (4-8 Hz): beginning stages of sleep
 - Alpha (8-13 Hz): principal resting rhythm
 - Beta (>13 Hz): background activity in tense and anxious subjects

Electroencephalogram (EEG)



● a: delta, b: theta, c: alpha, d: beta, e: blocking of alpha rhythm by eye opening, f: marker 50 μV , 1 sec

EEG - applications

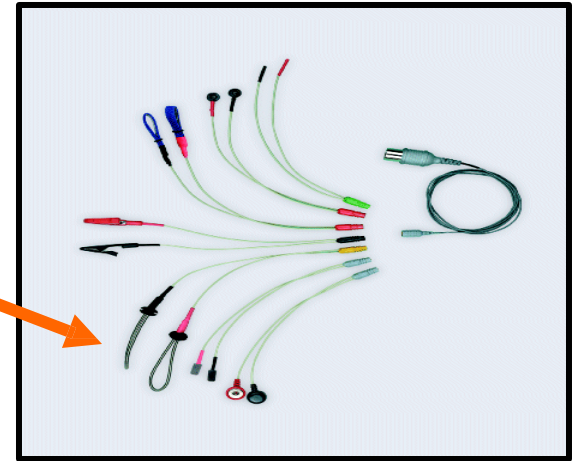
- **Diagnostics (Epilepsy, Oncology, ..)**
- **Cognitive Sciences**
- **Sleep Analysis**
- **Human Computer Interfaces (BCIs)**
- **Pharmacology**
- **Intensive Care, Monitoring**

Electromyography (EMG)

- Electromyography (EMG) is a technique for evaluating and recording the activation signal of muscles.
- EMG is performed by an **electromyograph**, which records an **electromyogram**.
- Electromyograph detects the electrical potential generated by muscle cells when these cells contract and relax.

ELECTRODE TYPES

Intramuscular -
Needle Electrodes

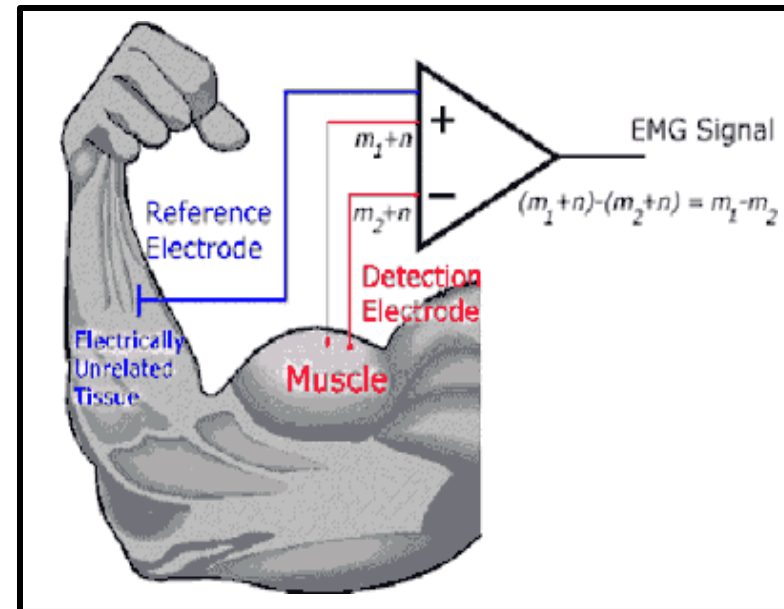


Extramuscular - Surface
Electrodes



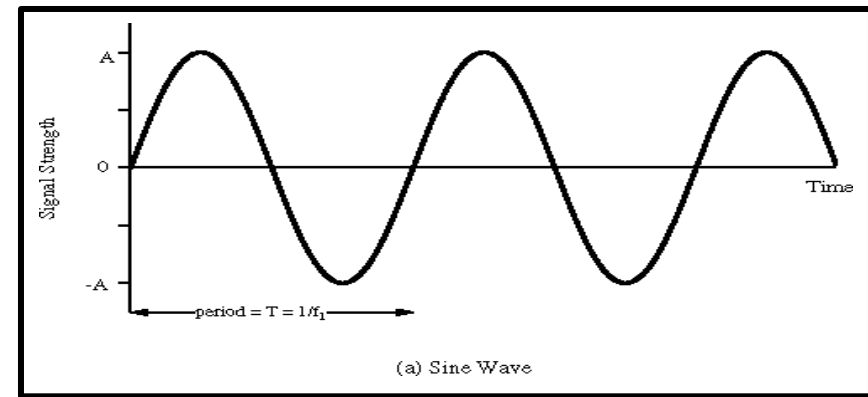
EMG PROCEDURE

- Clean the site of application of electrode;
- Insert needle/place surface electrodes at muscle belly;
- Record muscle activity at rest;
- Record muscle activity upon voluntary contraction of the muscle.

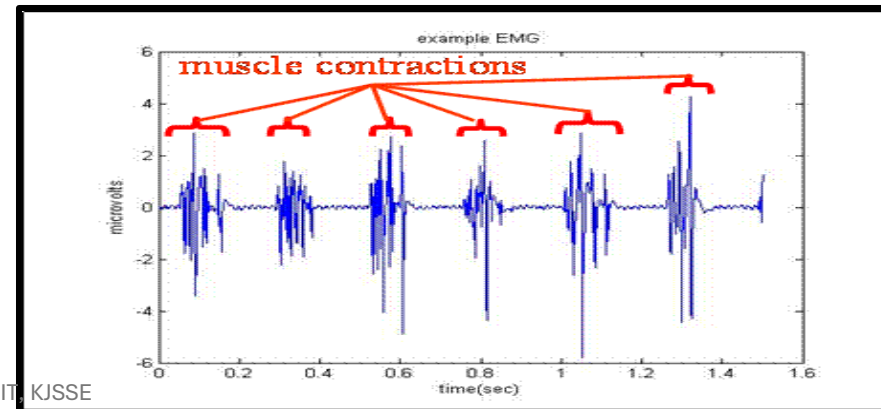


EMG Contd.

- Muscle Signals are Analog in nature.
- EMG signals are also collected over a specific period of time.

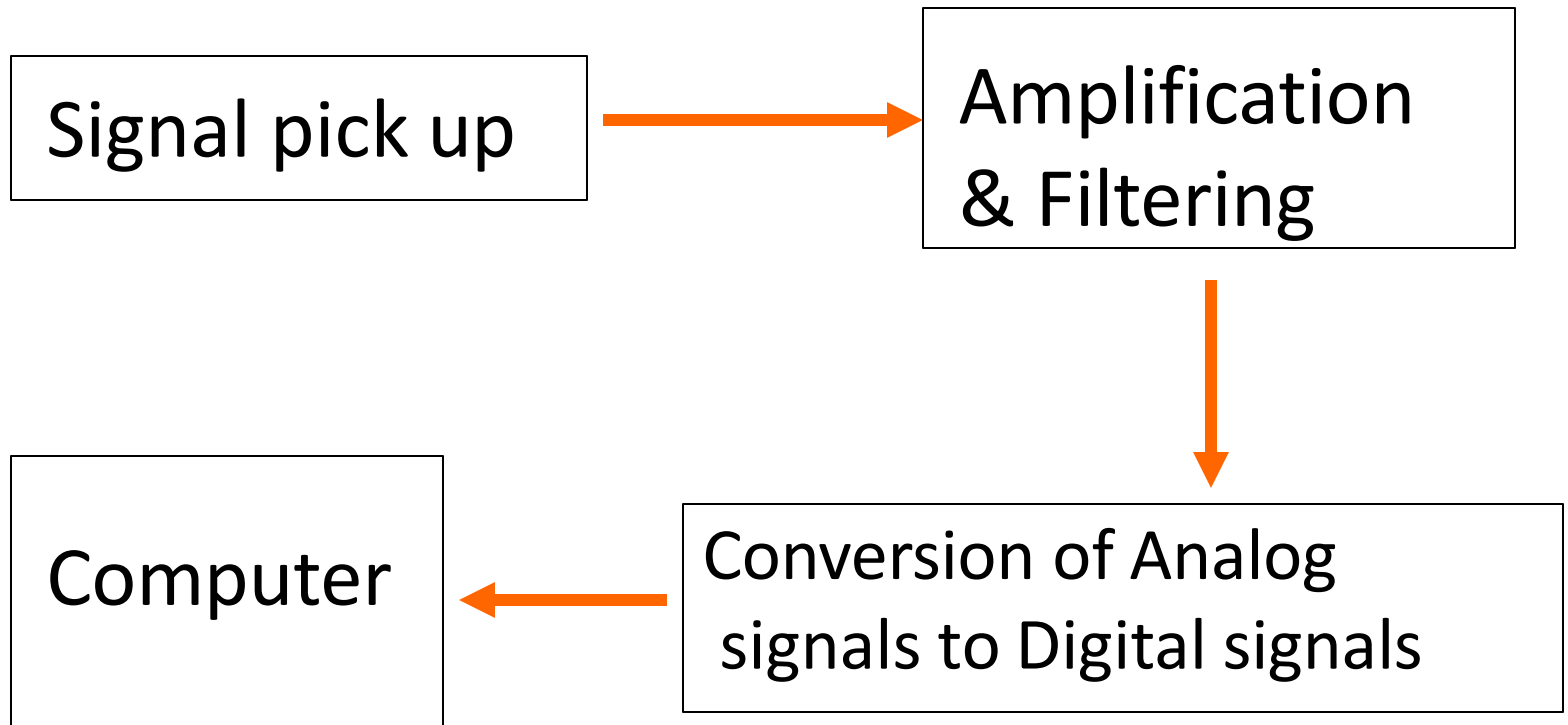


Analog Signal



EMG Contd.

EMG processing:



Factors Influencing Signal Measured

- Geometrical & Anatomical Factors
 - Electrode size
 - Electrode shape
 - Electrode separation distance with respect to muscle tendon junctions
 - Thickness of skin and subcutaneous fat
 - Misalignment between electrodes and fiber alignment
- Physiological Factors
 - Blood flow and temperature
 - Type and level of contraction
 - Muscle fiber conduction velocity
 - Number of motor units (MU)
 - Degree of MU synchronization

Factors That Influence the Signal Information Content of EMG

Factor	Influence
Neuroactivation	- firing rate of motor unit AP's - no. of motor units recruited - synchronization of motor units
Muscle fiber physiology	- conduction velocity of fibers
Muscle anatomy	- orientation & distribution of fibers - diameter of muscle fibers - total no. of motor units
Electrode size/orientation	- no. of fibers in pickup area

Factors That Influence the Signal Information Content of EMG

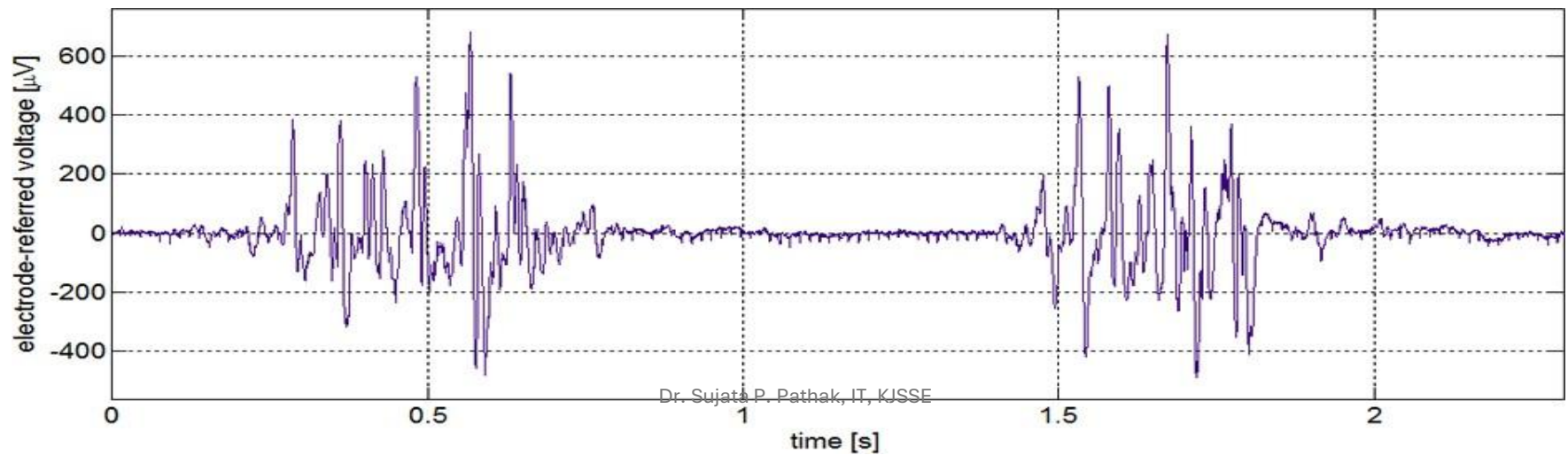
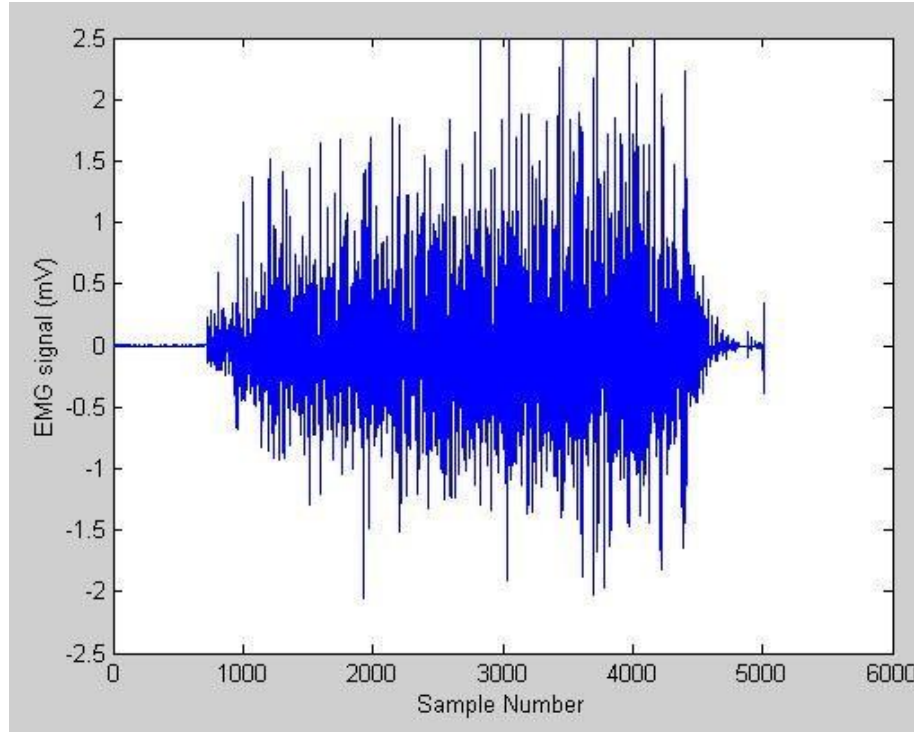
Factor

Influence

Electrode-electrolyte
Interface with
Bipolar electrode
configuration

- type of material and site
- electrode impedance decreases increasing frequency
- distance between electrodes
- orientation of electrodes relative to the axis of muscle fibers

EMG signal



What can be learned from an EMG?

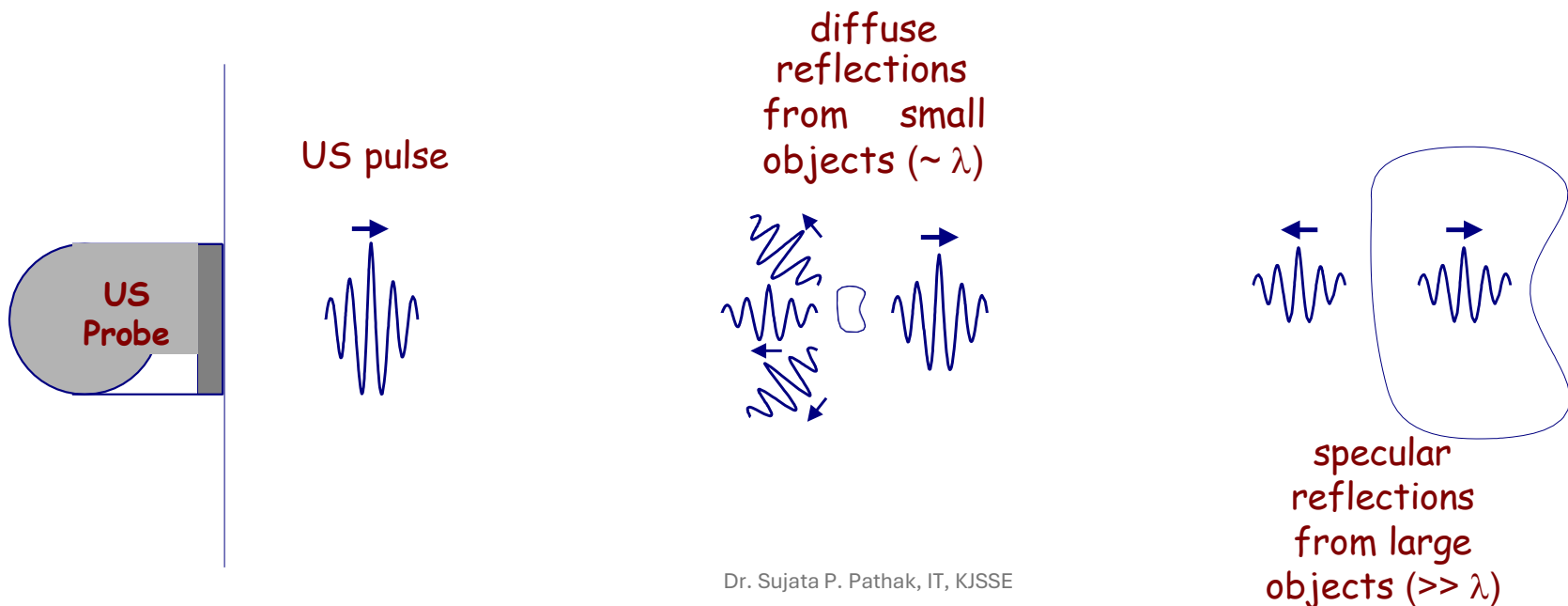
- Time course of muscle contraction
- Contraction force
- Coordination of several muscles in a movement sequence
 - These parameters are DERIVED from the amplitude, frequency, and change of these over time of the EMG signal
- Field of Ergonomics: from the EMG conclusions about muscle strain and the occurrence of muscular fatigue can be derived as well

APPLICATION OF EMG

- EMG can be used for diagnosis of Neurogenic or Myogenic Diseases.
- Rehabilitation
- Functional analysis
- Active Prothetics, Orthosis
- Biomechanics, Sports medicine

Ultrasound (US) Imaging

- US uses high frequency (> 1 MHz) ultra-sound waves (i.e., not electromagnetic) to create static and real time anatomical images
- contrast results from reflections due to sound wave impedance differences between tissues
- at diagnostic levels, no deleterious biological effects from US pulses
- technique similar to submarine ultrasound, a sound pulse is sent out, and the time delays of reflected "echoes" are used to create the image
- image texture results from smaller scatters (diffuse reflectors)
- boundaries result from specular reflections (large objects)



US Images

- by sending pulses out along different directions in a plane, slice images of anatomy are produced for viewing on monitor
- US does not work well through lung or bone, used mainly for imaging abdominal and reproductive organs
- one of the most well known US procedures is the examination of the living fetus within the mother's womb
- 3D imaging scanners now available (real time, so called 4D)

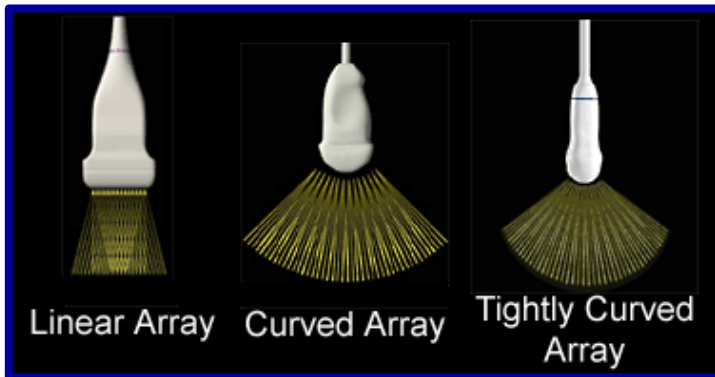




US Scanner

US Scanner

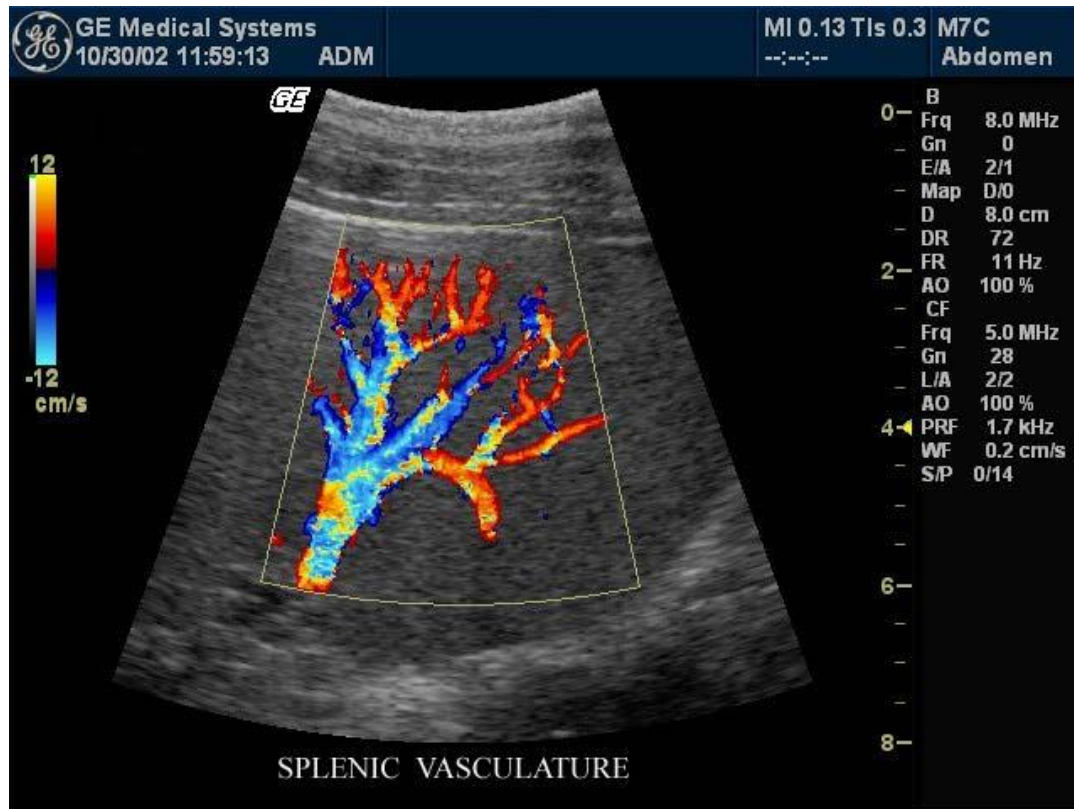
- scanner features probes, data processing computer, and image viewing monitor
- probes specialized for exam requirements
- modern probes feature phased transmit/receiver arrays to electronically steer and focus the US beam



US Probes

Doppler Ultrasound Measures Blood Flow

- using a special form of US called Doppler (just like police speed RADAR) the speed and direction of flowing blood can be measured and illustrated in color images
- Doppler US allows Radiologists to image vasculature and detect blocked blood vessels in the neck, and elsewhere



Transition to a Digital Imaging Environment

- modern radiology is making a transition to a digital imaging environment or **PACS** (picture archiving and communications system)
- advantages include efficient image distribution and reduced storage requirements
- integral to PACS, is digital image acquisition
- computer based modalities inherently digital
- film based modalities now being phased out by digital technologies

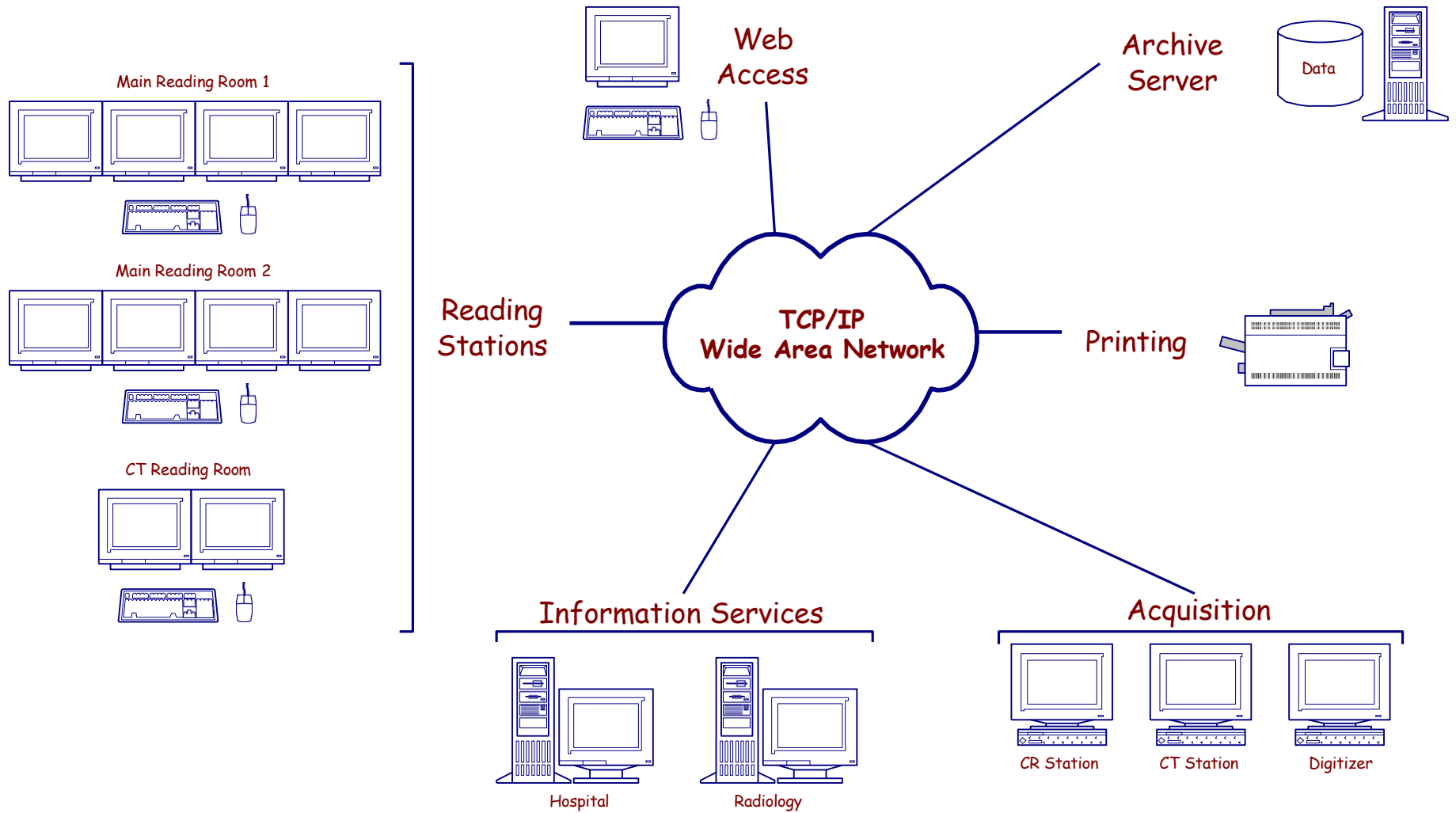


film (hard-copy) reading



PACS soft-copy reading

PACS Environment Example



Digital Imaging Technologies Replacing Film in Radiology

DR: digital radiography uses an x-ray imaging detector **electronically** connected to a readout system (e.g., flat panel detectors).



CR: computed radiography uses a **storage phosphor** x-ray imaging detector which is read out by a separate digital reading device

Role of the Physicist in Diagnostic Radiology (revenge of the nerds)

- Ensure equipment is producing high quality images
 - image quality control
 - periodic checks of equipment
 - supervise preventive maintenance
- Reduce dose to patients and personnel
 - monitor radiation dose records
 - evaluate typical doses for procedures
 - recommend equipment changes and/or dose reduction strategies

Suggested Reference Material

- Physics of Radiology, A. B. Wolbarst, Medical Physics Publishing (ISBN 0-944838-95-2)
- Search the Internet

Useful Links

- <https://www.youtube.com/watch?v=RYZ4daFwMa8>
- <https://www.youtube.com/watch?v=JSxd0UTt5gQ>

Any question

